The Effect of Public Sector Relocations on Regional Development in Germany

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Can governments create private sector jobs in lagging regions by relocating public sector employees? I study a program moving about 5,200 public sector jobs from thriving cities to lagging towns in one of Germany's largest states. Using linked employer-employee social security records, I compare municipalities affected by the relocations with similar municipalities outside the policy state. After the first relocation, private employment in receiving municipalities rises by 2.8% on average relative to matched municipalities, with no offsetting decline in sending municipalities. Following individual public sector workers affected by the relocations, I find that jobs move, but people do not.

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1 Introduction

Can governments narrow regional disparities in economic opportunities by moving public sector jobs from thriving to lagging regions? Governments have long paid subsidies to firms to locate in targeted lagging regions. Countries such as Germany and South Korea are now moving public sector jobs to these regions to stimulate local economic development instead. [1] In 2019, Germany's public wage bill amounted to 8.1% of GDP (Eurostat 2026b), whereas federal and state spending on the country's main place-based policy program[2] amounted to 0.03% of GDP (Deutscher Bundestag 2021; Eurostat 2026a). Because political and administrative power has historically been centralized, much of this wage bill is concentrated in national and subnational capitals (Galiani and Kim 2008; Dascher 2000). Relocating just 1% of the German public wage bill would move about €2.9 billion a year across space. How redistributing public employment as a place-based policy affects the lagging regions that gain this wage bill and the thriving regions that lose it is therefore a first-order policy question.

My paper answers this question with five contributions by studying an ongoing public employment reallocation policy in Bavaria, in the south of Germany. The policy is moving 5,196 public sector jobs and 1,170 paid training positions from thriving to lagging regions in two phases from 2015 to 2030. The jobs alone equal roughly 1.7% of the state's pre-policy direct workforce.[3] I examine *worker-level* and *place-level adjustment* in both *locations gaining and locations losing public jobs*, analyze the policy's *channels and welfare effects in general equilibrium*, and extend a quantitative spatial model to incorporate the *public sector's contribution to agglomeration economies* and its use of *floor space*.

I trace worker adjustment using the IAB's Integrated Employment Biographies, which link workers to establishments over time. Building on Goldschmidt and Schmieder (2017), I use establishment and worker identifiers to follow employment through changes in public employers. I combine public administration industry codes with the workplace postcodes of the program's origins and destinations to identify public sector agencies that close or substantially contract and agencies that open or substantially expand. I follow the original agencies' 2014 workforce to 2024, distinguishing workers who move, commute, remain employed elsewhere without changing residence, or leave regular employment. I then identify workers hired from outside the original workforce and compare the characteristics and wages of employees at closed and newly opened agencies. These comparisons describe how the workforce adjusts and how recruitment changes across locations.

To estimate place-level effects, I use the program's pre-announced eligibility criteria to construct comparison groups. Every municipality within the policy state is benchmarked

[1] See Freitas (2026) for a cross-country analysis of past and current public employment relocation programs.

[2] The Gemeinschaftsaufgabe Verbesserung der regionalen Wirtschaftsstruktur (GRW), or Joint Task for the Improvement of Regional Economic Structures.

[3] These figures describe intended final staffing. Phase I's scheduled implementation period ended in 2025, although staffing continued to build up. Phase II is still being implemented (Bavarian Ministry of Finance 2026).

against the state average on a socioeconomic distress index. Municipalities scoring below 90% of the state average are classified as lagging and become eligible to receive relocated public agencies. I reproduce the index for municipalities elsewhere in Germany and match receiving municipalities to those that would have been eligible had they been located in the policy state. I match sending municipalities, which the index classifies as thriving, to equally thriving municipalities outside the policy state. This approach yields never-treated comparison groups outside the policy state.[4] I combine a self-collected dataset of relocated agencies and the state's official public-agency register with a yearly municipal employment panel constructed from the Federal Employment Agency's linked employer-employee records for 2007 to 2024. I compare employment in sending and receiving municipalities with their matches before and after the relocations. I use the Sun and Abraham (2021) estimator to address staggered treatment timing.

I use a quantitative spatial model to evaluate the policy's channels and welfare effects throughout Germany, including regions that neither send nor receive public jobs. The model distinguishes productivity spillovers within the private sector from those between the public and private sectors and accounts for competition between public agencies and private firms for commercial floor space. These channels help explain why receiving municipalities gain private employment while sending municipalities show no evidence of losses. In the model, the government assigns public jobs to workplaces, public employees choose where to live and commute, and public agencies occupy commercial floor space. The model distinguishes agglomeration economies within the private sector from productivity spillovers between the public and private sectors. At the origins, an agency's departure removes public-to-private spillovers but releases space for private activity. At the destinations, the agency generates these spillovers but competes with private firms for commercial floor space. I match the model to the reduced-form private employment responses to quantify these competing forces.

My worker-level results show that public sector jobs move but public sector workers largely do not. Few incumbents change residence to follow their jobs, while about 10% work at the new locations without moving. Many remain employed near their previous residence, typically at another public agency, while others leave regular employment, largely through retirement. As the baby-boom generation retires, the policy state shifts its hiring from urban centers to small towns where the state's uniform pay schedule buys more qualified workers. Public sector agencies opening in these towns recruit almost entirely from outside the original workforce. Compared with the former sending workforce, their employees are younger, more often women, more educated, and more often in specialist and expert jobs, yet earn less per day. Many already have public sector experience. The program thus brings comparatively well-paid public employment to destination towns while changing who fills these jobs.

After the first relocation, private employment in receiving municipalities rises by 2.8% on average relative to matched controls. Public agencies opening in these municipalities employ a more female workforce, while the composition of private employment shifts

[4] I additionally exclude municipalities with the strongest commuting links to the policy state from the control pool.

toward men. Commercial rents rise modestly, while residential rents increase at longer horizons.

In sending municipalities, I find no evidence of local employment losses that offset the gains at the destinations. Residential and commercial rents also rise despite the release of public office space, as sending locations become more attractive to private firms. The government estimates that the program's first phase alone freed about 37,000 square meters of office space, mainly in the state capital (Bavarian Ministry of Finance 2026).

Matching the model to the reduced-form employment responses yields an estimated public-to-private productivity elasticity of 0.022. For a receiving municipality, a 10% increase in surrounding public employment density, weighted by proximity, raises private firms' productivity by about 0.21%, holding other determinants fixed. The model separates the consequences of these spillovers from those of reallocating commercial floor space. The program redistributes economic activity at a small national output cost. Among eligible destinations, the program's choices come close to minimizing that cost.

This paper contributes to the literature on worker adjustment to employer shocks, public employment as a place-based policy, equal public sector pay and unequal recruitment quality and the spatial allocation of economic activity.

My first contribution is to show how the workforce adjusts when public employers relocate. Gathmann et al. (2020) find that mass layoffs cause persistent local employment losses, while initially employed workers suffer smaller losses; geographic mobility protects younger workers in that initial workforce. I distinguish the adjustment of the original agency workforce from recruitment at receiving agencies. Goldschmidt and Schmieder (2017) find wage losses of about 10–15% after outsourcing, associated with the loss of firm-specific rents. In my setting, average public workplace wages decrease at receiving locations and increase at sending locations. The worker records show that receiving agencies recruit a lower-paid workforce, while many incumbents remain at sending locations and earn higher nominal wages. I follow the workers employed at relocated agencies and show that public sector jobs move but public servants largely do not. Few incumbents change residence, and receiving agencies hire mostly from outside the original workforce.

My second contribution is to the literature on place-based policies, which asks whether spatially targeted interventions improve economic outcomes in lagging regions (Moretti 2012; Kline and Moretti 2014b; Ehrlich and Seidel 2018; Duranton and Venables 2018; Ehrlich and Overman 2020). Most of this work studies subsidies, tax incentives, and infrastructure investment. Relocating public sector jobs differs from these instruments because the government moves its own activity rather than altering private firms' incentives. The closest literature to my work studies how public employment affects private employment, using expansions, contractions, and relocations of the public sector (Faggio and Overman 2014; Senftleben-König 2014; Jofre-Monseny et al. 2020; Auricchio et al. 2020; Freitas 2026). Much of the relocation evidence concerns large interventions concentrated in a single destination, such as the German government's moves to Bonn and back to Berlin and the BBC's move from London to Salford (Becker et al. 2021; Faggio, Schlüter, et al. 2025; Nathan et al. 2024). In another study (Freitas 2026), I review public employment decentralization programs and find gains of about 7 private

jobs per 10 relocated public jobs, concentrated in non-tradable activities (Freitas 2026). This paper studies public employment relocation as a place-based policy targeting many lagging locations. It is therefore closer to studies of programs targeting multiple locations in the United Kingdom, South Korea, and Denmark (Faggio 2019; Lee et al. 2024; Jensen et al. 2026). I add evidence from a program whose eligibility rules were announced before implementation, and I estimate effects in the municipalities that send the jobs as well as in the lagging municipalities that receive them. I also contribute evidence on the gendered effects of place-based policies. Women are overrepresented in public employment (Gornick and Jacobs 1998; Gomes and Kuehn 2026) and disproportionately affected when the public sector contracts (Glasmeier and Lee-Chuvala 2011). In receiving municipalities, the relocated agencies become more female, but the additional private jobs that their spillovers create in the private sector go mostly to men. Gender differences in sector specialization therefore relocate with the public agencies.

My third contribution concerns recruitment under uniform public sector pay across different local labor markets. Propper and Van Reenen (2010) and Britton and Propper (2016) show that stronger private outside wage opportunities worsen hospital and school performance under nationally regulated pay in England. Dal Bó, Finan and Rossi (2013) find that higher public wages attract more able applicants without reducing public-service motivation and help fill positions in less attractive municipalities. Di Porto, Dustmann, Giannetto and Incoronato (2026) show that salary caps induce disproportionately able public managers to leave where private sector opportunities are available. I add the location of public employment as a policy margin. When administrative functions can serve users from different locations, the government can change the labor markets in which it recruits while retaining a common pay schedule. My findings suggest that relocation can strengthen the state's position relative to competing employers, although access to specialized workers may constrain recruitment in smaller labor markets.

My fourth contribution is to bring the public sector into quantitative spatial models (Ahlfeldt, Redding, et al. 2015; Redding and Rossi-Hansberg 2017). Earlier work models public employment in spatial equilibrium in a two-city economic geography model of Bonn (Becker et al. 2021), in a search model of the average Spanish city (Jofre-Monseny et al. 2020), and in models of public sector pay and local public goods (Guillouzouic et al. 2021; Guillouzouic et al. 2024). Rothenberg et al. (2025) study the relocation of Indonesia's capital in a dynamic quantitative spatial model with public employment. I build a quantitative spatial model in which the government assigns public sector jobs to workplaces, public employees choose where to live and commute, and public agencies occupy floor space. A novel feature of the model is that public agencies occupy floor space, allowing me to quantify how their relocation affects real estate markets and private activity. Matching the model to a quasi-experiment, I estimate agglomeration economies between the public and the private sector separately from those within the private sector, and find them to be small. In their two-city model of Bonn, Becker et al. (2021) find positive amenity spillovers from public employment, while the fitted productivity spillover to the private sector is close to zero and slightly negative.

My fifth contribution is to the debate on whether place-based policies trade efficiency for equity (Kline and Moretti 2014b; Kline and Moretti 2014a; Busso et al. 2013; Gaubert

et al. 2025). Relocating public employment may be a case in which a place-based policy removes a misallocation instead of creating one. Public employment is concentrated not only in capitals but also at their centers. The location of city hall is even a common proxy for a city's central business district (Holian 2019). Political accountability benefits from capitals that are close to their citizens (Campante and Do 2014), but many administrative jobs need neither proximity to political decisions nor proximity to the people they serve.[5] Faggio (2019) describes such jobs as providing tradable public services. They resemble the tradable private sector more than the usual view of the public sector suggests. Yet the literature mostly studies the size of the public sector or the provision of public goods (Alesina et al. 2001; Algan et al. 2002; Guillouzouic et al. 2021). The program I study moves the state's back office, such as tax processing, payroll, land registries, geodata, and IT services, out of thriving centers. These functions serve users statewide without face-to-face contact. Many of these jobs occupy prime locations, the hyper-dense clusters of economic activity where commercial rents and productivity peak (Ahlfeldt, Albers, et al. 2026; Ahlfeldt, Heblich, Seidel, and Yin 2026). By moving these jobs out of prime locations and making room for more productive private firms, the program may correct a historical misallocation of this type of back office public employment.

Section 2 describes the policy program and Section 3 the data. Section 4 shows how the workforce adjusts when agencies relocate. Section 5 presents the empirical strategy and Section 6 the effects in receiving and sending municipalities. Section 7 develops the quantitative spatial model and evaluates the relocations for the economy as a whole. Section 8 concludes.

2 Institutional Setting

The public sector is a major employer in Germany and accounts for a substantial share of government expenditure.[6] By choosing where to locate public jobs, governments can direct employment and recurring wage income toward disadvantaged places. The state of Bavaria used this instrument to address spatial inequality within its borders.

Bavaria lies in southeastern Germany, bordering Austria and the Czech Republic. Its 13.25 million inhabitants in 2024 made it Germany's second most populous state, with a population comparable to Pennsylvania's (Bavarian Statistical Office 2025a; U.S. Census Bureau 2025). It is also Germany's largest state by area (Bavarian Statistical Office 2025b).

Economic opportunities differ sharply across Bavaria. Wealthy municipalities around Munich and in the south contrast with struggling areas in the north and along the

[5] Studying cuts in Italian public employment, which includes local public good providers such as schools and health services and has long served as a regional subsidy (Alesina et al. 2001), Auricchio et al. (2020) find that an inflated public sector crowds out private employment, mostly in tradable sectors.

[6] According to the OECD, general government employment accounted for approximately 11% of total employment in Germany, 15% in the United States, 21% in France, and 17% in the United Kingdom in 2021 (OECD 2023, Figure 12.1). In the same country order, the World Bank reports wage bills of 17.4%, 25.8%, 22.1%, and 23.4% of public expenditure in 2019 (World Bank 2024).

Czech border.[7] The state thus faces a familiar regional-policy problem: how to bring jobs to places that have fallen behind its economic centers. In 2015, Bavaria began reallocating public employment from its major agglomerations to these places as part of its *Heimatstrategie*, a regional development strategy.

To identify eligible destinations, the government benchmarked all municipalities within the state in terms of economic and demographic performance against the state average of an index. This index consists of five indicators of population forecasts, unemployment, employment density, income, and migration among young adults. Municipalities that fall behind the state average form the eligible set for the reallocation policy. A municipality qualifies through either its own index value or its district's index value, with a threshold of 90 relative to a state average of 100 (Bavarian Ministry of Finance 2015). Panel (a) of Figure 1 shows the municipalities meeting these eligibility criteria. Appendix A reports the index components and their construction.

Within the eligible set, the government favored towns officially designated as service centers for surrounding communities in the state's regional development plan and places that had not already received substantial support through other place-based higher-education policies (Bavarian Ministry of Finance 2025). Ministries assessed whether potential destinations offered suitable premises and infrastructure for the agencies they proposed to relocate. They selected sites in coordination with the finance ministry; the cabinet approved the program, and the responsible ministries implemented it (Kaltenhauser 2023). Panel (b) adds the service-center designations, while panel (c) shows the selected destinations and origins. Some selected destinations lie outside the index-defined eligible set.

The agencies selected for relocation perform functions that are not tied to local public good provision. Tax processing, information technology, and other specialized administrative functions can serve the same users from a different municipality. Ordinary schools and local police stations are therefore excluded from the relocation program (ibid.). The program includes whole and partial agency moves, as well as the establishment or expansion of offices. For agency moves, Munich is an important origin: relocating an office shifts public jobs away from the state's largest employment center while adding them to a receiving town.

I study both phases of the public employment reallocation program. Together, they are moving 5,196 jobs and 1,170 paid training positions, equivalent to about 1.7% of Bavaria's pre-policy direct state workforce when counting jobs alone.[8] The training positions include public-service apprenticeships organized as degree programs: participants receive full remuneration during their training, so these positions also bring wage income to receiving places (Freistaat Bayern 2026, Art. 75). I assembled a relocation dataset that

[7] Before the program, average annual wages in high-wage municipalities, concentrated around Munich and Starnberg, were more than twice those in low-wage municipalities, concentrated in Upper Franconia and Lower Bavaria (Bavarian Statistical Office 2015).

[8] The government's March 2026 statement reports intended final staffing of 2,511 jobs and 730 training positions in phase I, and 2,685 jobs and 440 training positions in phase II. Phase II remains under way, with implementation extending to 2030 (Bavarian Ministry of Finance 2026). The employment benchmark uses the direct state workforce before the program; Appendix A gives the calculation.

records agencies, destinations, origins, staffing, and implementation dates. Appendix A documents the program's administrative components and intended final staffing.

Most workers in Bavaria's direct state administration are civil servants. Before the program, civil servants and judges accounted for 68.4% of this workforce; the remaining workers held employment contracts (Statistical Offices 2014; Bavarian Statistical Office 2014). Civil servants have a distinct legal status and are absent from the social-security employment records used in the empirical analysis[9].

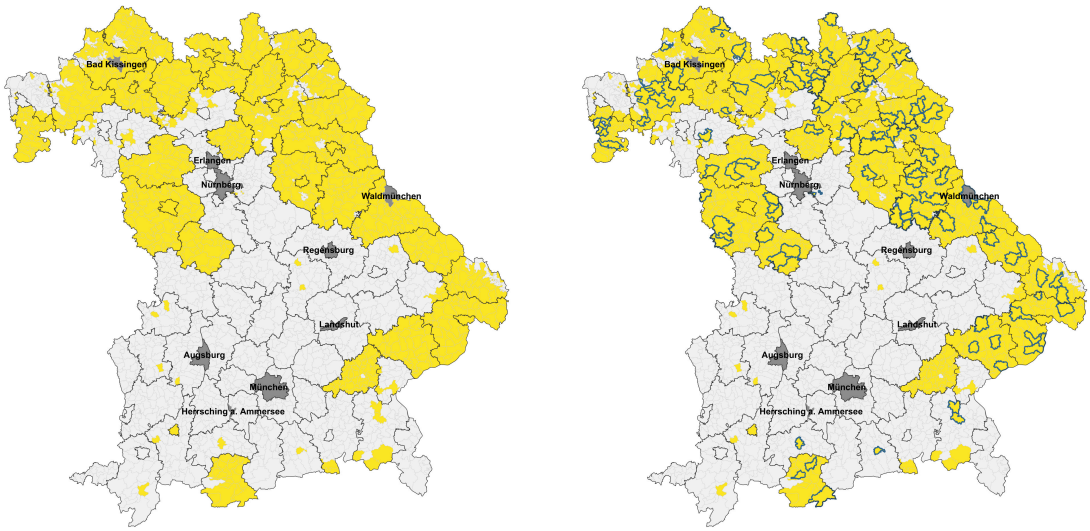
Public base pay does not vary across the policy state's municipalities for a given position and experience level. Contractual employees are paid under the collective agreement for state-government employees which Bavaria negotiates jointly with the other participating German states. Wage setting is thus coordinated nationally rather than determined by individual agencies or municipalities. Civil-service pay is set by Bavarian law. Both systems use schedules in which the job determines the pay grade and experience determines progression within it, much as grades and steps organize pay in the US federal General Schedule (TdL 2025; Freistaat Bayern 2026; OPM 2026).

The schedules provide a benchmark for earnings in the relocated jobs. Under the February 2025 schedule, qualified administrative work accessible without a university degree, illustrated by grade E9a, starts at about €3,520 gross per month. A position requiring a master's-level qualification, illustrated by E13, starts at about €4,630 (LfF 2025b). Appendix Table 11 gives the corresponding civil-service grades and pay ranges, distinguishing career level from experience.[10]

Agencies build up employment at their destinations gradually. An office may open with a small team, recruit additional staff, and move into permanent premises only later. Positions can shift as workers leave or retire at the origin, allowing employment at the destination to grow over several years (Bavarian Ministry of Finance 2026). I date treatment to the government press release announcing an office's opening. This convention captures responses that occur while staffing is still being built up, and the event studies allow me to check for responses before the announced opening.

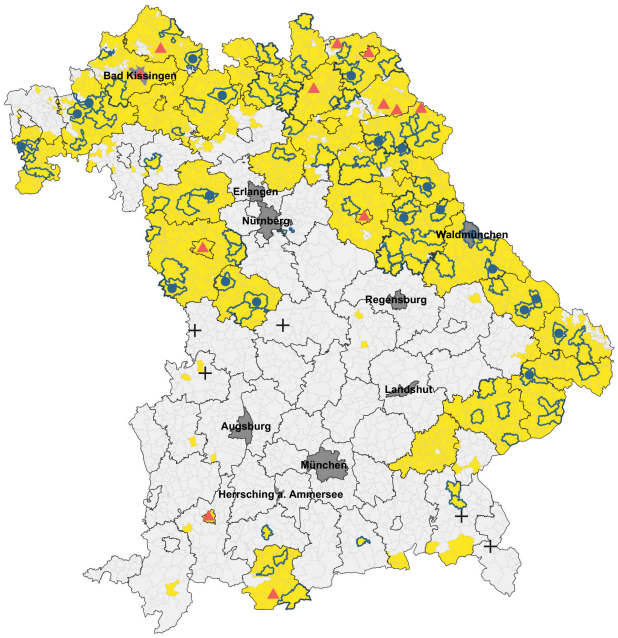
[9] Municipal civil-service employment data from the statistical offices will allow me to extend the analysis to both groups of public employees in the future.

[10] A public-sector wage premium also depends on the private-sector alternative. Using German survey data, Bonaccolto-Töpfer et al. (2022) find a wage advantage for women but a pay penalty for men after accounting for worker characteristics and selection. These estimates refer to Germany as a whole.



(a) Municipal eligibility

(b) Service-center designations



(c) Receiving and sending locations

Figure 1. Geographical targeting and site selection. Panel (a) marks municipalities meeting the index-based eligibility criteria in yellow. Panel (b) outlines eligible service centers in teal. Panel (c) adds receiving municipalities: blue circles mark service centers, red triangles other municipalities, and black crosses municipalities outside the index-defined eligible set. Sending locations appear in gray throughout.

Employees do not have to move when their agency relocates. The personnel framework rules out compulsory permanent transfers, relocation-related dismissal, and involuntary downgrading. Workers can seek reassignment, while those who follow an agency can receive training and support for moving or commuting, including a conditional €3,000 mobility payment. Telework is available where duties permit (Bavarian Ministry of Finance 2021); in practice, two work-from-home days per week are common. Incumbents can therefore stay, commute, or move, while agencies can also recruit at the destination. Public jobs therefore need not bring an equal number of new residents. Section 4 analyzes how these conditions are translated into realized worker responses.

Relocations also change the use of office space. Receiving agencies rent, adapt, or construct premises, often before reaching full staffing. At the origins, the government estimates that phase I freed about 37,000 square meters of office space, mainly in Munich, the state capital (Bavarian Ministry of Finance 2026). Vacated premises create scope for other uses, including private-sector offices. Sending locations thus lose public jobs while potentially releasing space for other employers.

Providing new premises can be a substantial part of relocation costs. For three completed tax-office relocations with matched cost and staffing figures, construction spending averaged about €180,000 per public job (Kaltenhauser 2023). Firm-specific investment incentives provide a benchmark. Approved direct grants for Northvolt's Heide battery plant in the North of Germany amounted to about €230,000 per planned job, while proposed support for Intel's Magdeburg semiconductor plant in the center of Germany amounted to €3.3 million per planned job.[11]

3 Data

Studying public employment relocation requires knowing where agencies move and what happens to the people and firms around them. I assemble a record of the Bavarian program, use successive editions of the Bavarian public-agency register to document agency locations, and measure employment with linked employer–employee data from the Institute for Employment Research (IAB). The empirical analysis follows municipalities over time. The same records provide the baseline for the quantitative model, which I quantify for all German municipalities.

3.1 Data on the Relocation Program

I construct the relocation dataset by combining official program records. Starting from the BayernAtlas records, I collect the agency name, responsible ministry, origin and destination, intended staffing, program phase, and implementation status (State Office for Digitization, Broadband and Geodetic Measurement 2024). I then follow each project through the annual *Heimatstrategie* reports, government announcements, agency websites,

[11] The relocation ratio covers construction spending per public job accommodated. The Northvolt and Intel ratios use announced support and planned direct employment; Intel cancelled its project in July 2025 (Intel 2025). These figures illustrate spending magnitudes, not relative cost-effectiveness. Appendix A documents the calculations and other reported relocation costs.

and local newspaper coverage. These sources turn a list of planned relocations into a record of where the policy was implemented and when employment could begin to change.

Implementation often unfolds over several years. An agency may open a branch before moving into its permanent building, while staffing grows toward the announced target. I therefore distinguish the start of operations from construction milestones and intended final staffing. I use the official state government sources to establish implementation dates. The program totals in the preceding section describe the intended scale of program rollout phases I and II.

I complement these project histories with a new source for studying the geography of public employment: the Bavarian public-agency register (Bavarian State Statistical Office 2026). Its records identify agencies and their addresses, administrative affiliations, municipality codes, and headquarters or branch status. I reconcile the January 2014, March 2023, and June 2026 editions to follow agencies across changes in names, identifiers, and addresses. Comparing these records with the policy histories helps distinguish a move to another municipality from an address correction or an organizational change. It also reveals why a partial relocation may leave the parent agency at its original address. The register and IAB data serve separate purposes: the former documents agency locations, while the latter measures employment at the establishment level. Appendix B describes the registry reconciliation.

3.2 Employment Data

The main employment source is the IAB's Integrated Employment Biographies (IEB), from which I use linked worker and establishment records for 2007–2024. The IEB combine employers' social-security notifications with administrative records on benefit receipt, job search, and labor-market programs (IAB FDZ 2026; IAB 2017). This source offers a combination essential for the analysis: broad coverage of insured employment, detailed locations, and worker histories that continue across employers. It allows me to study small local labor markets and to connect changes in local employment with the movements of individual workers.

The records distinguish the municipality where a person works from the municipality where that person lives. I use this distinction to construct annual workplace and residence employment measures and to observe commuting across municipal boundaries. The same person and establishment identifiers link employment spells over time; the records also report industry, earnings, education, and demographic characteristics. These dimensions make it possible to ask whether employment grows around a receiving agency, whether local residents benefit, and whether the adjustment involves changes in residence or commuting. County-level aggregates provide the corresponding measures for the model comparison.

Table 1 describes employment growth and the administration share in receiving and sending municipalities. Workplace employment grew by 6.0% in receiving municipalities and 14.9% in sending municipalities between 2014 and 2024; employment among residents grew by 10.4% and 15.8%, respectively.

Industry and employment status answer different questions. Industry codes identify

	Receiving municipalities		Sending municipalities	
	2014	2024	2014	2024
Workplace employment (2014 = 100)	100	106.0	100	114.9
Administration workplace employment (2014 = 100)	100	120.0	100	122.8
Other-industry workplace employment (2014 = 100)	100	105.3	100	114.6
Employed residents (2014 = 100)	100	110.4	100	115.8
Administration share of workplace employment (%)	4.68	5.27	4.24	4.58
Establishments	715	659	14,742	14,213
Municipalities	34	34	6	6

Table 1. Employment growth and composition in treated municipalities. Employment indices set the corresponding 2014 mean to 100. Administration shares are unweighted mean municipal percentages; establishments are unweighted municipal means. Administration denotes WZ 2008 division 84.

public administration, defense, and social security, but do not identify every publicly owned employer. Public schools and hospitals, for example, fall in other industries. Civil servants are also outside the IEB employment records because their jobs are exempt from social-security contributions, so relocated civil-service positions are not observed in the employment data. Self-employment is not covered either.

The worker histories also provide evidence on how agencies adjust their workforce. I identify candidate establishments in two steps. First, I use the workplace postcodes recorded for program origins and destinations to locate public-administration establishments in the IAB data. Second, I look for establishment openings, closures, and unusually large employment changes between 2014 and 2024. At destinations, I retain establishments that are absent in 2014 but present in 2024, or that expand by at least two baseline standard deviations. At origins, I retain establishments that disappear by 2024 or contract by at least one baseline standard deviation. The baseline standard deviation is calculated across establishments' employment changes from 2010 to 2013, separately for origin and destination postcode areas. This procedure captures both whole-establishment changes and substantial expansions or contractions within continuing establishments.

I then use worker identifiers to follow employees of candidate sending establishments in 2014 through 2024. Among those subsequently employed at candidate receiving establishments, changes in recorded residence distinguish movers from workers who keep their residence and commute. I also identify employees at receiving establishments who were not in the original sending cohort. Section 4 presents this evidence. Appendix B gives the sample definitions, including the treatment of postcodes that occur at both ends of the program.

The model uses employment from the same records in 2014, the year before the first relocation. Workplace and residence counts by municipality distinguish public administration, defense, and social security (WZ 2008 division 84) from all other industries, which define private employment. They supply the baseline cross-section for the model,

while the annual panel supplies the empirical changes. Because civil servants are outside these records, baseline public employment in the model covers employees subject to social-security contributions. For the version of the model on Census towns (2011) in Appendix E, I use employment from the 2011 German Census instead. Its reference date, May 9, 2011, precedes the program, and its employment coverage includes civil servants as well as workers outside social-security employment (Statistical Offices' Research Data Centres 2026).

Table 2 reports the administration shares of municipal employment in 2014. Administration accounts for an average of 6.2% of workplace employment across German municipalities and 5.0% in the policy state. Weighted by employment, the shares are 4.9% and 4.2%. Workplace and residence measures describe different sides of local labor markets and provide the baseline for the model.

	Germany		Bavaria	
	Mean	SD	Mean	SD
Administration share of workplace employment (%)	6.2	8.1	5.0	4.3
Administration share of employed residents (%)	6.1	2.3	4.8	1.3
Aggregate share of workplace employment (%)	4.9		4.2	
Municipalities	10,637		2,055	

Table 2. Administration shares of municipal employment in 2014. Unweighted means and sample standard deviations across the municipalities of the model. Sector shares are percentages calculated within each municipality; the aggregate share divides administration employment by total workplace employment of the group. Administration comprises public administration, defense, and compulsory social security (WZ 2008 division 84). Employment subject to social-security contributions; civil servants are not covered. 1,131 municipalities, 104 of them in Bavaria, have no administration workplace employment.

3.3 Controls Used in Mahalanobis Matching

For matching, I assemble the indicators used to assess regional disadvantage: the population outlook, unemployment, workplace employment density, income per taxpayer, and net migration of young adults. I use administrative data from the German Regional Database and Federal Employment Agency, together with the government's index documentation, to reproduce the policy benchmark outside Bavaria (German Statistical Offices 2026). I also account for municipal size and pre-policy outcome trajectories.

3.4 Travel Costs

Commuting connects the local effects of the policy across space. I use origin–destination flows to measure these connections and road travel times to characterize the cost of reaching a workplace. The model's commuting decay comes from flows of private employees between municipalities in the 2014 IAB records; Federal Employment Agency

flows of Bavarian residents provide an untargeted check. For the model, the travel-time matrix describes car journeys between municipalities, in minutes. I compute road travel times with OpenStreetMap routing through OSRM.

3.5 Rents, Housing Prices, and Commercial Floor Space

To measure the price of residential space, I use the micro-geographic rent indices of Ahlfeldt, Heblich, and Seidel (2023). I take the municipal rent measures for 2014, the baseline year of the model; the version on Census towns (2011) uses those for 2011. These inputs describe the housing costs households face in the model and allow location choices to depend on more than wages and commuting.

Office rents provide a separate measure of the cost of space used by firms. I use the municipal office-rent series associated with Ahlfeldt, Heblich, Seidel, and Yin (2026) and take the corresponding municipal measures for 2014. Developable land in each municipality comes from the CORINE 2012 land-cover data, and the floor area of the office buildings the state constructed for the program calibrates the office space of public jobs.

4 Worker Adjustment and Public Capacity

The program moved public jobs, not public workers. As the baby-boom generation retired, Bavaria shifted its rehiring from urban centers to small towns where its uniform pay buys more qualified workers. The new public sector agencies are staffed almost entirely by new employees who are younger, mostly women, more educated, and more often in specialist and expert jobs, yet earn less per day. Destination towns gained a stream of stable, well-paid public jobs that locals, especially women, can take without commuting to the city or leaving.

4.1 Tracing Public Sector Agencies and Their Workers

Linked employer-employee records at the postcode level let me locate public sector agencies that closed and opened and follow workers through the relocation.

Goldschmidt and Schmieder (2017) trace outsourced jobs through clustered flows of workers who move together between establishment identifiers in German social-security records. I build on their idea of following jobs through establishment and worker identifiers in the Integrated Employment Biographies of the IAB (IAB FDZ 2026; IAB 2017). I first select public administration establishments[12] at the workplace postcodes of the program's origins and destinations. I then keep those whose employment changes unusually between 2014 and 2024. At the origins, these are public sector agencies that close or shrink by at least one standard deviation of the 2010 to 2013 employment changes. At the destinations, they are public sector agencies that open or grow by at least two standard deviations. Section 3 and Appendix B give the details.

[12] WZ 84 is division 84 of the German classification of economic activities (WZ 2008), public administration and defence and compulsory social security.

I follow the 2014 workforce of the closed and shrinking public sector agencies to 2024 and compare it with the workforce that replaced it, distinguishing six groups.

- *Incumbents* are all regular employees of closed or shrinking origin public sector agencies in 2014, 2,352 workers in total.
- *Movers* are incumbents who work at an opened or expanding destination public sector agency in 2024 and have changed their municipality of residence.
- *Commuters* are incumbents who work at an opened or expanding destination public sector agency in 2024 and still live in the same municipality as before the relocation.
- *Stayers* are incumbents in regular employment outside these public sector agencies in 2024 who still live in the same municipality.
- *Incumbents no longer observed* are not in regular employment in 2024, because they exited the social security system (civil service or retired).
- *New employees* are workers at opened or expanding destination public sector agencies in 2024 who were not incumbents.

To compare the old and the new workforce, I contrast the workers of closed public sector agencies in 2014 with those of opened public sector agencies in 2024. The 2014 averages of the sending and receiving municipalities serve as local benchmarks. Appendix B shows in detail which sectors stayers moved into and which sectors new employees came from, and the public sector agency profiles as figures.[13]

4.2 Worker Adjustment

Few incumbents followed their jobs. Most kept working near their previous place of residence or left regular employment, largely through retirement.

Only 1.8% of incumbents moved their place of residence and work at a destination public sector agency in 2024, and another 10.3% work at the new locations without having moved (Table 3). In total, incumbents make up 0.9% of the destination workforce.

Following was a choice between two equally paid jobs. Followers keep their pay grade and experience step, and the personnel framework rules out compulsory transfers (personnel framework, Section 2). Wages could not compensate for a longer commute either. In Danish firm relocations, wages rise by about 0.15% per extra kilometre of commuting three years after the move (Mulalic et al. 2014), but the uniform German

[13]The public sector agency groups count public administration workers, while the municipal benchmarks count all sectors, so each gap partly compares public administration with the whole economy. Opened public sector agencies are measured in 2024 and the benchmarks in 2014, so any general shift toward specialist and expert jobs over the decade may explain part of the gaps. Civil servants are not in the IAB social-security data, so the public sector agencies' expert shares are understated and their skilled shares overstated, and this bias may differ between closed and opened public sector agencies. Closures and openings may include changes of establishment identifiers or local reorganizations unrelated to the program. The occupation shares cover regular employees with a valid occupation code, which leaves out apprentices and marginal jobs.

Table 3. Worker adjustment and recruitment shares

Worker outcome	Share (%)
Original sending workforce, observed in 2014	
Receiving employment, changed residence	1.8
Receiving employment, unchanged residence	10.3
Employment elsewhere, unchanged residence	40.7
No longer observed in regular employment	40.0
Other or unclassified outcomes	7.2
Receiving workforce, observed in 2024	
Workers from outside the original sending workforce	99.1
Workers from the original sending workforce	0.9

Notes: The first panel reports 2024 outcomes as shares of the 2014 workforce at closed or substantially contracting sending establishments. Residence refers to the recorded municipality of residence. Receiving employment refers to employment at opened or substantially expanding receiving establishments. The second panel reports shares of the 2024 receiving workforce.

public sector pay schedule rules this out. Adjustment therefore had to run through following, switching jobs, moving home, or leaving employment. Leaving was about as common as switching. In 2024, 40.7% of incumbents kept working near home and 40.0% were no longer in regular employment.

Retirement explains much of the leaving. The staff of closed public sector agencies were older than the local workforce, with a mean age of 47.3 years against 41.8 in the sending municipalities (Table 4). More than a quarter of them (28%) were aged 55 or older, against 16% locally, so they reached retirement age while the program ran from 2015 to 2030.

For those who switched jobs, the origin's public sector absorbed the shock. In 2024, 78% of stayers still worked in the private sector after the relocation (Figure 19 in Appendix B). Especially Munich is a thick public labour market, and no one was legally displaced. After mass layoffs, Gathmann et al. (2020) find that workers under 50 avoid employment losses through geographic mobility. Here, mobility within the local public sector does the same job.

Those who gain are destination residents, a majority of them women. Women hold 64% of the jobs at opened public sector agencies, against 48% of all jobs in the receiving municipalities in 2014 (Table 5). Women value commuting time about 20% more than men (Le Barbanchon et al. 2021), so public jobs in small towns may reach a female labour pool with high commuting costs. These jobs also pay well by local standards, about 10 euros per day more than the receiving municipalities' average.

4.3 Public Capacity

The retirement wave let the policy state redraw its map of public jobs at low cost. It lost experience that retirement would have taken anyway and moved its hiring to towns

Table 4. Closed public sector agencies and sending municipalities

	Closed public sector agencies (2014)	Sending municipalities (2014 avg.)	Closed − sending
Mean worker wage (gross EUR/day)	108.7	108.1	+0.6
Share women	0.48	0.50	−0.02
Mean education (1–3)	2.16	2.18	−0.02
Mean age (years)	47.3	41.8	+5.5
Share under 35	0.15	0.32	−0.17
Share 55+	0.28	0.16	+0.12
Share foreign	0.01	0.13	−0.12
Share helper	0.09	0.14	−0.05
Share skilled	0.71	0.53	+0.18
Share specialist or expert	0.20	0.33	−0.13

Notes: Closed public sector agencies are public administration establishments (WZ 84) at sending postcodes that exist in 2014 but not in 2024. Their values pool all their regular employees in 2014. Sending municipalities are unweighted averages of 2014 municipal values across all sectors. Wages are deflated gross daily wages. Education is coded 1 for no vocational qualification, 2 for vocational training, and 3 for an academic degree. Occupation shares use the requirement level of the KldB 2010 occupation among workers with a valid code. Differences are computed from the rounded values.

where it competes less for skilled workers against the private sector.

In other relocation programs like in Denmark, workers do not move (Rigsrevisionen 2018). The one moment it becomes cheap to redraw the map of public employment is when workers leave voluntarily. The program ran from 2015 to 2030, as the baby-boom generation retires, and positions could shift as workers left or retired at the origin (Section 2). Without layoffs, origin positions had to empty on their own, and an older workforce provided that attrition. The state did not move individual workers. It moved the place where the next generation is hired.

The new public sector agencies are staffed by a different workforce. Compared with closed public sector agencies in 2014, opened public sector agencies in 2024 employ younger workers, with a mean age of 44.3 against 47.3 years and more workers under 35 (26% against 15%, Table 6). Their staff are more often women (64% against 48%), more educated, and more often in specialist or expert jobs (32% against 20%). Yet they earn 5.2 euros less per day.

Part of the pay gap could be explained through a gap in experience. Under § 16 of the German public sector pay schedule, a hire without relevant experience starts at step 1. Progressing in the scale happens through more years of experience and reaching step 6, for example, takes 15 years. A gap of 3 to 5% in daily wages is consistent with lower seniority, although part-time work, which these records do not separate, could also contribute to it. Opened public sector agencies have more specialist and expert jobs and more young workers, yet lower daily pay, consistent with hiring at low steps of the German public sector pay schedule. Read this way, the gap is the price of lost experience, not of lower grades. The experience was lost because many workers retired, which would have happened without the relocation as well.

New public sector agencies have more demanding jobs in towns where expert labour

Table 5. Opened public sector agencies and receiving municipalities

	Opened public sector agencies (2024)	Receiving municipalities (2014 avg.)	Opened − receiving
Mean worker wage (gross EUR/day)	103.5	93.4	+10.1
Share women	0.64	0.48	+0.16
Mean education (1–3)	2.30	2.05	+0.25
Mean age (years)	44.3	42.6	+1.7
Share under 35	0.26	0.29	−0.03
Share 55+	0.27	0.18	+0.09
Share foreign	0.04	0.08	−0.04
Share helper	0.10	0.16	−0.06
Share skilled	0.58	0.62	−0.04
Share specialist or expert	0.32	0.21	+0.11

Notes: Opened public sector agencies are public administration establishments (WZ 84) at receiving postcodes that exist in 2024 but not in 2014. Their values pool all their regular employees in 2024. The other definitions follow Table 4.

Table 6. Opened public sector agencies and closed public sector agencies

	Opened public sector agencies (2024)	Closed public sector agencies (2014)	Opened − closed
Mean worker wage (gross EUR/day)	103.5	108.7	−5.2
Share women	0.64	0.48	+0.16
Mean education (1–3)	2.30	2.16	+0.14
Mean age (years)	44.3	47.3	−3.0
Share under 35	0.26	0.15	+0.11
Share 55+	0.27	0.28	−0.01
Share foreign	0.04	0.01	+0.03
Share helper	0.10	0.09	+0.01
Share skilled	0.58	0.71	−0.13
Share specialist or expert	0.32	0.20	+0.12

Notes: Definitions follow Tables 4 and 5.

is less common. Their skill mix is close to that of the large origin cities, with 32% of workers in specialist or expert jobs, against 33% in the sending municipalities and 21% in the receiving municipalities (Tables 4 and 5). Public sector wages are more or less the same everywhere, but local private sector wages differ, and the opened public sector agencies pay about 10 euros per day more than the receiving municipalities' average. By moving its hiring to these towns, the state faces less competition from private employers for skilled workers. What the state gained is a better draw of new staff per euro. The public sector agencies employ experts at 1.6 times the share in their host towns (16% against 10%).[14]

Less competition for the state means more competition for local firms. Whether the public sector agencies pull the best-educated workers out of the local private sector remains open. Answering it requires following the firm side over the long run, once the programs are fully implemented.

5 Empirical Methodology

I estimate how local labor markets adjust when public jobs arrive and when they leave. For each receiving municipality, I construct a comparison with a similar untreated municipality; I repeat the procedure separately for sending locations. Event studies trace adjustment around the first relocation.

5.1 Matching and comparison groups

The policy directs public jobs toward places classified as disadvantaged relative to the state average. I reproduce the eligibility indicators for municipalities outside the policy state, identifying places that would qualify under the same benchmark. I match receiving municipalities to these comparison places and sending municipalities to similarly thriving locations. The matching accounts for the economic and demographic dimensions of the policy rule, municipal size, and pre-policy outcome trajectories. Within each comparison pool, I select a nearest neighbor by Mahalanobis distance, without replacement, separately for receiving and sending municipalities.

I choose comparison municipalities outside Bavaria to limit contamination by policy spillovers within the state. Even municipalities that neither receive nor send an agency may be affected as workers, firms, and expenditure adjust across nearby locations. As an additional precaution, I exclude municipalities in the 20 non-Bavarian counties with the strongest commuting links to Bavaria.[15]

A causal interpretation requires that, without the relocation, treated municipalities would have followed the same outcome trends as their matched controls. Matching on pre-policy levels and changes addresses observed differences relevant to this comparison. The

[14] Part of the higher expert share may also reflect contract type. If expert posts at new public sector agencies were filled with employees rather than civil servants, which are not observed, the social-security records would show more experts even if the true job mix had not changed.

[15] The exclusion uses the counties identified by their shares of commuters to Bavaria in 2013. Appendix C shows their geography and documents the matching procedure.

design also requires that anticipation does not change outcomes during the comparison's pre-treatment period. Agencies may prepare their premises and employees may adjust before operations begin, so I examine pre-treatment coefficients.

5.2 Staggered Difference-in-Differences

I use the estimator of Sun and Abraham (2021) as the main staggered difference-in-differences specification. It traces employment before and after treatment while allowing the response to differ across cohorts of municipalities first treated in different years. I estimate receiving and sending effects separately, using their matched never-treated municipalities as the comparison groups.

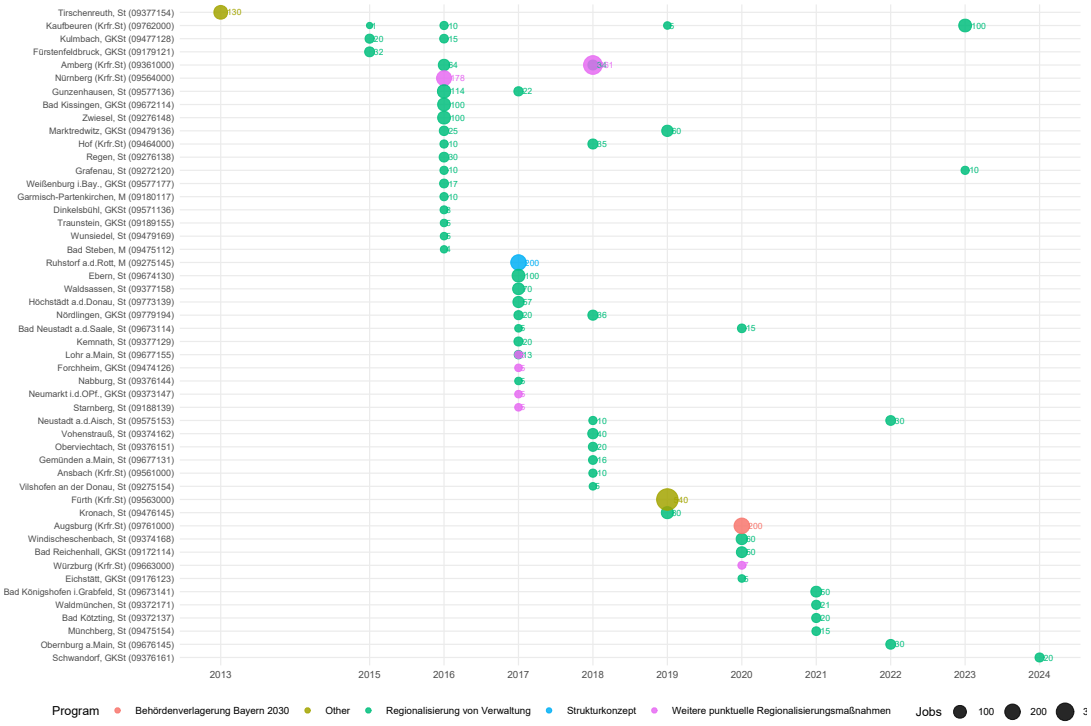
Treatment begins in the year of the first recorded relocation into a receiving municipality or out of a sending municipality. I use the official treatment year where it is available and the recorded completion year otherwise. Some municipalities experience further relocations, so first treatment marks the start of exposure rather than the completion of every project. Figure 2 shows the timing of individual projects by municipality and program; Table 19 reports first- and last-treatment cohorts in the matching roster. The distinction is especially relevant for places receiving several agencies over time.

Let G_i denote municipality i's first treatment year. I estimate cohort-specific event-time effects using

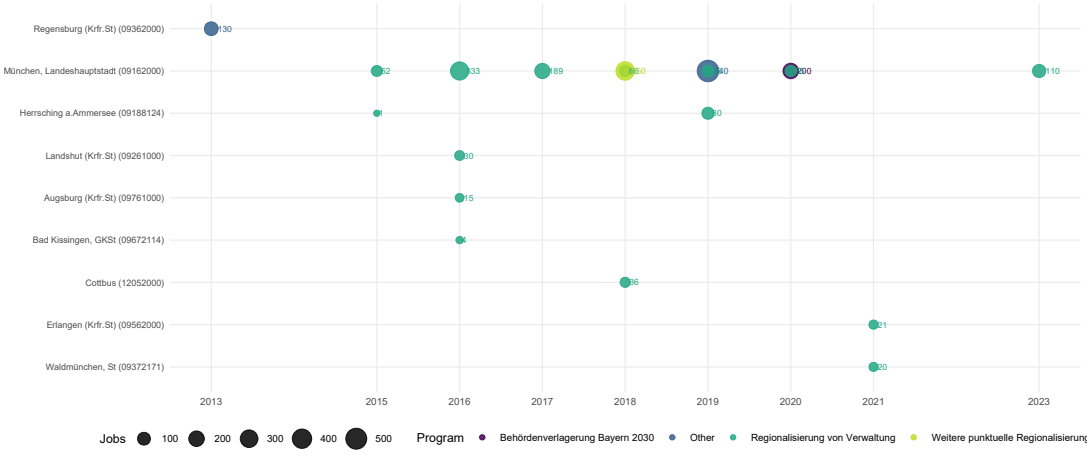
$$Y_{it} = \alpha_i + \delta_{p(i),t} + \sum_{g \in \mathcal{G}} \sum_{\ell \neq -1} \theta_{g\ell} \, \mathbf{1}\{G_i = g\} \, \mathbf{1}\{t - g = \ell\} + u_{it}, \tag{1}$$

where α_i absorbs time-invariant municipal characteristics and $\delta_{p(i),t}$ absorbs shocks common to a matched pair in a given year. The year before first treatment is the reference period. I aggregate the cohort-specific coefficients at each event time to obtain the reported event-study effects. The main employment-count event studies use levels, so their coefficients measure employees. Confidence intervals allow for clustering by municipality.

The next section presents the municipal responses at both ends of the relocations.



(a) Receiving locations



(b) Sending locations

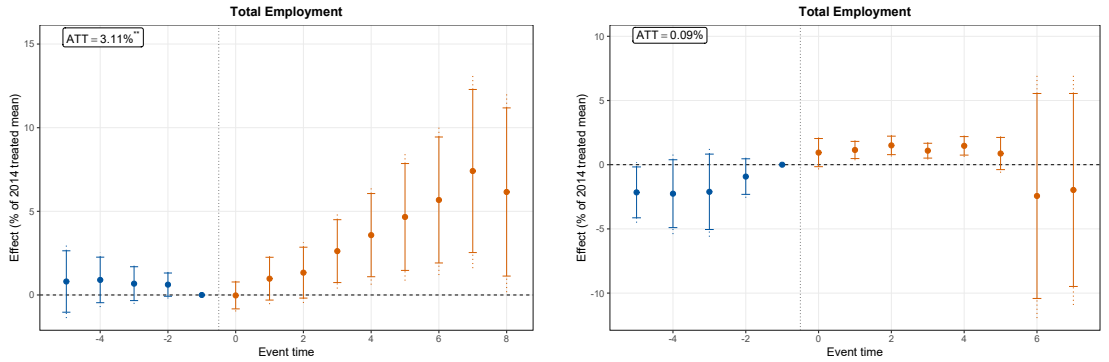
Figure 2. Timing of relocation projects by municipality and program. The figures show the broader program inventory; the matched-sample first- and last-treatment cohorts are reported in Table 19.

6 Place-Level Effects

Receiving municipalities gain private sector jobs as public employment expands, while sending municipalities show no evidence of offsetting private employment losses. Five years after the first relocation, the receiving private employment gain amounts to 4.3% relative to matched controls. I trace how population, employment, commuting, and factor prices adjust at both ends of the program. The figures report Sun–Abraham event studies around each municipality's first relocation, comparing treated municipalities with matched controls outside the policy state. I express employment and population responses as percentages compared to 2014 treated means for each outcome. The appendix reports additional outcomes.

6.1 Employment Outcomes

Receiving municipalities gain workplace employment gradually after the first agency arrives. Total employment, including public and private jobs, rises by 4.7% after five years and 7.4% after seven years (Figure 3). The average post-treatment gain is 3.1%. Sending municipalities also have positive employment estimates in the first six years after the first relocation, averaging 0.1% over the post-treatment period.



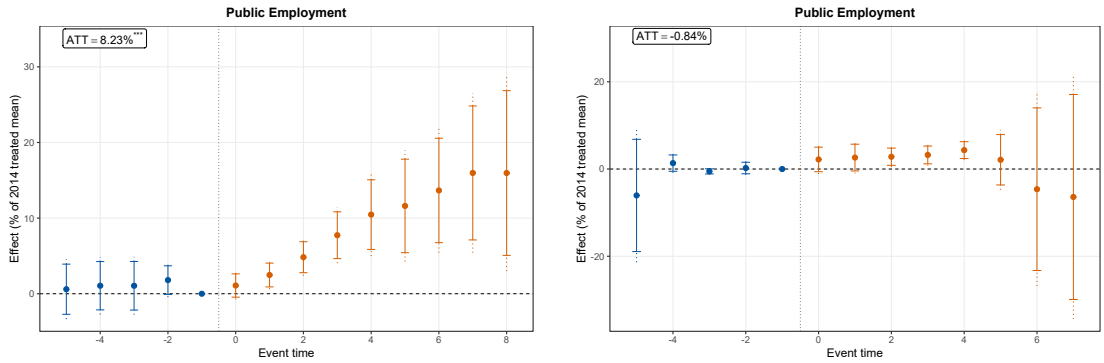
(a) Receiving. The 2014 treated mean is 8,857 employees.

(b) Sending. The 2014 treated mean is 167,362 employees.

Figure 3. Total employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public and private employment. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

I next separate the total employment response into its public and private components. Public employment builds up as receiving municipalities gain relocated agencies and those agencies fill their positions. It rises by 4.8% two years after the first relocation and by 13.7% after six years (Figure 4). The average post-treatment increase is 8.2%. This gradual expansion fits the phased implementation described in Section 2. In sending municipalities, municipal public employment does not fall immediately. The estimates remain positive in the first six years after the first relocation and turn negative afterwards,

averaging −0.8%. The trajectories suggest that the receiving public sector expands before a contraction appears in municipal employment at the origins.



(a) Receiving. The 2014 treated mean is 450 employees. **(b)** Sending. The 2014 treated mean is 5,658 employees.

Figure 4. Public employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public administration, defense, and compulsory social security. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

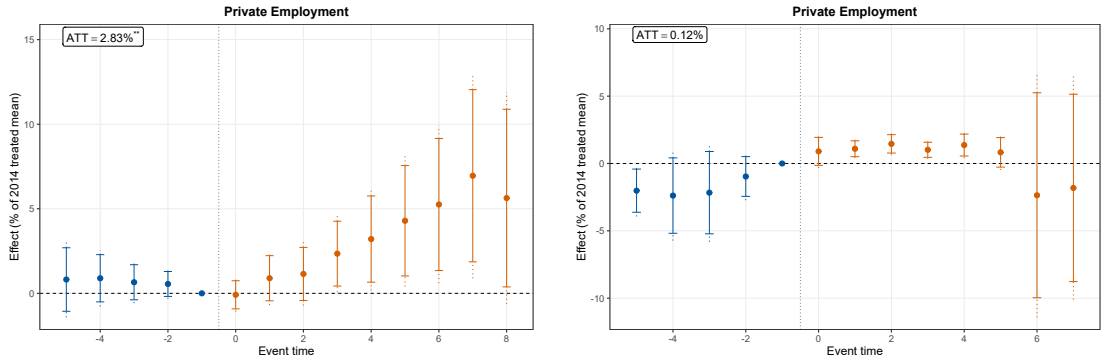
Private employment expands alongside public employment in receiving municipalities. The response is small during the first four years, then grows to 3.2% after four years, 4.3% after five years, and 7.0% after seven years (Figure 5). Averaged across the post-treatment period, the increase is 2.8%. Receiving municipalities thus gain jobs beyond those located in the public agencies themselves, with the private response taking four years to develop. Section 7 uses the long difference in $\log(1 + E)$ from 2014 to 2024 for receiving municipalities relative to their controls as a target.

Sending municipalities' private employment estimates point to expansion rather than to losses that offset the gains at the destinations. They are positive in the first six post-treatment years, averaging 0.1% and reaching 1.5% after two years (Figure 5).

Figure 6 reports local employment.[16] Local private employment in receiving municipalities rises after the first relocation, by 1.5% on average across the post-treatment period. Over the full period from 2014 to 2024, its long difference in $\log(1 + E)$ is 0.038 (standard error 0.014) relative to controls, the second target of the model in Section 7.

Receiving municipalities draw more workers from surrounding places as their employment grows. In-commuting rises by 2.4% one year after the first relocation and by 5.9% after five years (Figure 7). The average post-treatment increase is 3.8%. The additional jobs therefore reach workers who live outside the receiving municipality, consistent with the distinction between jobs and residence in the worker evidence. Sending municipalities also have a positive average in-commuting response of 9.2%, suggesting that they continue to attract workers from surrounding places. However, the results are not statistically significant.

[16]Local employment is the employment of residents who also work in their municipality of residence.



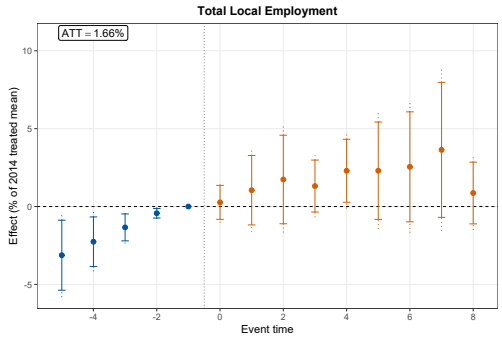
(a) Receiving. The 2014 treated mean is 8,407 employees.

(b) Sending. The 2014 treated mean is 161,704 employees.

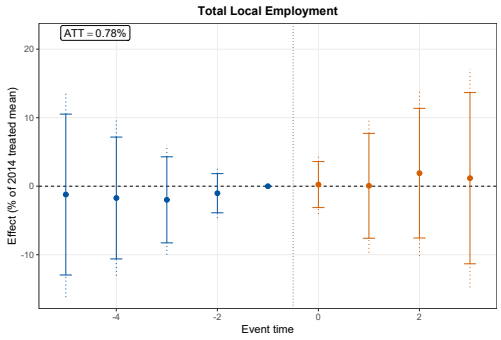
Figure 5. Private employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in industries outside public administration. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Out-commuting initially remains unchanged in both receiving and sending municipalities (Figure 7). In receiving municipalities, it begins to rise six to seven years after the first relocation, reaching 6.7% after six years and 37.6% after seven years. One possible interpretation is that people who move with agency employees commute to jobs outside the receiving municipality. Further study would be needed to establish whether such co-movers explain the increase.

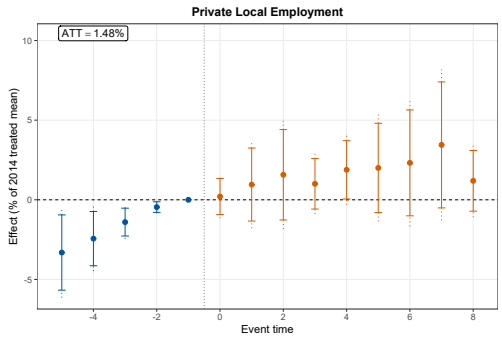
The municipal employment gains leave open how far the benefits extend and what they imply for the economy as a whole. More in-commuting means that some of the workers filling jobs in receiving municipalities live elsewhere, and local employment growth may draw activity from nearby places. The quantitative spatial model accounts for adjustments across locations and evaluates how public-to-private agglomeration economies and the use of floor space translate local employment responses into aggregate output and welfare.



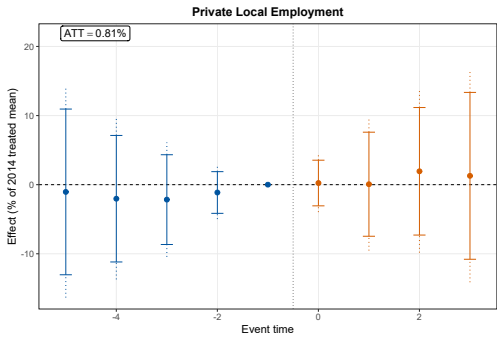
(a) Total: Receiving. The 2014 treated mean is 10,686 employees.



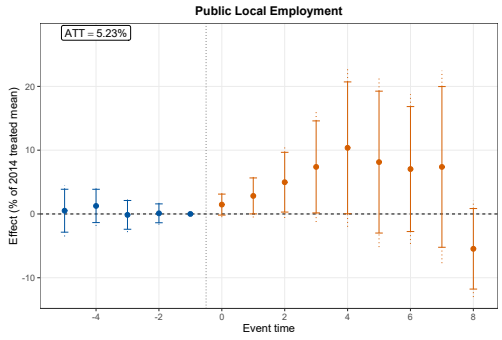
(b) Total: Sending. The 2014 treated mean is 122,400 employees.



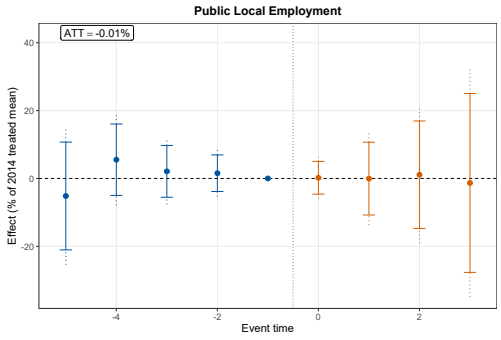
(c) Private: Receiving. The 2014 treated mean is 10,167 employees.



(d) Private: Sending. The 2014 treated mean is 117,383 employees.

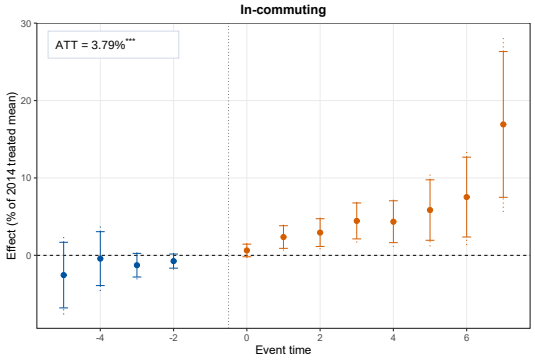


(e) Public: Receiving. The 2014 treated mean is 519 employees.

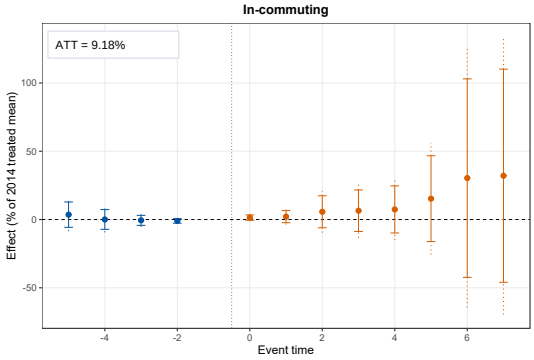


(f) Public: Sending. The 2014 treated mean is 5,017 employees.

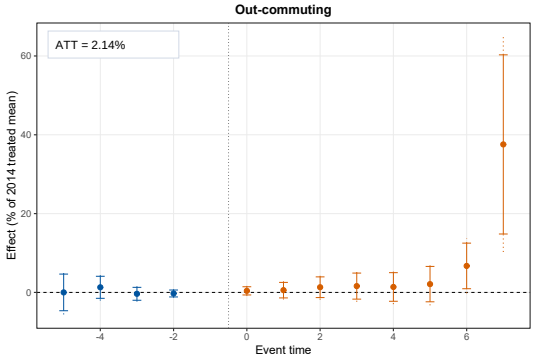
Figure 6. Local employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees who live and work in the same municipality. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



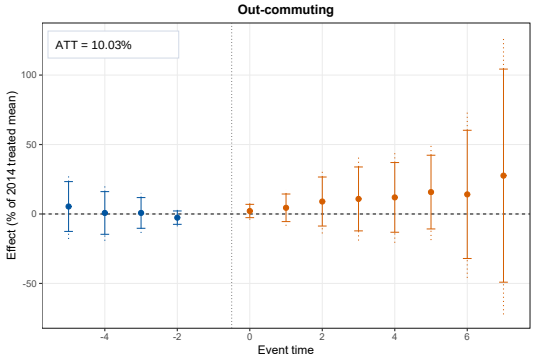
(a) In-commuting: Receiving. The 2014 treated mean is 5,172 commuters.



(b) In-commuting: Sending. The 2014 treated mean is 115,581 commuters.



(c) Out-commuting: Receiving. The 2014 treated mean is 2,891 commuters.



(d) Out-commuting: Sending. The 2014 treated mean is 47,419 commuters.

Figure 7. Commuting around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Commuters across municipal boundaries. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7 Two-Sector Quantitative Spatial Model

Can moving public jobs to lagging regions reduce regional disparities while improving aggregate efficiency? A common concern with place-based policies is that redistributing activity away from productive regions sacrifices the benefits of agglomeration. When agglomeration elasticities are positive and constant across locations, the marginal gains from concentrating activity are greater in more productive regions. An efficiency rationale for directing activity toward lagging regions therefore requires additional forces, such as congestion or differences in spillovers across locations and worker types (Fajgelbaum and Gaubert 2026).

The allocation of public employment raises a distinct question because the location of government activity reflects the historical concentration of political and administrative power. Public offices occupy central locations that also attract private firms. City halls are sufficiently close to central business districts to serve as proxies for their location (Holian 2019). Public offices in these dense locations occupy space that private firms could use, so congestion from public occupancy can hinder private expansion. Relocating public administration could therefore resolve this inefficiency by freeing up space at dense central locations and, at the same time, help increase density in lagging regions. This raises the possibility that reallocating public employment can improve the spatial allocation of private production even when moving private activity away from productive locations would be costly.

To examine this possibility, I extend the quantitative spatial framework of Ahlfeldt, Redding, et al. (2015) to distinguish public and private employment. The central extension allows public workplaces to generate productivity spillovers for private firms separately from the agglomeration economies generated within the private sector. The relative marginal contributions of public and private employment depend on their relative densities in and around each location. An additional public job can therefore have different effects across locations with different employment compositions. Public agencies also occupy floor space, so their relocation changes the space available to private firms and households at both origins and destinations. These mechanisms interact as employment, residence choices, wages, and rents adjust throughout the economy.

I implement the relocation program as a counterfactual change in the geography of public employment. My model contributes to the evaluation of place-based policies by accounting jointly for effects in locations losing and locations gaining jobs, the rest of the policy state, and the rest of Germany. Counterfactual comparisons separate the roles of public-to-private productivity spillovers, private agglomeration, and the reallocation of floor space. I use the resulting equilibria to evaluate changes in aggregate private output, its spatial distribution, and welfare. The distinction between public and private activity also provides a framework that can be adapted to other local shocks, including private establishment closures and public agency closures, or both at the same time.

7.1 Model Setup

Locations are German municipalities, with i indexing residences and j workplaces. Subscripts M and R denote workplace and residence quantities. Each location has land area $K_i > 0$ and effective floor-space stock $L_i > 0$. The national masses of private and public workers, $\bar{H}^P$ and $\bar{H}^G$, are fixed. Each worker supplies one unit of labor. Whether a worker is employed in the private or the public sector is exogenous. The equilibrium reallocates workers across locations.

The public sector is introduced as follows. The government assigns $\bar{H}^G_{Mj}$ public jobs to workplace j, with $\sum_j \bar{H}^G_{Mj} = \bar{H}^G$. Public workers choose where to live conditional on that assignment, making public resident employment, H^G_{Ri}, endogenous. Private workers choose both residence and workplace, generating private workplace employment H^P_{Mj} and private resident employment H^P_{Ri}. Total workplace and resident employment are

$$H_{Mj} = H^P_{Mj} + \bar{H}^G_{Mj}, \qquad H_{Ri} = H^P_{Ri} + H^G_{Ri}.$$

Resident employment refers to individuals living in a location, wherever they work. Public agencies occupy office space at their workplace in proportion to the public jobs assigned there. This space is not available to private firms or households.

Private firms pay the endogenous gross nominal wage w_j at workplace j, determined by private labor demand. At baseline, public employers pay the same local wage, $w_j^{G,0} = w_j^0$. In Germany, public pay follows collective agreements and statutory pay scales, so a given grade and experience step carries the same scheduled base pay in every Bavarian municipality (Appendix A). Public pay therefore does not respond to local labor-market conditions. All wages in this section are nominal wages.

I implement the relocation program as a counterfactual change in the assignment of public jobs. Relocated jobs take their office space with them, public workers choose where to live given the new workplaces, and private workers choose new residences and workplaces. Each public job keeps the wage assigned to it at its previous agency location. An incumbent job at j keeps $w_j^{G,0}$, and a job relocated from origin s keeps $w_s^{G,0}$. The average wage at a workplace after the policy is therefore a weighted average of the private wage, which responds endogenously, and public wages, which combine the unchanged local public wage of incumbent jobs and the origin wages of relocated jobs. Formally, w_j^G is the resulting mean public wage at j, $WB_j^G = w_j^G \bar{H}^G_{Mj}$ the public wage bill paid there, and the average wage at j is $(w_j H^P_{Mj} + WB_j^G)/(H^P_{Mj} + \bar{H}^G_{Mj})$. The government finances public payroll and office rents with a common proportional tax on labor income, which adjusts to keep its budget balanced.

7.1.1 Workers

Workers value consumption, housing, amenities, and access to work. They spend share β of disposable labor income on a freely traded numeraire good and share $1 - \beta$ on housing. Let Q_i^R denote residential rent, t the common proportional labor-income tax, and z^m_{ijo}

the preference draw of a worker o in sector $m \in \{P, G\}$. Indirect utility is

$$u_{ijo}^{m} = \frac{z_{ijo}^{m} B_i (1-t) w_j^{m}}{(Q_i^R)^{1-\beta} d_{ij}}, \qquad d_{ij} = \exp(\kappa \tau_{ij}), \tag{2}$$

where $w_j^P = w_j$ and w_j^G is the public wage of the job, B_i is the amenity level at residence i, τ_{ij} is road travel time in minutes, and κ governs commuting costs per minute. Commuting reduces utility in non-monetary terms. Independent Fréchet preference draws have common shape $\epsilon > 1$ and scales $T_i E_j$ for residence–workplace pairs. Public workers draw only over residences, with scale T_i.

Public workers compare residences for their exogenously assigned public sector job. With $D_{ij} = \exp(-\kappa \epsilon \tau_{ij})$, their conditional probabilities and flows are

$$\pi_{i|j}^{G} = \frac{T_i B_i^{\epsilon} (Q_i^R)^{-\epsilon(1-\beta)} D_{ij}}{\sum_r T_r B_r^{\epsilon} (Q_r^R)^{-\epsilon(1-\beta)} D_{rj}}, \qquad N_{ij}^{G} = \bar{H}_{Mj}^{G} \pi_{i|j}^{G}, \qquad H_{Ri}^{G} = \sum_j N_{ij}^{G}. \tag{3}$$

The workplace wage and common tax cancel from this residential comparison because they are the same for every residence available to the exogenously assigned public sector job. They still determine income, housing demand, and utility. Moving a public workplace changes its commuting access without mechanically moving its workers' residences. This distinction carries the worker evidence into the model.

Private workers compare all residence–workplace pairs. Private choice probabilities and bilateral worker flows are

$$\begin{aligned} \phi_{ij}^{P} &= T_i B_i^{\epsilon} (Q_i^R)^{-\epsilon(1-\beta)} D_{ij} E_j [(1-t) w_j]^{\epsilon}, \qquad \Phi^{P} = \sum_{r,s} \phi_{rs}^{P}, \\ \pi_{ij}^{P} &= \frac{\phi_{ij}^{P}}{\Phi^{P}}, \qquad N_{ij}^{P} = \bar{H}^{P} \pi_{ij}^{P}. \end{aligned} \tag{4}$$

Summing N_{ij}^P across residences gives workplace employment, $H_{Mj}^P = \sum_i N_{ij}^P$. Summing across workplaces gives resident employment, $H_{Ri}^P = \sum_j N_{ij}^P$. Higher wages make workplaces more attractive. Higher housing costs and longer commutes make the corresponding residence–workplace pairs less attractive. The diagonal flow N_{jj}^P is local private employment in municipality j.

The choice sets determine expected utility. With $\Gamma_\epsilon = \Gamma(1 - 1/\epsilon)$,

$$\bar{U}^{G|j} = \Gamma_\epsilon (1-t) w_j^{G} \Big[\sum_r T_r B_r^{\epsilon} (Q_r^R)^{-\epsilon(1-\beta)} D_{rj} \Big]^{1/\epsilon}, \qquad \bar{U}^{P} = \Gamma_\epsilon (\Phi^{P})^{1/\epsilon}. \tag{5}$$

Public expected utility is conditional on an assigned workplace. Private expected utility values access to the full set of private residence–workplace pairs. Both incorporate disposable wages, housing costs, amenities, and commuting. Appendix D derives the choice probabilities and employment margins.

7.1.2 Private Firms

Private firms produce the freely traded good using labor and private commercial floor space L_j^P. Their output is

$$Y_j^P = A_j^P (H_{Mj}^P)^{\alpha} (L_j^P)^{1-\alpha}. \tag{6}$$

Firms take productivity A_j^P, wages, and commercial rent q_j as given. Competition and constant returns imply

$$w_j H_{Mj}^P = \alpha Y_j^P, \qquad q_j L_j^P = (1-\alpha) Y_j^P. \tag{7}$$

Public offices are separate from L_j^P. Vacated offices can therefore increase the space available for private production or housing, with rents and demand determining its use.

7.1.3 Within- and Between-Sector Productivity Spillovers

The key tradeoff of relocating public jobs runs through private sector productivity and floor space. A public agency can raise the productivity of nearby private firms, but its offices occupy space those firms could use. The central extension of the model, and its main contribution, is that private sector productivity depends separately on nearby private sector and public sector workplace density. Productivity at workplace j is

$$A_j^P = a_j (\Upsilon_j^P)^{\lambda_{PP}} (\Upsilon_j^G)^{\lambda_{PG}}, \tag{8}$$

$$\Upsilon_j^m = \sum_s \exp(-\delta_A \tau_{js}) \frac{H_{Ms}^m}{K_s}, \qquad m \in \{P, G\}, \quad H_{Ms}^G = \bar{H}_{Ms}^G. \tag{9}$$

The productivity fundamental a_j is fixed across policy scenarios. The elasticities λ_{PP} and λ_{PG} govern private-to-private and public-to-private spillovers, while δ_A governs their decay with travel time in minutes. Each source contributes employment per unit of its own land area. The travel-time weights are not normalized. In the spillover kernels, the travel time within a municipality is zero, so a municipality's own density enters with weight one. Moving public jobs changes the public density term directly. The subsequent response of private sector firms and workers changes the private sector density term. I maintain the private sector elasticity from Ahlfeldt, Redding, et al. (2015) and estimate the public sector elasticity from the relocation responses.

7.1.4 Amenities, Floor Space, and Market Clearing

Amenities depend on where workers live. They are given by

$$B_i = b_i \Omega_i^{\eta}, \qquad \Omega_i = \sum_s \exp(-\delta_B \tau_{is}) \frac{H_{Rs}^P + H_{Rs}^G}{K_s}. \tag{10}$$

Both worker types contribute to surrounding resident density. The amenity fundamental b_i is fixed, η is the amenity elasticity, and δ_B governs spatial decay per minute. A relocation

can consequently change workplace productivity more than residential amenities when workers commute to the new jobs from their existing homes.

Public offices occupy $\bar{L}_i^G$, leaving private firms and households to share the residual stock. Floor space therefore satisfies

$$L_i = L_i^P + \bar{L}_i^G + L_i^R, \qquad F_i = L_i - \bar{L}_i^G > 0. \tag{11}$$

Office requirements are assigned with public employment. At baseline, a public job uses fraction γ_G of the floor space used per private worker in the same location. Baseline office space per public job and per location is

$$\ell_i^{G,0} = \gamma_G \frac{1-\alpha}{\alpha} \frac{w_i^0}{q_i^0}, \qquad \bar{L}_i^{G,0} = \ell_i^{G,0} \bar{H}_{Mi}^{G,0}. \tag{12}$$

I set $\gamma_G = 0.292$. The median floor area per workplace in the five office buildings the state constructed for the program is 30 m^2, and the model-implied floor space per private worker in Munich, the main origin, is 102.8 m^2. Office requirements remain fixed during the equilibrium adjustment. A transferred position carries its origin's baseline requirement to its destination.

Income earned at workplaces finances housing at residences. Gross resident income is $I_i = \sum_j (w_j N_{ij}^P + w_j^G N_{ij}^G)$, and disposable income is $v_i = (1-t) I_i$. Housing demand satisfies $Q_i^R L_i^R = (1-\beta) v_i$. Every municipality is in mixed use, and a fixed wedge ξ_i relates commercial and residential rents through $q_i = \xi_i Q_i^R$. I use the observed municipal rent ratio. Floor-space market clearing gives

$$q_i = \frac{(1-\alpha) Y_i^P + \xi_i (1-\beta) v_i}{L_i - \bar{L}_i^G}, \qquad Q_i^R = \frac{q_i}{\xi_i}. \tag{13}$$

Public occupancy reduces the space available to private uses. Rents also respond to private production and resident income, so releasing public offices does not mechanically determine the direction of the rent response.

The private share of the residual stock differs from the commercial share of the total stock. The two shares are

$$\vartheta_i = \frac{L_i^P}{L_i - \bar{L}_i^G} = \frac{(1-\alpha) Y_i^P}{q_i (L_i - \bar{L}_i^G)}, \qquad \theta_i^M = \frac{L_i^P + \bar{L}_i^G}{L_i}.$$

The first measures how private firms and households divide the space left after public offices. The second includes public offices as commercial use. A relocation can change both shares by changing public occupancy and the demand for private space.

7.1.5 Government Budget

The government finances public payroll and office costs with a common tax on labor income. The private wage bill is $W^P = \sum_j w_j H_{Mj}^P$, the public wage bill $W^G = \sum_j W B_j^G$, and the public office bill $C^G = \sum_j q_j \bar{L}_j^G$. The balanced budget implies

$$t (W^P + W^G) = W^G + C^G, \qquad t = \frac{W^G + C^G}{W^P + W^G}. \tag{14}$$

Public wages are taxed on the same basis as private wages. The office bill values both leased and state-owned premises at their commercial rental equivalent. Rents accrue to absentee landlords and are not returned to resident households. The budget covers the modeled administration sector. Because every public job keeps its wage, a relocation moves the public wage bill across workplaces without changing W^G. The office bill and the tax base change with equilibrium rents and private wages.

An equilibrium makes worker choices, private factor demand, assigned public employment, floor-space use, spillovers, and the government budget mutually consistent. Private labor supplied to each workplace equals firms' demand, public assignments are filled, and each sector's workplace and residence totals equal its national mass. Private firms, households, and public offices use no more floor space than each location has, and the tax rate lies between zero and one, which requires the public office bill to be smaller than the private wage bill. Expected utility adjusts endogenously with the allocation of the fixed national workforce.

7.2 Quantification

I quantify the model for 10,637 municipalities that cover Germany, 2,055 of them in Bavaria. Baseline employment by workplace and residence municipality comes from the IAB employment records for 2014, the year before the first relocation (Section 3). Public employment is employment in public administration, defense, and social security (WZ 2008 division 84). All other industries define private employment.[17] Residential and office rents for 2014 come from the municipal indices of Ahlfeldt, Heblich, and Seidel (2023) and Ahlfeldt, Heblich, Seidel, and Yin (2026), land K_i is developable land in CORINE 2012, and road travel times in minutes come from OSRM routing between municipalities. The commuting time within a municipality is the mean distance between two points in a disk of the municipality's area, at 30 km per hour with a circuity factor of 1.25. I rescale the workplace counts of each sector to that sector's national residence total, so that $\sum_j H^m_{Mj} = \sum_i H^m_{Ri} = \bar{H}^m$.

I take the production and consumption shares, the private productivity elasticity, the amenity elasticity, and the spatial decay rates from Ahlfeldt, Redding, et al. (2015). The commuting decay comes from a gravity equation for bilateral commuting flows of private employees between municipalities in 2014. Flows from residence i to workplace j are

$$N^P_{ij} = \exp\left(\mu_i + \chi_j - \kappa\epsilon\,\tau_{ij} + \rho\,\hat{v}_{ij}\right) u_{ij},$$

where μ_i and χ_j are residence and workplace fixed effects and $\hat{v}_{ij}$ is a control function for travel time. It is the residual from a regression of τ_{ij} on straight-line distance with residence and workplace fixed effects. I estimate the equation by Poisson pseudo-maximum likelihood, which gives $\kappa\epsilon = 0.0397$ per minute (standard error 0.0022).

In a private-sector commuting inversion, I choose the Fréchet shape ϵ to match the cross-municipality variance of log private workplace wages in 2014. This gives $\epsilon = 5.334$

[17] Of the 10,742 municipalities in the 2014 employment data, I exclude 53 without developable land in the CORINE 2012 land-cover data and 52 without private workplace employment.

and $\kappa = 0.00745$ per minute. The observed commercial-to-residential rent ratio has median $\xi_i = 1.20$ and ranges from 0.53 to 6.58. It is below one in 1,734 municipalities. The tax rate that balances the baseline government budget is 5.30%. Table 7 reports the parameters.

Table 7. Model parameters, all municipalities (2014)

Symbol	Description	Value	Source
Estimated			
λ_{PG}	Public-to-private productivity elasticity	0.0222	Indirect inference
ϵ	Fréchet shape	5.334	Variance of log municipal private wages
$\kappa\epsilon$	Commuting decay per minute	0.0397	Municipal commuting gravity
Calibrated or derived			
κ	Commuting cost per minute	0.00745	$\kappa\epsilon/\epsilon$
ξ_i	Commercial-to-residential rent ratio	1.200	Median observed rent ratio
γ_G	Public office intensity relative to private	0.292	State-built offices relative to private office space use in Munich
t^0	Baseline labor-income tax rate	0.0530	Balanced government budget
Maintained			
λ_{PP}	Private-to-private productivity elasticity	0.071	Ahlfeldt, Redding, et al. (2015)
δ_A	Productivity decay per minute	0.3617	Ahlfeldt, Redding, et al. (2015)
η	Amenity elasticity	0.1553	Ahlfeldt, Redding, et al. (2015)
δ_B	Amenity decay per minute	0.7595	Ahlfeldt, Redding, et al. (2015)
α	Labor share in private production	0.80	Ahlfeldt, Redding, et al. (2015)
β	Consumption expenditure share	0.75	Ahlfeldt, Redding, et al. (2015)

Notes: 10,637 municipalities. Travel times are in minutes, so decay parameters are per minute. The rent ratio varies by municipality. The table reports its median.

Given the parameters, I invert the model to recover productivity and amenity fundamentals that rationalize the baseline. The inversion jointly matches private workplace employment and total resident employment. The split into private and public residents follows from their respective choice rules. I scale the recovered nominal wages to the geometric mean of observed municipal private workplace wages in 2014, and public jobs earn the recovered local wage at baseline. Wages and rents recover private productivity and factor demand. Public office requirements and housing demand then complete the

effective floor-space stock. Appendix D gives the inversion equations and validation checks.

I estimate λ_{PG} by indirect inference (Ahlfeldt, Baum-Snow, et al. 2023). For each candidate value, I recover the baseline fundamentals, solve a zero-shock equilibrium that must reproduce the baseline, solve the relocation counterfactual, and compute model-implied long differences in the transformation of the reduced form, $g_j^o(\lambda_{PG}) = \log\left(1 + E_j^{o,\mathrm{cf}}\right) - \log\left(1 + E_j^{o,0}\right)$. I target two receiving outcomes. These are private workplace employment, $E_j^P = H_{Mj}^P$, and local private employment, $E_j^{LW} = N_{jj}^P$. I apply the matched-set regression of the reduced form to these simulated outcomes in the receiving sample $\mathcal{S}$. The auxiliary regression is

$$g_j^o(\lambda_{PG}) = \mu_{p(j)} + \beta_o^M(\lambda_{PG})\mathrm{Treat}_j + e_j, \qquad j \in \mathcal{S}, \quad o \in \{P, LW\}, \tag{15}$$

where $\mu_{p(j)}$ is a matched-set fixed effect, so that β_o^M is the mean treated-minus-control difference. The targets $\widehat{\beta}_o^{RF}$ are the reduced-form long differences for 2014–2024 on the 55 receiving matched sets of the all-relocations design, with standard errors s_o^{RF} clustered by matched set. Treated minus control, private workplace employment rises by $\widehat{\beta}_P^{RF} = 0.0153$ ($s_P^{RF} = 0.0209$) and local private employment by $\widehat{\beta}_{LW}^{RF} = 0.0378$ ($s_{LW}^{RF} = 0.0137$). Control municipalities receive their model-implied outcomes. I minimize the weighted distance between the simulated and reduced-form coefficients. The estimator is

$$\widehat{\lambda}_{PG} = \arg\min_{\lambda_{PG}} J(\lambda_{PG}), \qquad J(\lambda_{PG}) = \sum_{o \in \{P, LW\}} \left[\frac{\beta_o^M(\lambda_{PG}) - \widehat{\beta}_o^{RF}}{s_o^{RF}}\right]^2, \tag{16}$$

with fixed weighting scales s_o^{RF} and a search over $\lambda_{PG} \in [-0.1, 1]$. The local private employment moment adds information on the residential side. Its model counterpart, $N_{jj}^P = \bar{H}^P T_j B_j^{\epsilon} (Q_j^R)^{-\epsilon(1-\beta)} D_{jj} E_j [(1-t) w_j]^{\epsilon} / \Phi^P$, responds to both the local wage and residential attractiveness. I report the sending municipalities and total residence employment as untargeted outcomes.

The estimated elasticity is $\widehat{\lambda}_{PG} = 0.0222$, with $J = 6.86$ at the minimum (Figure 8). A 1% increase in travel-time-weighted public workplace density raises private productivity by about 0.02%, holding private density and fundamentals fixed. The objective rises steeply for larger values. It is 30.0 at $\lambda_{PG} = 0.1$ and exceeds 140 from $\lambda_{PG} = 0.2$. Estimated from the private workplace moment alone, the elasticity is 0.0144. I hold $\lambda_{PP} = 0.071$ fixed because the relocation responses reflect both mechanisms together. The estimated public elasticity applies to the weighted public density index in equation (9).

7.3 Policy Counterfactual

I simulate the relocations completed between 2015 and 2024 as a single shock to the 2014 baseline. The shock covers all relocation programs, as does the main receiving design of Section 6. It comprises 80 relocations with 3,856 positions. Let $n_{rs} \geq 0$ denote positions transferred from origin s to destination r. The records name the sending municipality for 2,223 positions. The other 1,633 positions have no single sending municipality, for

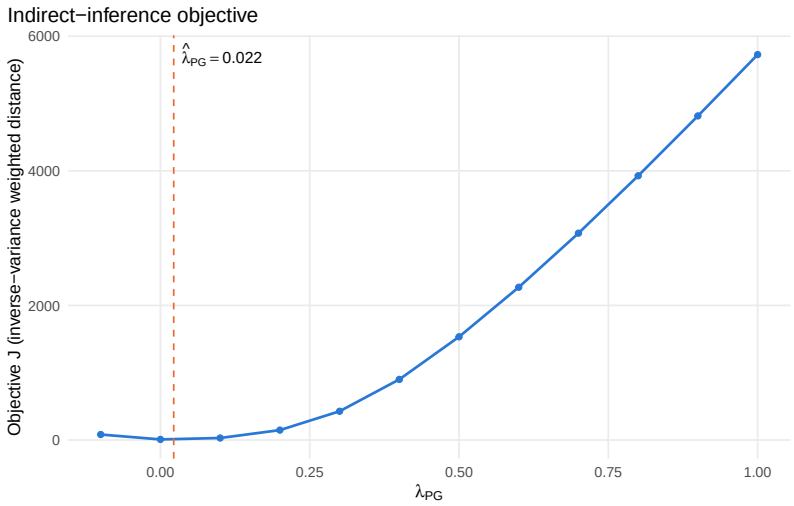


Figure 8. Indirect-inference objective. $J(\lambda_{PG})$ from equation (16) on a grid with step 0.1 over the search range. The dashed line marks the minimum found by a line search, $\widehat{\lambda}_{PG} = 0.0222$ with $J = 6.86$. The targets are receiving private workplace and local private employment, log changes 2014–2024, 55 matched sets. All municipalities (2014).

example because they belong to newly founded agencies. Let n_r^{pool} denote such positions arriving at r. I draw them from the national public workforce in proportion to baseline public employment, $o_j = (\sum_r n_r^{\text{pool}})\bar{H}_{Mj}^{G,0}/\bar{H}^G$. The net change in public workplace employment is $\Delta\bar{H}_{Mj}^G = \sum_s n_{js} + n_j^{\text{pool}} - \sum_r n_{rj} - o_j$, giving

$$\bar{H}_{Mj}^{G,\text{cf}} = \bar{H}_{Mj}^{G,0} + \Delta\bar{H}_{Mj}^G, \tag{17}$$

$$\sum_j \Delta\bar{H}_{Mj}^G = 0. \tag{18}$$

National public and private worker masses remain fixed. The shock has 55 receiving and three sending municipalities. The sending municipalities are Munich (1,753 positions), Herrsching am Ammersee (71), and Erlangen (31).

Each transferred position carries its origin's baseline office requirement. Office space at j in the counterfactual is

$$\bar{L}_j^{G,\text{cf}} = \bar{L}_j^{G,0} + \sum_s n_{js}\ell_s^{G,0} - \sum_r n_{rj}\ell_j^{G,0} + n_j^{\text{pool}}\bar{\ell}^G - o_j\ell_j^{G,0}. \tag{19}$$

Positions from the national pool carry the average requirement $\bar{\ell}^G = \sum_j \bar{H}_{Mj}^{G,0}\ell_j^{G,0}/\bar{H}^G$ and the average public wage. Vacated offices return to the space available for private production and housing. Arriving offices generally occupy the destination's existing stock. For 408 positions in seven destinations, the state completed new office buildings by 2024. I add the corresponding 13,847 m^2 to the destination's total stock as well as its public office occupancy. Office requirements and pay follow the origin. The government

continues to value every office at its commercial rental equivalent, and the tax rate adjusts to balance the budget.

I re-solve for wages, residential and commercial rents, private workplace and residence choices, public residence choices, floor-space use, productivity, amenities, and the tax rate. Productivity and amenity fundamentals, taste scales, rent wedges, land, travel times, and national worker masses remain at their baseline values. Existing floor-space supply is fixed apart from the documented additions due to the relocation program. The comparison describes the difference between the relocation equilibrium and the 2014 baseline equilibrium without the relocations.

7.4 Counterfactual Results

Private firms respond endogenously to the relocated public jobs, and receiving municipalities gain private workplaces. Private workplace employment in the 55 net receiving municipalities rises by 0.72% relative to the no-relocation baseline, about 9,660 jobs for 3,488 net positions received, or 2.8 private jobs per position (Figure 9). The response varies widely across destinations, from -1.0% to 16.5%, with a median of 1.1%. Employment among residents of these municipalities falls by 0.73%. Private residents decline by 0.77%, while public residents rise by 0.28%. Receiving places therefore gain workplaces but lose residents. Commuters fill the additional jobs, and the arriving offices and the expanding private firms compete with housing for space.

Private firms in receiving municipalities become more productive and expand production. Private productivity rises by 0.12%, and private output rises by 0.65% (Figure 10). Holding fundamentals fixed, the productivity response is

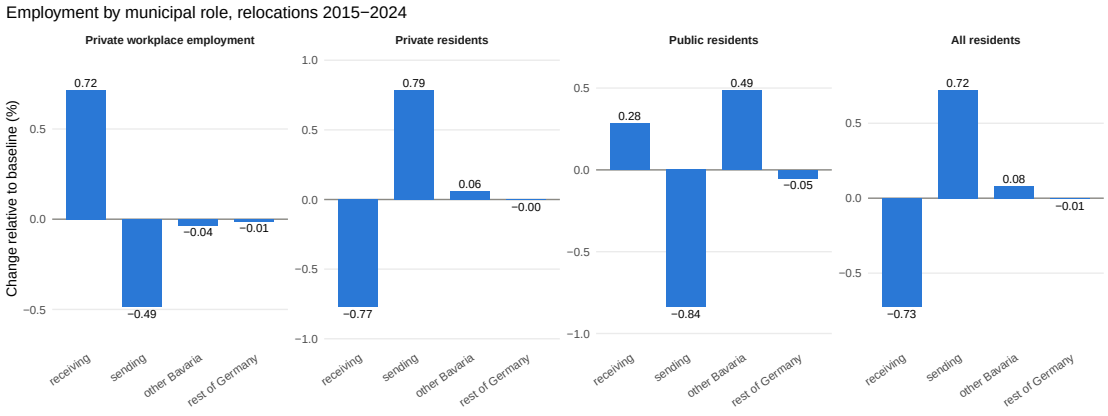
$$\Delta \log A_j^P = \lambda_{PP} \Delta \log \Upsilon_j^P + \lambda_{PG} \Delta \log \Upsilon_j^G. \tag{20}$$

The relocated public workplaces increase nearby public density, and the induced private expansion adds private agglomeration. These productivity effects allow the receiving private sector to grow even though the agencies also occupy floor space.

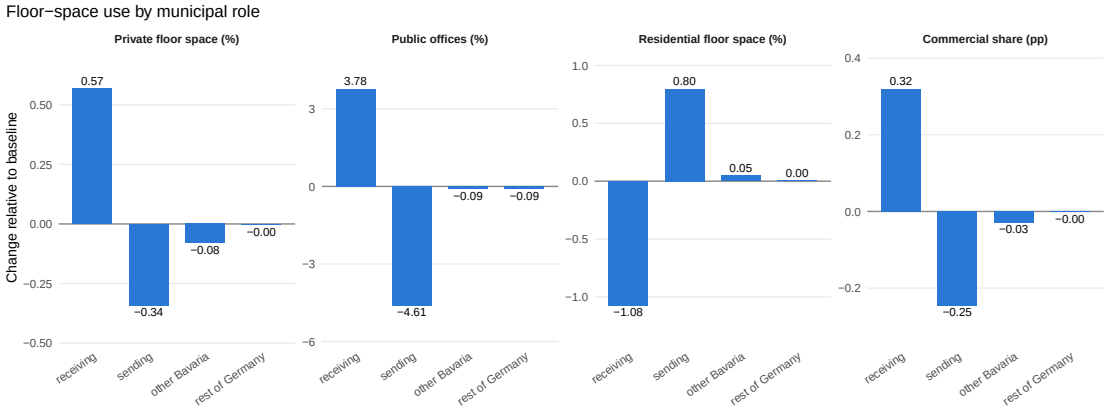
The receiving private expansion increases demand for space. Public offices in receiving municipalities expand by 3.8% and private commercial floor space by 0.57%, while residential floor space falls by 1.1%. The commercial share of floor space rises by 0.32 percentage points. Commercial and residential rents rise by 0.15% and 0.18%, respectively, alongside a 0.11% increase in the private wage. Amenities fall by 0.08% as resident density declines. The gains in private activity thus come with higher housing costs and fewer residents.

At the origins, the effects run in the opposite direction. Private workplace employment in the three net sending municipalities falls by 0.49%, about 4,760 jobs for 1,855 net positions sent, or 2.6 private jobs per position. Munich loses about 3,430 private jobs, 0.39% of its private workplace employment. Herrsching am Ammersee, which sends 71 of its 81 baseline public jobs in the records, loses 26% of its private workplace employment. Public office space in the sending municipalities falls by 4.6%, and the released space goes to housing. Residential floor space expands by 0.80%, employment among residents rises by 0.72%, and both rents fall by about 0.24%. In the model, the loss of public density at

the origins outweighs the space released for private firms. Local private employment in the sending municipalities changes by [XX]%.



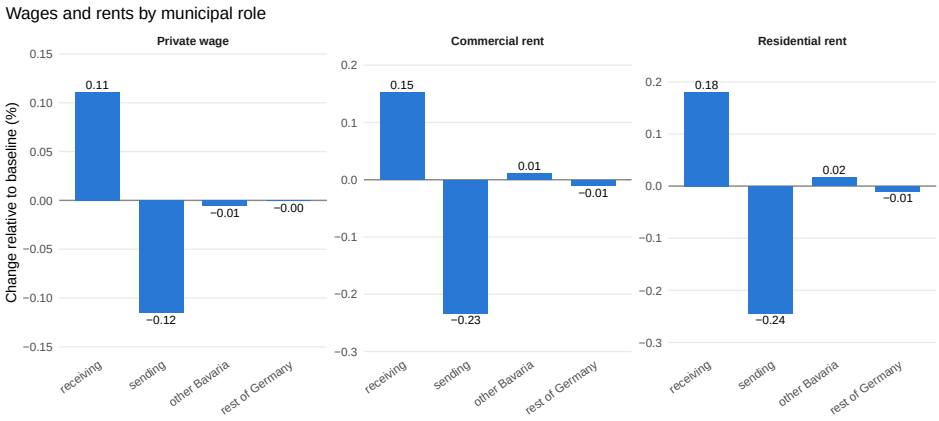
(a) Employment



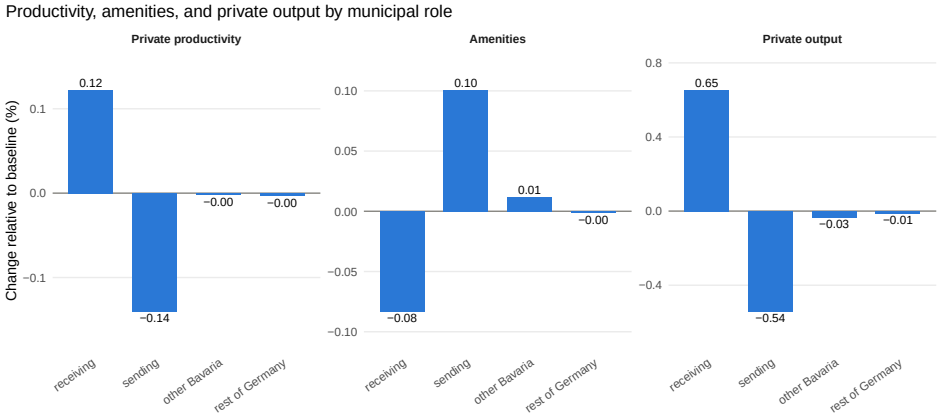
(b) Floor-space use

Figure 9. Employment and floor space by municipal role. Changes between the relocation equilibrium and the no-relocation baseline, in percent. The commercial share of floor space is in percentage points. Each bar is a ratio of group sums. Roles follow net positions. There are 55 receiving and 3 sending municipalities, 1,997 other Bavarian municipalities, and 8,582 municipalities in the rest of Germany. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

The gains at the destinations leave national aggregates nearly unchanged (Table 8 and Figure 11). Private output in Germany falls by 0.0090%, about €175 million per year, while private output in Bavaria rises by 0.0087%. Private workers' expected utility rises by 0.0020%. The concentration of private output across municipalities, measured by the Herfindahl–Hirschman index, falls by 0.14% relative to its baseline value. Private workplace employment falls by 0.04% in the other Bavarian municipalities and by 0.01% in the rest of Germany. Public payroll is unchanged by construction. The public office bill falls by 0.12% as offices move to locations with lower rents, and the tax rate changes by less than 0.0001 percentage points. The program spreads activity more widely with



(a) Wages and rents



(b) Productivity, amenities, and private output

Figure 10. Prices, productivity, and output by municipal role. Changes between the relocation equilibrium and the no-relocation baseline, in percent. The private wage is weighted by baseline private workplace employment, commercial rents by baseline private floor space, residential rents by baseline residential floor space, productivity by baseline private workplace employment, and amenities by baseline resident employment. Public wages are fixed per job. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

a small aggregate output cost. Private workers' living standards reflect the combined adjustment of disposable wages, housing costs, amenities, and commuting, rather than output alone.

Table 8. Private activity and aggregate effects of the relocations

Outcome	Change (%)
Receiving municipalities (55)	
Private workplace employment	0.72
Local private employment	[XX]
Employed residents, both sectors	−0.73
Private productivity	0.12
Private output	0.65
Residential rent	0.18
Sending municipalities (3)	
Private workplace employment	−0.49
Local private employment	[XX]
Employed residents, both sectors	0.72
Private output	−0.54
Germany	
Private output	−0.0090
Private output, Bavaria	0.0087
Private workers' expected utility	0.0020
Concentration of private output (HHI)	−0.14

Notes: Changes compare the equilibrium with relocations completed by 2024 with the no-relocation baseline at $\widehat{\lambda}_{PG} = 0.0222$, all municipalities (2014). Roles follow net positions. Employment and output changes refer to group totals. Productivity is weighted by baseline private workplace employment and the residential rent by baseline residential floor space. Resident employment includes private and public employed residents. HHI denotes the Herfindahl–Hirschman index of municipal shares in national private output. Its change is relative to the baseline index.

7.4.1 Model and Reduced Form

Figure 13 compares the model's matched-set coefficients with the reduced form. All coefficients are treated-minus-control differences in $\log(1 + E)$ between 2014 and 2024 on the same matched sets. They are long differences, while Section 6 reports event-study estimates. The model matches the targeted private workplace coefficient (model 0.024, reduced form 0.015, standard error 0.021) but falls short of the local private employment coefficient (0.002 against 0.038, standard error 0.014). Total residence employment is not targeted. The model predicts a decline of 0.020 at receiving municipalities, while the reduced form shows an increase of 0.029 (standard error 0.009). In the model, workplace and residence employment move in opposite directions at the destinations. In the data, both rise. The model's coefficient for public workplace employment, 0.158, reflects the shock itself and exceeds the reduced-form estimate of −0.005 (standard error 0.036). Wage responses are small in both. For the three net sending municipalities of the all-relocations

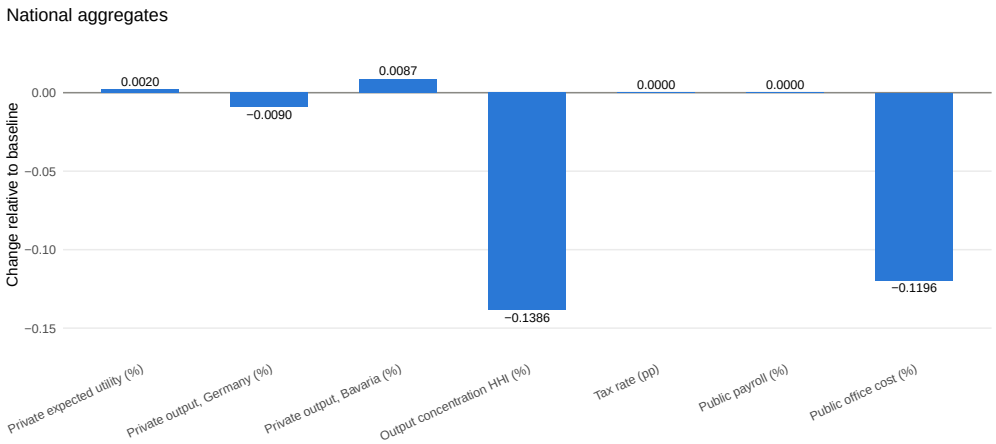


Figure 11. National aggregates. Changes between the relocation equilibrium and the no-relocation baseline. Private expected utility, private output in Germany and in Bavaria, the Herfindahl–Hirschman index of municipal shares in national private output, public payroll, and the public office bill are in percent. The tax rate is in percentage points. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

[FIGURE. Model effects in sending municipalities]

Figure 12. Model effects in sending municipalities. [XX]

design, the model predicts a decline of private workplace employment (-0.103), whereas the reduced form shows an increase (0.045, standard error 0.010). Section 6 reports the six sending municipalities of the main sending design, whose private employment estimates are also positive. Herrsching am Ammersee drives the model coefficient. For sending municipalities, the model predicts a change of local private employment of [XX], against [XX] (standard error [XX]) in the reduced form.

The model implies 2.3 private jobs per relocated position in the reduced form's matched-set multiplier regression for receiving municipalities, against a reduced-form estimate of 8.3 with standard error 7.7 (Figure 14). For sending municipalities, it implies -1.9 private jobs per position sent, against a reduced-form estimate of 6.8. Estimated from the receiving private workplace moment alone, $\lambda_{PG} = 0.0144$ fits that moment exactly but predicts a local private employment coefficient of 0.001 and a residence coefficient of -0.013, both below the reduced form (Appendix Figure 28). Because the model computes the effect on every municipality, it also measures how much the relocations affect the matched controls. The mean log change of private workplace employment is -0.0001 in receiving controls, against 0.024 in the treated municipalities, so effects on the controls change the matched-set coefficient by 0.5% of the effect on the treated (Appendix Figure 29). The model concentrates the effects in the receiving municipalities. Relative to Bavarian municipalities 60 to 90 minutes from the nearest receiving site, workplace employment net of relocated jobs rises by 2.3% in the receiving municipalities and changes by less than 0.4% in every band in between. For residence employment, the reduced form shows declines of 3 to 4% within 30 minutes of a receiving site relative to

the far band, which the model does not reproduce (Appendix Figure 30).

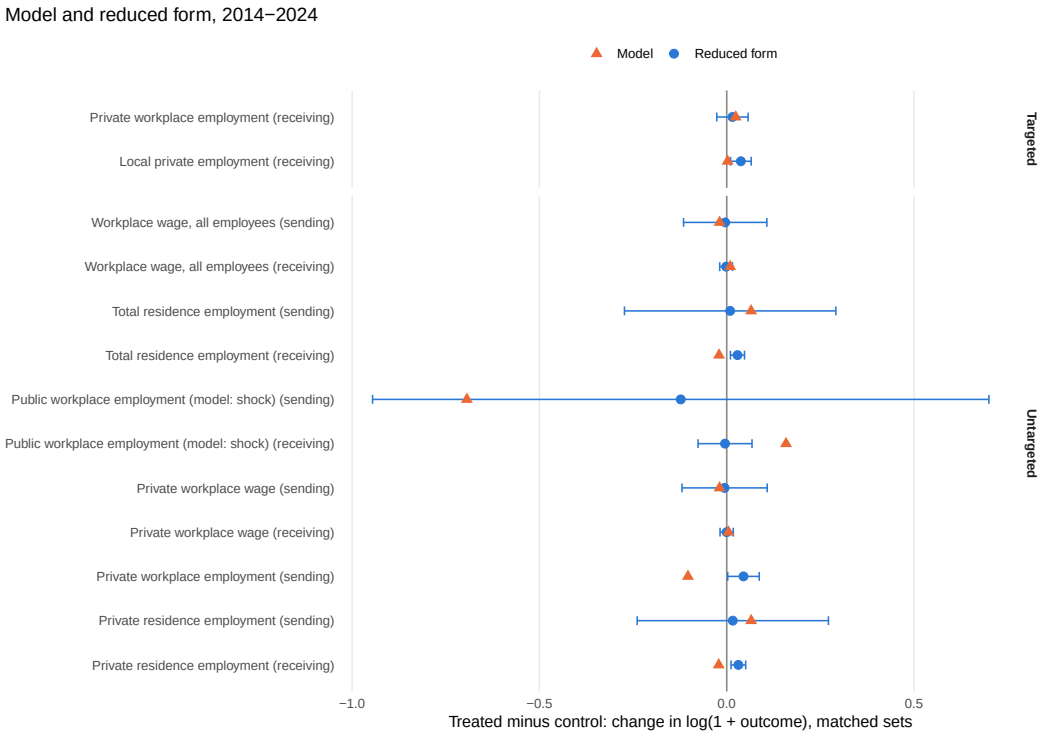


Figure 13. Model and reduced form. Reduced-form coefficients (circles) with 95% intervals based on standard errors clustered by matched set and $t(G-1)$ critical values, and model coefficients (triangles) from the same regression on model outcomes. Treated minus control, $\log(1+E)$ changes 2014–2024, 55 receiving and 3 sending matched sets of the all-relocations design. The targeted outcomes are receiving private workplace and local private employment. Public workplace employment in the model is the relocation shock. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

7.4.2 Channels

I measure each channel by comparing the full equilibrium with the equilibrium in which that channel is switched off. These comparisons allow the other channels to adjust together. When an elasticity is set to zero, I re-invert the productivity fundamentals so that the baseline remains an equilibrium. Table 9 and Figures 15 and 16 report the results.

Public-to-private productivity spillovers account for the private employment response at both ends. Without them, private workplace employment in receiving municipalities falls by 0.04% instead of rising by 0.72%, and in sending municipalities it rises by 0.05% instead of falling by 0.49%. Private agglomeration amplifies both responses. Setting $\lambda_{PP} = 0$ leaves a receiving gain of 0.47% and a sending loss of 0.34%. Moving the offices restrains private employment at the destinations and supports it at the origins, by about 0.02 percentage points in each case. New state-built floor space adds 0.002 percentage

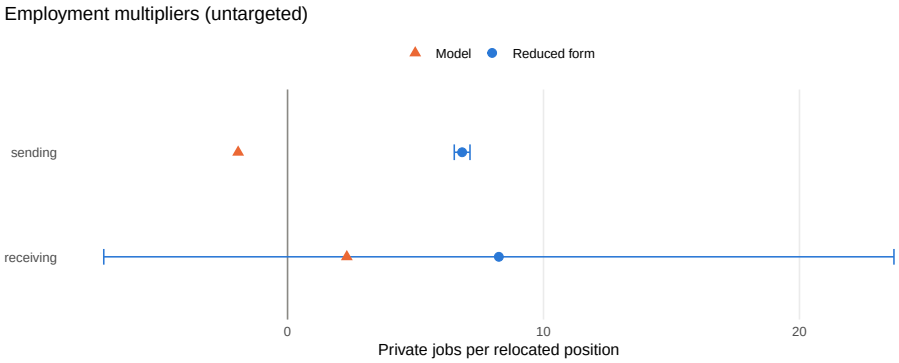


Figure 14. Employment multipliers. Private jobs per relocated position from a regression without intercept of the within-set level effect on private workplace employment (treated minus control) on the relocated positions of the set. Reduced form (circles) with 95% intervals based on standard errors clustered by matched set. Model (triangles) on the same matched sets. Not targeted. All municipalities (2014).

points to the receiving response, and the tax adjustment is negligible. The contributions of single channels do not add up to the full change. The interactions between channels and the change with all channels switched off, −0.02% in receiving municipalities, account for the remainder. In sending municipalities, public-to-private spillovers account for [XX] percentage points of the change in private workplace employment, private agglomeration for [XX], and the office relocation for [XX].

The public spillover channel also accounts for the national output loss. Without it, national private output rises by 0.0017% instead of falling by 0.0090%. The public density lost in Munich lowers productivity across a large private employment base, while the gains accrue to small destinations. Moving the offices adds 0.0007 percentage points to national output, because private firms and households take over space released at the origins. Private workers' expected utility rises by 0.0020% in the full model and falls by 0.0003% without public spillovers.

7.4.3 Allocation of Public Jobs

The small national output loss also depends on where the public jobs go. I first compute the marginal value of an agency. It is the change in national private output when 100 public positions move from Munich to a candidate municipality, carrying Munich's baseline office space and pay per job, with nothing newly built. The candidates are the 54 municipalities that qualify for the program as central places but never received a relocation, and the 55 actual net receivers. Each candidate requires its own equilibrium solve.

Moving the agency lowers national private output for all candidates but one. The loss ranges from €4.5 million per year for Ingolstadt to €24.8 million for Hallbergmoos. Traunreut is the only candidate with a gain, of €0.6 million. The loss is smallest in destinations with high baseline wages (Figure 17). The actual receivers have higher marginal values than the central places that were passed over. Their median rank among

Table 9. Channel decomposition of the relocation effects

Scenario	Private employment (%) Receiving	Sending	Local, sending	Output, Germany (%)	HHI (points)	Private utility (%)
Full model	0.715	−0.487	[XX]	−0.0090	−0.184	0.0020
No public spillover ($\lambda_{PG} = 0$)	−0.044	0.046	[XX]	0.0017	0.025	−0.0003
No private agglomeration ($\lambda_{PP} = 0$)	0.469	−0.344	[XX]	−0.0071	−0.132	0.0014
Offices stay	0.733	−0.509	[XX]	−0.0097	−0.193	0.0015
No state-built floor space	0.713	−0.487	[XX]	−0.0090	−0.183	0.0020
Tax held at baseline	0.715	−0.487	[XX]	−0.0090	−0.184	0.0021
All channels off	−0.022	0.019	[XX]	0.0007	0.013	−0.0008

Notes: Changes relative to each scenario's own zero-shock equilibrium. Every scenario applies the same relocation of public positions. "Offices stay" keeps every municipality's baseline public office space and builds no new floor space. "No state-built floor space" moves the offices but rents every arriving office from the existing stock. "Tax held at baseline" leaves the budget unbalanced. "All channels off" combines $\lambda_{PG} = \lambda_{PP} = 0$, offices that stay, no new floor space, and the baseline tax rate. HHI is the change in the Herfindahl–Hirschman index of municipal shares in national private output, in points of 10,000. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

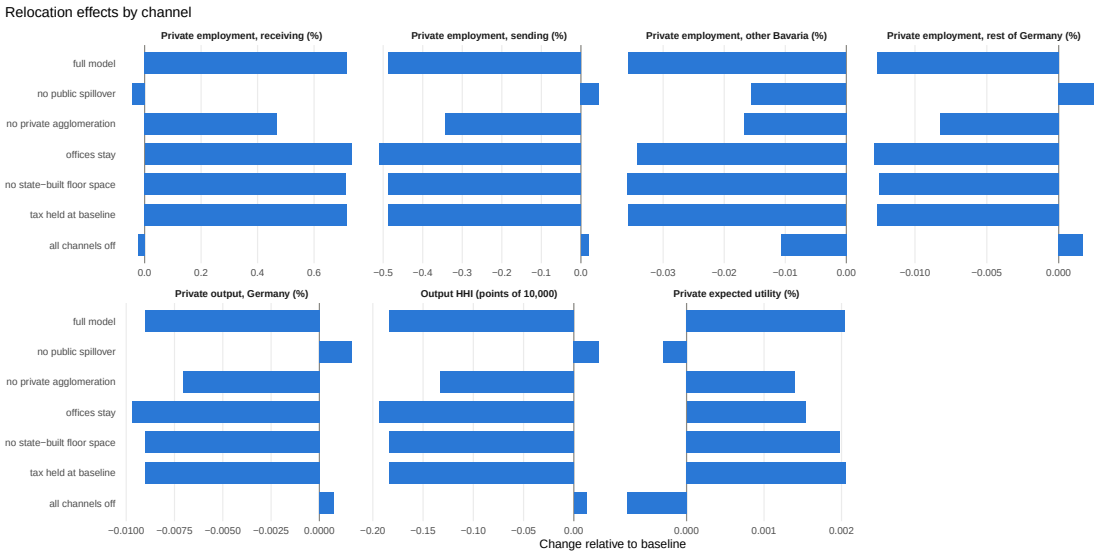


Figure 15. Relocation counterfactual by channel. Each bar is the change between the relocation and the zero-shock equilibrium of a scenario in which one channel, or all of them, is switched off. Private employment by municipal role, private output in Germany, and private expected utility in percent. The Herfindahl–Hirschman index of municipal output shares is in points of 10,000. All municipalities (2014).

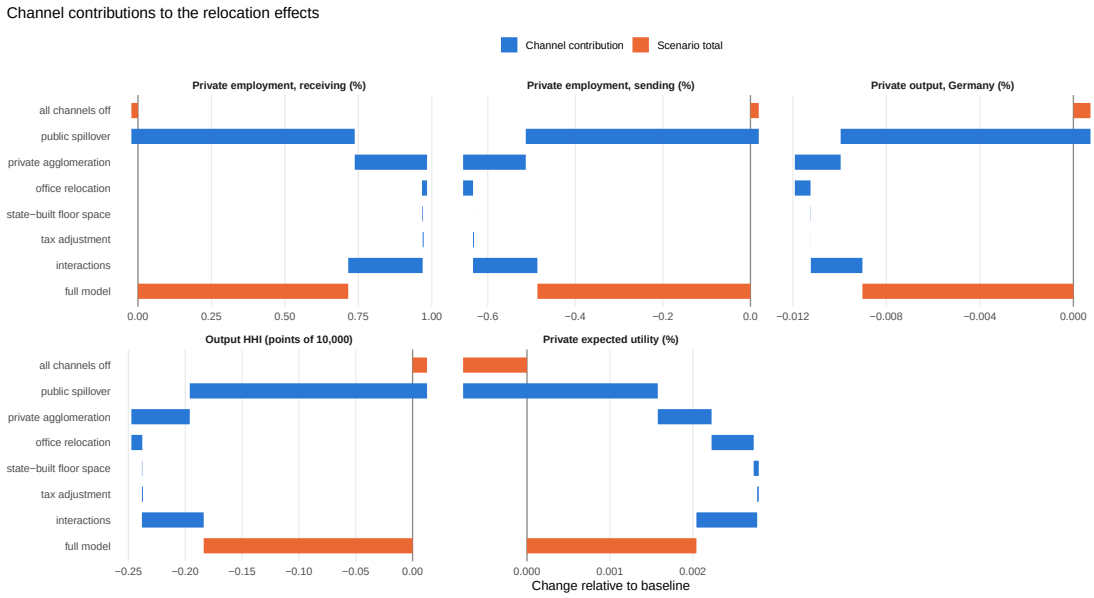


Figure 16. Channel contributions. The contribution of a channel is the full-model change minus the change without that channel. The bars run from the change with all channels off through each contribution and the interactions between channels to the full-model change. All municipalities (2014).

the 109 candidates is 38, against 64.5. In a linear probability model, a one-standard-deviation higher marginal value raises the probability of being an actual receiver by 11.9 percentage points (standard error 5.4). Conditional on the log baseline wage, the marginal value no longer predicts receipt (−7.7 percentage points, standard error 6.6), while a one-standard-deviation higher log wage raises the probability by 24.7 percentage points (standard error 6.1). The program thus chose destinations with higher wages, and in the model these are the destinations where public jobs cost least national output.

I then keep the actual origins and number of positions and change the destinations (Table 10 and Figure 18). Both alternatives use 55 destinations drawn from the candidates, split the positions equally, and give every destination the average office space and pay of the moved positions. For comparability, I solve the actual allocation without new state-built floor space. Sending the positions to the 55 candidates with the highest marginal values reduces the national output loss only slightly, from 0.0090% to 0.0089%. Of these destinations, 34 are actual receivers. Sending them to the 55 candidates with the lowest baseline wages more than doubles the loss, to 0.0198%, while private workplace employment in these destinations rises by 8.1%. Private workers' expected utility does not follow the output ranking. It rises by 0.0020% under the actual allocation, 0.0029% under the highest-value allocation, and 0.0086% under the lowest-wage allocation. Among the candidate destinations, the actual allocation was close to the one that minimizes the national output loss, and directing the jobs to the lowest-wage places would have raised that loss.

Table 10. Same positions, different destinations

Allocation	Private output (%) Germany	Bavaria	HHI (points)	Private utility (%)	Destination employment (%)
Actual destinations	−0.0090	0.0081	−0.183	0.0020	0.71
Highest marginal value	−0.0089	0.0211	−0.206	0.0029	0.81
Lowest baseline wage	−0.0198	0.0430	−0.215	0.0086	8.08

Notes: All allocations move the actual 3,856 positions from the actual origins to 55 destinations, without new state-built floor space. The alternatives draw their destinations from the 109 candidates of the marginal-value experiment and split the positions equally. Changes relative to the no-relocation baseline. HHI is the change in the Herfindahl–Hirschman index of municipal shares in national private output, in points of 10,000. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

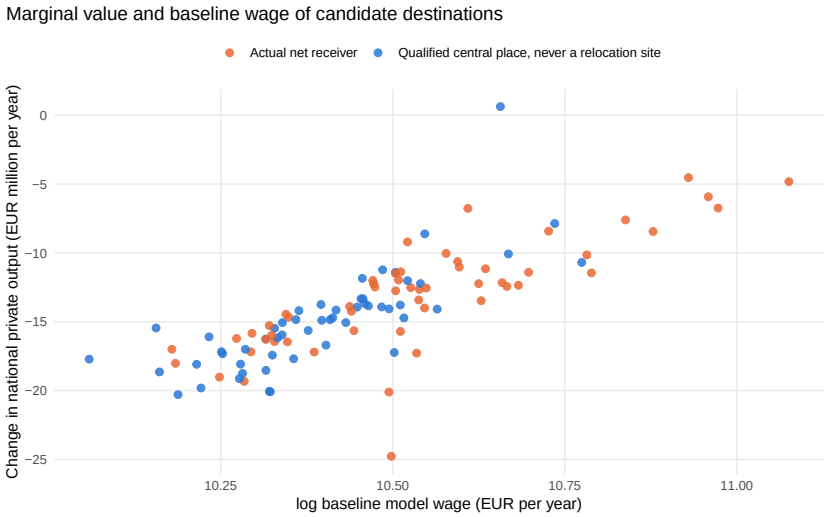


Figure 17. Marginal value and baseline wage of the candidate destinations. Change in national private output, in € million per year, when 100 public positions move from Munich to the candidate, against the log baseline model wage. Actual net receivers and qualified central places that never received a relocation. All municipalities (2014).

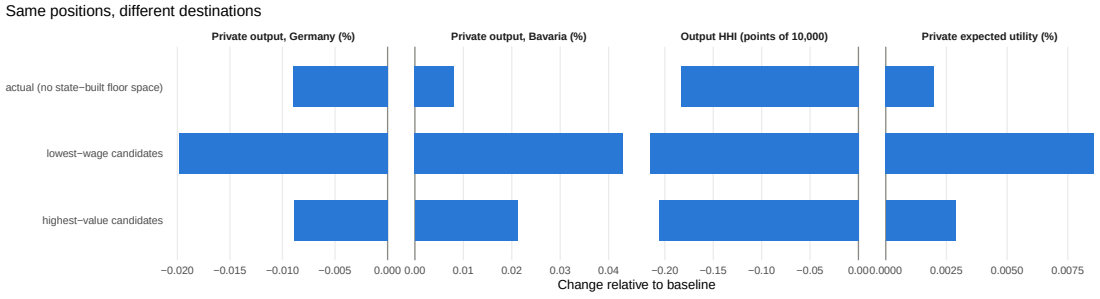


Figure 18. Alternative allocations. The actual origins and positions sent to the actual destinations without state-built floor space, to the 55 candidates with the lowest baseline wages, or to the 55 candidates with the highest marginal values. Private output and private expected utility in percent. The Herfindahl–Hirschman index of municipal output shares is in points of 10,000. All municipalities (2014).

8 Conclusion

Relocating public employment can bring private jobs to lagging places. Studying a program that moves public agencies from thriving to lagging municipalities, I find that private employment in receiving municipalities rises by 2.8% on average after the first relocation relative to matched controls. Sending municipalities show no evidence of offsetting local employment losses.

Public jobs move much more than the original public workforce. Few incumbents change residence and take jobs at receiving agencies, while many retain their residence and continue working in public administration. Receiving agencies recruit almost their entire workforce from outside the original sending cohort. They become more female, while the female share of private employment in receiving municipalities decreases.

The quantitative spatial model connects these local responses to national private production and private workers' living standards. Public workplaces generate productivity spillovers, and public offices compete with private firms and households for floor space. Estimated on all German municipalities, the public-to-private productivity elasticity is small, 0.022. The relocations raise private output in receiving municipalities by 0.65%, while national private output falls by only 0.009% and private workers' expected utility rises by 0.002%. Receiving municipalities gain workplaces but lose residents, and private production becomes slightly less concentrated across municipalities.

The location of public jobs shapes the aggregate cost of regional redistribution. Among the eligible destinations, the program chose places where public jobs cost comparatively little national output: assigning the same jobs to the destinations with the highest marginal contribution to private output would barely reduce the national output loss, while assigning them to the lowest-wage destinations would more than double it. Governments can therefore improve the allocation of their workplaces by considering both the private activity that public jobs support and the space that public offices occupy.

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Appendix

Appendix A documents institutions and costs. Appendix B describes the data and worker samples. Appendix C gives the empirical specifications and diagnostics. Appendix D derives the model and reports its quantification on all municipalities. Appendix E reports the model on Census towns. Appendix F reports additional empirical specifications. Appendix G reports residence and local employment results.

Appendix A: Institutional Setting

Public Employment and Pay

Germany's June 2024 public-service workforce comprised approximately 527,000 federal employees, 2.67 million state employees, 1.80 million municipal employees, and 381,000 employees of social insurance institutions (Destatis 2025; Destatis 2026). In the World Bank series, the public wage bill remained close to one-sixth of total public expenditure from 2000 to 2019. Its share declined in 2020 as total spending rose during the pandemic (World Bank 2024).

Bavaria's direct state administration employed 311,580 people in 2014, including 213,090 civil servants and judges. The count includes full-time and part-time staff and excludes municipal employers and legally separate public institutions (Statistical Offices 2014; Bavarian Statistical Office 2014). I use this population to benchmark the policy's intended size.

	Monthly gross base pay (€)	
Grade (published steps)	First step	Last step
E9a (1, 6)	3,520.10	4,490.04
A9 (2, 10)	3,294.99	4,108.12
E13 (1, 6)	4,629.74	6,580.44
A13 (5, 11)	5,247.58	6,168.25

Table 11. Contractual and civil-service pay schedules, effective February 1, 2025. Amounts are full-time monthly base pay at each grade's first and last published steps. The endpoints do not represent equal experience across employment statuses. Annual special payments and other compensation are excluded. Source: LfF (2025b) and LfF (2025a).

Bavarian budget law pairs E9 with A9 and E13 with A13 for the purpose of comparing budgeted posts; this is not a rule assigning employees to grades (Freistaat Bayern 2024, Art. 6a). E9a covers administrative duties accessible without a university degree. In the non-graduate civil-service career, workers normally enter at A6 or A7 and reach A9 through promotion. A9 can also be an entry grade for the degree-based career track. E13 and A13 illustrate master's-level work rather than all university-qualified employment (TdL 2025; Freistaat Bayern 2026).

A non-graduate who reaches A9 through promotion will already have accumulated service. The first published A9 step therefore does not provide an equal-experience comparison with an E9a entrant. Civil servants and contractual employees also face

different pension and insurance arrangements, so gross base pay alone does not determine their relative take-home pay (Freistaat Bayern 1992). A net-pay comparison would require a specified career history, tax situation, and health-insurance arrangement. For the policy, the common feature is that a given grade and experience step carries the same scheduled base pay across Bavarian municipalities.

Program Boundaries and Position Counts

Phase	Administrative designation	Coverage in this paper
I	*Regionalisierung von Verwaltung*	Included
I	*Strukturkonzept – Chancen im ganzen Land*	Included
II	*Behördenverlagerung Bayern 2030*	Included; implementation under way
Other measures	Individual projects outside phases I and II	Not included

Table 12. The two phases of the relocation program. The empirical sample covers projects with an observed opening within the outcome-data window. Source: Bavarian Ministry of Finance (2025) and Kaltenhauser (2023).

Phase	Intended jobs	Paid training positions
I	2,511	730
II	2,685	440
Total	5,196	1,170

Table 13. Intended final staffing across both phases. Source: Bavarian Ministry of Finance (2026, pp. 6, 16). These are the government's final staffing figures, not current employment or the number of incumbents who move. Training positions include remunerated public-service degree programs.

The intended jobs equal 1.668% of the pre-policy direct state workforce (5,196/311,580). Including paid training positions gives 2.043% (6,366/311,580). Section 2 reports the jobs-only ratio to keep the two categories distinct.

Geographical Targeting

The local reconstruction identifies 829 municipalities through the district index and 392 through the municipal index. Their union contains 885 of Bavaria's 2,056 municipalities, with overlap counted once. The reconstruction uses an index value of 90 or below; the government describes eligibility as below 90, and published scores are rounded to one decimal place. Service-center designations and ministerial site selection operate in addition to these index-based eligibility criteria. Figure 1 illustrates these geographical selection layers.

Weight	Indicator	District reference period
30%	Projected population change	2010–2030
30%	Unemployment rate	2007–2011
10%	Employment density	2007–2011
20%	Disposable household income per resident	2005–2009
10%	Net migration of ages 18–29 per 1,000 in that age group	2006–2010

Table 14. Composition of the structural index. Weights and periods refer to the district index underlying the 2014 expansion of eligibility. The Bavarian mean is 100 and the expanded threshold is 90. Municipal qualification uses a comparable index with municipal proxies, including taxable income. Source: Bavarian Ministry of Finance (2015) and StMWi (2014).

Implementation Costs

Reported category	Amount (€)
Budgeted temporary staffing costs, 2016–2022	16,623,849.00
Construction expenditure, four sites	74,923,320.64
Renovation expenditure	3,708,003.37
Standardized moving-cost estimates, 2015–2021	450,713.00
Standardized rental-cost estimates, 2015–2022	18,188,015.00
Sum of reported categories	113,893,901.01

Table 15. Reported cost categories for completed relocation projects. Source: Kaltenhauser (2023, pp. 13–14), response dated May 23, 2023. The sum combines actual construction and renovation spending, budgeted costs of temporary posts, and standardized estimates. Categories cover different periods. It is neither a measure of total cash expenditure nor the full program's lifetime net cost.

For three tax-office relocations, the same report provides construction spending and the number of jobs established by completion. Höchstädt an der Donau accounts for €9,712,273.52 and 53 jobs, Waldmünchen for €4,358,424.71 and 21 jobs, and Bad Kötzting for €2,691,643.64 and 20 jobs (Kaltenhauser 2023, pp. 8, 13). Summing spending and jobs across these sites gives €16,762,341.87/94, or €178,323 per public job, rounded to €180,000 in the main text. This construction-only ratio excludes other transition costs, savings at origins, and the value of the resulting assets. It does not measure the fiscal cost per additional private job.

Regional investment subsidies under the German GRW program provide a further benchmark. Using regional variation in subsidy rates, Grunau et al. (2025) estimate a cost of about €25,000 per additional job. Their causal employment denominator differs from the public positions accommodated by the three relocation projects.

For Northvolt's planned battery plant in Heide, the European Commission approved €700 million in direct grants and €202 million in guarantees in January 2024 (European Commission 2024). The government announced 3,000 direct jobs (German Federal Government 2024). Dividing the grants by planned direct employment gives €233,333

per job, rounded to €230,000 in the main text. This calculation excludes guarantees, whose face value is not a grant expenditure, and projected indirect jobs.

For Intel's planned semiconductor plant in Magdeburg, the federal government proposed €9.9 billion in support in June 2023, subject to European Commission approval, alongside about 3,000 direct jobs (German Federal Government 2023). This implies €3.3 million per planned direct job, excluding temporary construction and projected indirect employment. Intel cancelled the project in July 2025 (Intel 2025). This ratio therefore describes the original subsidy offer, not a realized cost per job.

Both firm-specific benchmarks measure announced support per planned direct job, not a causal cost per additional job. All amounts are in nominal euros at their respective source dates. A cost-effectiveness comparison would require net policy costs and additional employment measured over comparable horizons.

Structured News Search

I conducted a structured German-language news search in Nexis Uni using the search string

> **Behördenverlagerung OR Behördenumzug OR "Umzug von Behörden" OR "Umzug vom Amt" OR "Amtsumzug"**.

I restricted the search period to articles published from 1991 onward and limited the sample to news articles, excluding private-company publications, individual webpages, and legal texts.

The query yielded 1,531 results. I identified news reports on 22 federal and 87 state agencies between 2005 and 2032. In total, the reports cover 13,395 relocated jobs. The largest number of relocations occurs in 2019, with 5,843 jobs, followed by 2030, with 2,300 jobs, and 2021, with 1,070 jobs. Table 16 reports the number of relocated jobs and study places by destination state, and Table 17 lists all sending and destination cities.

The search identifies both single-agency relocations and broader relocation programs. The three main programs are the federal program for coal regions, the Bavarian *Heimatstrategie*, and the Saxon *Lausitzprogramm 2038*. I use this search to document the broader institutional context of public employment reallocation in Germany.

State	N of Relocated Jobs	N of Relocated Study Places
Bavaria	9,681	1,730
Brandenburg	510	
Hesse	100	
Saxony	2,214	
Saxony-Anhalt	850	
Thuringia	40	

Table 16. Number of public-sector jobs and study places relocated to each state according to the Nexis Uni news search.

Sending Cities	Destination Cities
Ammersee	Amberg, Ansbach, Augsburg
Berlin, Bonn	Bad Kissingen, Bad Königshofen im Grabfeld,
Bad Kissingen	Bad Kötzting, Bad Neustadt an der Saale, Bad Reichenhall,
	Bad Steben, Bautzen, Berlin, Bischofswerda, Borna
	Cottbus
Dresden	Dessau, Dinkelsbühl, Dresden
Erlangen, Eschborn, Essen	Ebern, Eichstätt
	Freital, Freyung, Fürstenfeldbruck, Fürth
Großenhain	Garmisch-Partenkirchen, Gemünden am Main, Gera,
	Görlitz, Grafenau, Gunzenhausen
Herrsching a. Ammersee	Halle (Saale), Höchstädt a.d.Donau, Hof
	Ingolstadt
	Kaufbeuren, Kemnath, Kitzingen, Kronach, Kulmbach
	Lauterbach, Leipzig, Lohr am Main
München	Marktredwitz, Meißen, Merseburg, Mittenwald
	Mühldorf a. Inn, Münchberg
Nürnberg	Nabburg, Neustadt an der Aisch, Nördlingen, Nürnberg
	Obernburg am Main, Oberviechtach
	Pegnitz
Regensburg	Regen, Rosenheim
Starnberg	Schwandorf, Schweinfurt
	Tirschenreuth, Traunstein
	Vilshofen an der Donau, Vohenstrauß
Würzburg	Waldkirchen, Waldmünchen, Waldsassen, Weiden in der Oberpfalz
	Weißenburg i. Bay., Weißwasser, Windischeschenbach, Wunsiedel
	Zittau, Zwiesel

Table 17. Sending and destination cities identified in the Nexis Uni news search, listed alphabetically. Some cities, such as Berlin and Dresden, appear both as sending and destination cities.

Appendix B: Data

The project records distinguish intended staffing, reported implementation dates, and construction milestones. Source links document the evidence behind each entry. A program entry may represent an entire agency, a branch, or part of an existing organization. Neither an agency's continued presence at its origin nor the absence of a separately named destination in the register establishes that no employment was relocated.

I reconcile the 2014, 2023, and 2026 register editions first by agency identifier. Among unmatched records, I use normalized agency names that occur only once in each edition and then normalized addresses that occur only once in each edition. Normalization removes differences in capitalization, punctuation, and spacing. I review location changes against agency and program sources, distinguishing changes within a municipality from moves between municipalities. Shared addresses and renamed units require particular care. The register records administrative units and locations; it is not an employment census. Its coverage includes state agencies and schools, as well as other institutions with state personnel. These cross-year links remain separate from the establishment identifiers in the IAB data.

IAB descriptive samples and candidate establishments

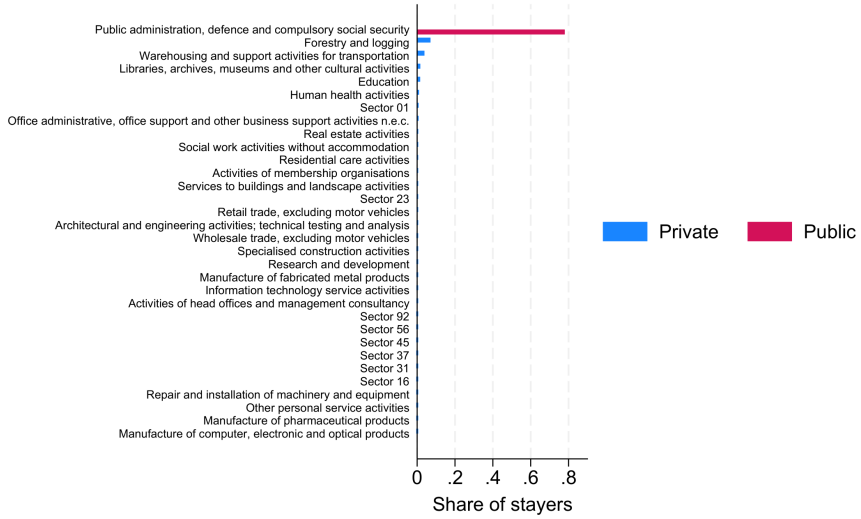
Table 1 reports the treated municipalities in the one-to-one matched samples used for the municipal analysis. The comparison uses controls outside Bavaria. Treatment dates come from the official year where available and otherwise the recorded completion year. The sample retains *Regionalisierung von Verwaltung* projects with a treatment date by 2024 and excludes projects that are still being planned. The receiving sample contains 34 municipalities and the sending sample six. The table reports municipal means, rather than pooled worker totals.

For the worker analysis, workplace postcodes identify public-administration establishments near program origins and destinations. An establishment is classified by its identifier and postcode. A postcode can appear among both origins and destinations and is retained in both sets. Establishments observed in both 2010 and 2013 provide the baseline distribution of employment changes over that interval. I calculate its standard deviation separately for the origin and destination groups. Among continuing establishments, the change from 2014 to 2024 must be at least twice the destination-group standard deviation to qualify as a candidate expansion, or at most minus the origin-group standard deviation to qualify as a candidate contraction. Establishments observed only in 2024 qualify as candidate openings, and those observed only in 2014 as candidate closures. These rules define the establishments used in the descriptive worker comparisons.

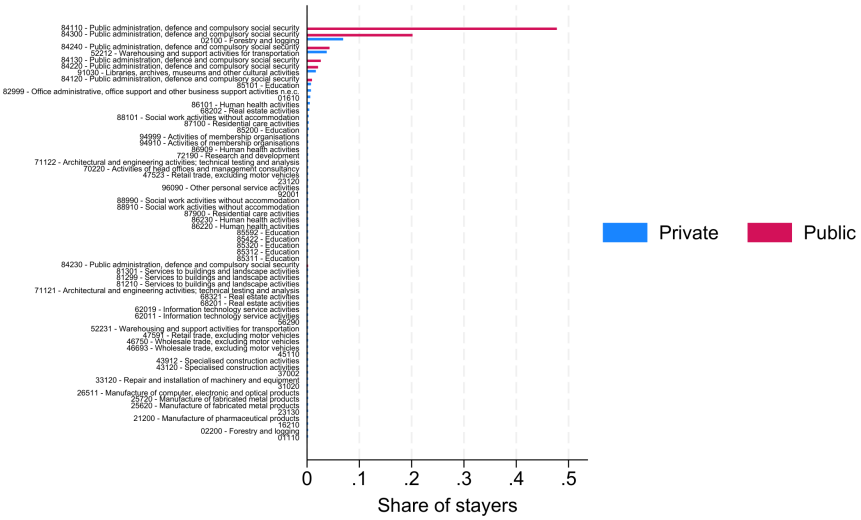
The source cohort consists of workers employed at candidate sending establishments in 2014. Residential movers and commuters are distinguished among those employed at candidate receiving establishments in 2024 using changes in recorded residence. Employees at candidate receiving establishments who were not in the source cohort are additional employees relative to that cohort, not necessarily new labor-market entrants. Workers no longer observed in regular employment in 2024 are very likely retirees.

The agency-profile comparisons in Figures 21 and 22 use closed establishments in 2014 and opened establishments in 2024. The worker classification also includes substantial contractions and expansions, as defined above. For the model on Census towns (Appendix E), the Census inputs select total workplace and residence employment and the industry O components, keeping published totals separate from their component breakdowns.

The figures below show the sectors of stayers in 2024 and of new employees in 2014, and the female share and educational attainment of closed and opened agencies against their municipalities' 2014 averages.

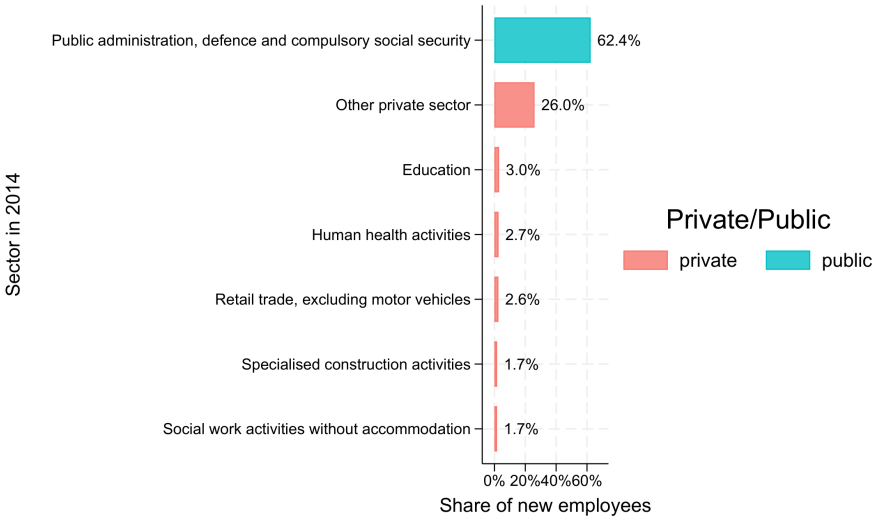


(a) Broad industries

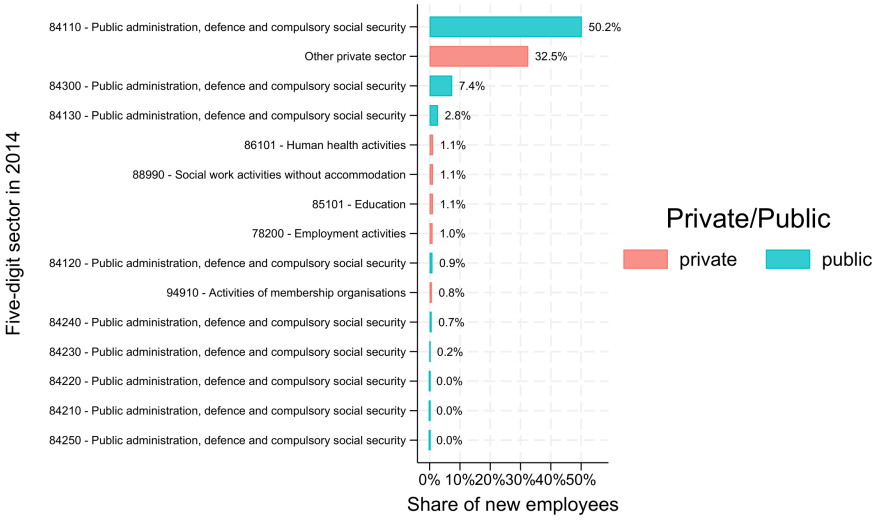


(b) Detailed activities

Figure 19. Employment sectors of stayers in 2024. Shares among original sending workers who retain their municipality of residence, work outside receiving establishments, and have a recorded sector in 2024.

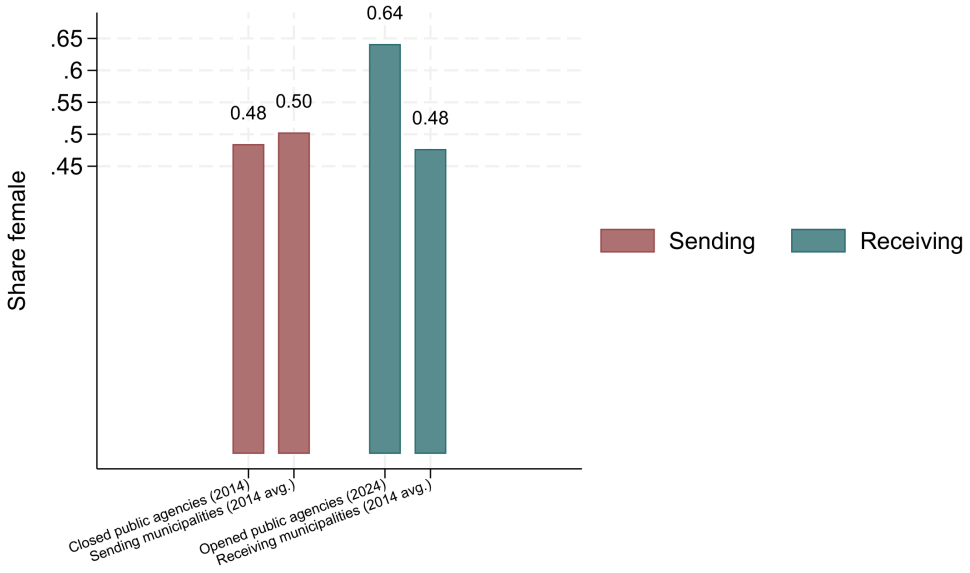


(a) Broad industries



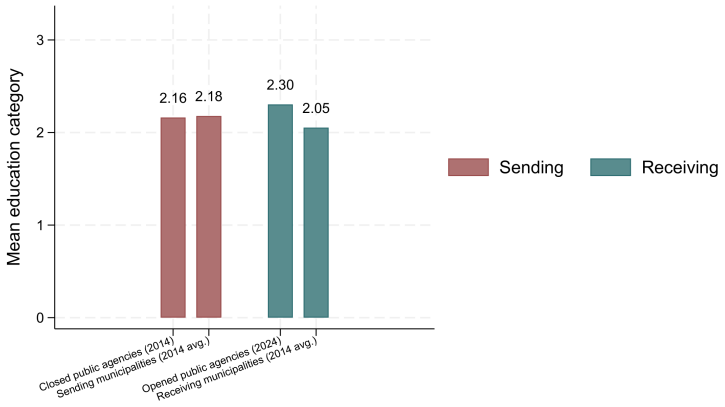
(b) Detailed activities

Figure 20. New hires' employment sectors before the program. Shares among workers at receiving establishments in 2024 who were outside the original sending workforce and have a recorded sector in 2014.

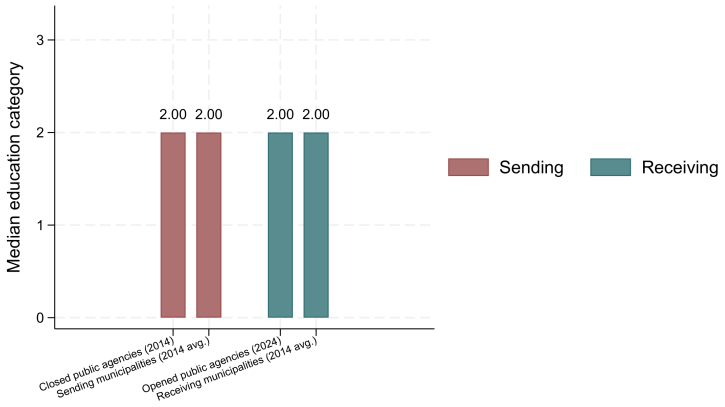


(a) Female employment shares

Figure 21. Female employment shares at sending and receiving agencies. Closed sending agencies in 2014 and newly opened receiving agencies in 2024 are compared with their respective municipalities' 2014 averages.



(a) Mean educational attainment



(b) Median educational attainment

Figure 22. Educational attainment at sending and receiving agencies. The panels compare closed sending agencies in 2014 and newly opened receiving agencies in 2024 with their municipalities' 2014 averages. Categories distinguish no formal vocational qualification, vocational training, and tertiary education.

Appendix C: Empirical Methodology

Matching and comparison groups

I construct receiving and sending matches separately. Let X_i contain the matching characteristics described in Section 5. The squared Mahalanobis distance between a treated municipality and a potential control is

$$d_{ij}^2 = (X_i - X_j)' S^{-1} (X_i - X_j), \tag{21}$$

where S is the covariance matrix of the matching variables. Nearest-neighbor matching without replacement selects one control for each treated municipality. The covariance adjustment accounts for the different scales and correlations of the matching variables.

The main comparison pool excludes Bavarian municipalities to limit exposure to policy spillovers within the state. It also excludes municipalities in 20 non-Bavarian counties selected for their commuting links to Bavaria. Figure 23 shows the top and bottom 20 non-Bavarian counties by their share of commuters to Bavaria in 2013. The top-20 group defines the additional exclusion; the bottom-20 group is shown for comparison and does not define the donor pool.

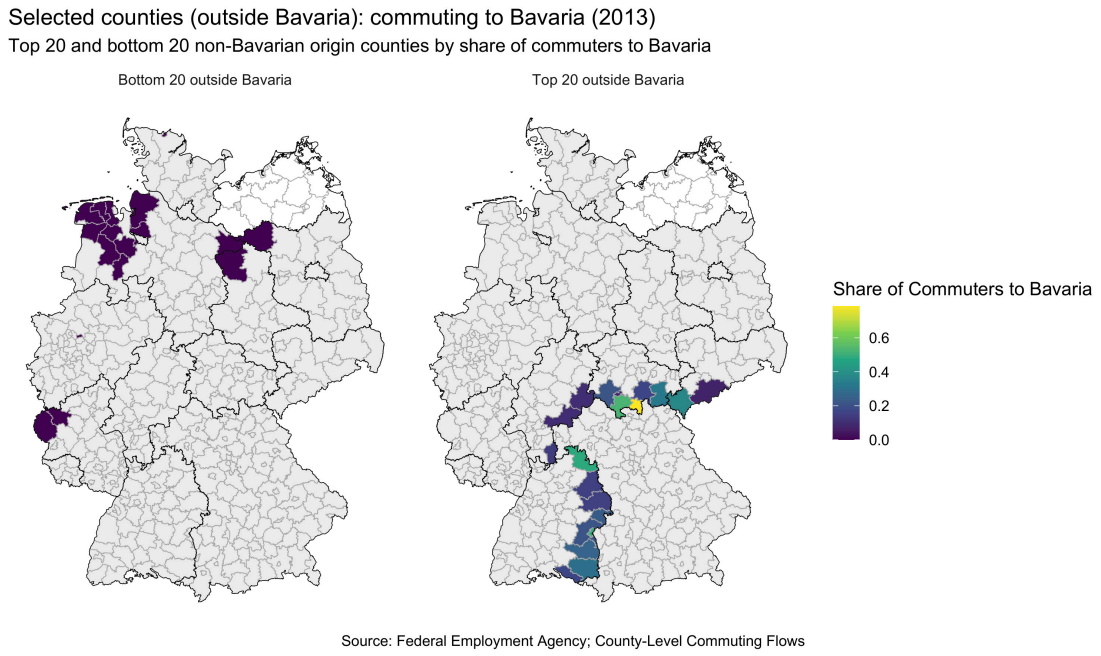


Figure 23. Commuting links used to restrict the comparison pool. Counties outside Bavaria with the lowest and highest shares of commuters to Bavaria in 2013. Municipalities in the top-20 counties are excluded from the main control pool, in addition to municipalities within Bavaria. Source: Federal Employment Agency county commuting flows.

For the auxiliary specifications that match on pre-policy levels and trends, Table 18 reports absolute standardized mean differences after matching with one and three outside-Bavaria controls. The receiving and sending columns use their respective treated-group

standard deviations. Figure 24 also shows the unmatched comparison. These statistics describe balance on the matching variables; the event-study leads provide separate evidence on pre-treatment outcome dynamics.

	Receiving		Sending	
Covariate	1:1	3:1	1:1	3:1
Population, 2014	0.001	0.060	0.058	0.197
Residence employment, 2014	0.011	0.064	0.019	0.221
Unemployment, 2014	0.113	0.005	0.318	0.060
Income per taxpayer, 2014	0.028	0.076	0.330	0.322
Population aged 18–30, 2014	0.015	0.077	0.021	0.210
Population change	0.099	0.001	0.445	0.256
Residence-employment change	0.089	0.085	0.051	0.096
Unemployment change	0.023	0.047	0.737	0.330
Income change	0.111	0.102	0.348	0.040
Population aged 18–30 change	0.043	0.074	0.962	0.681

Table 18. Balance after Mahalanobis matching with outside-Bavaria controls. Entries are absolute standardized mean differences for one and three controls per treated municipality, corresponding to Figure 24. Levels refer to 2014; changes are log differences from 2009 to 2014, except income per taxpayer, which uses 2010–2014.

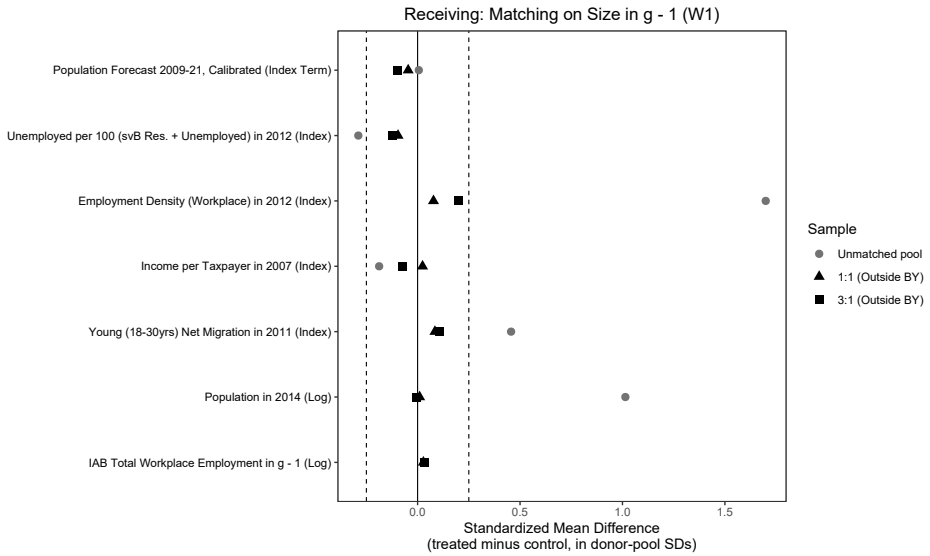
I examine matching ratios of three controls per treated municipality. These checks vary the available comparisons while retaining the distinction between receiving and sending locations. I also examine sending-location estimates without Munich, whose scale differs sharply from that of the other origins.

Treatment cohorts and repeated relocations

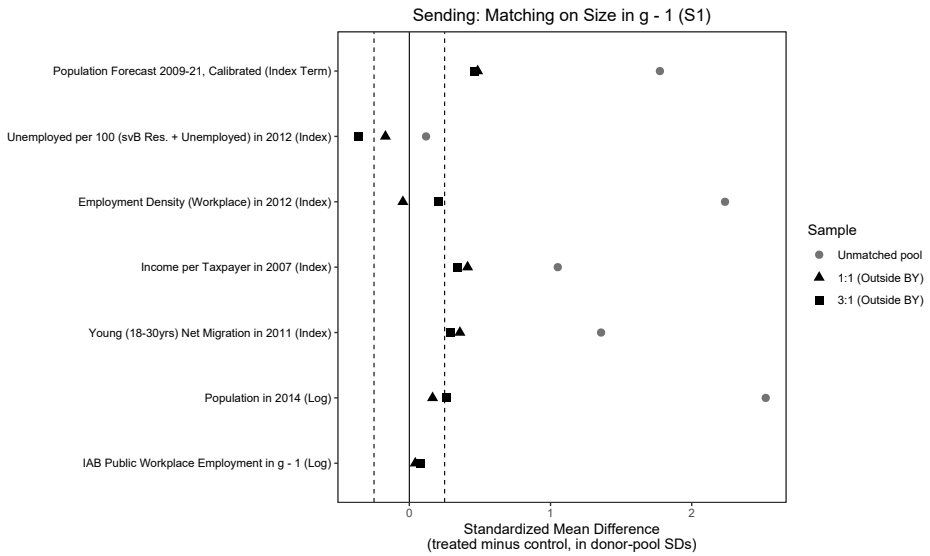
The baseline treatment year is the first recorded relocation for a municipality among the projects included in the analysis. The last-treatment year is the latest such date. Both use the official treatment year when available and otherwise the recorded completion year. Repeated dates at a location therefore describe further exposure to the policy, rather than additional independent treated municipalities. Table 19 gives the first- and last-treatment distributions underlying Figure 2.

Event-time aggregation and inference

In the Sun–Abraham specification, each cohort's event-time effects are measured relative to the year before first treatment. I aggregate cohort-specific coefficients at common event times using the estimator's cohort weights. Cohorts contribute only where their outcomes are observed, so the composition of the aggregate effect can vary across event times. The municipal panel is allowed to be unbalanced. Municipality-clustered standard errors allow serial correlation within municipalities.



(a) Receiving municipalities



(b) Sending municipalities

Figure 24. Covariate balance for Mahalanobis matching on 2014 levels and pre-policy log changes. The figures compare the unmatched pool with one-to-one and three-to-one outside-Bavaria matches. Symbols distinguish the three comparisons. The horizontal axis reports absolute standardized mean differences.

	Receiving		Sending	
Year	First	Last	First	Last
2015	3	1	2	1
2016	15	9	2	2
2017	7	8	1	1
2018	3	5	1	1
2019	0	1	0	0
2020	3	4	0	0
2021	2	2	2	2
2022	1	3	0	0
2023	1	2	0	1
2024	0	0	0	0
Municipalities	35	35	8	8

Table 19. First and last treatment years in the common matching roster. Each municipality appears once in each timing distribution. First and last refer to the earliest and latest recorded treatment among included projects, using the official year when available and the completion year otherwise.

Synthetic difference-in-differences for sending municipalities

Because only five municipalities are net senders, I complement the matched design with synthetic difference-in-differences (Arkhangelsky et al. 2021). I estimate one model per sending municipality. Donors are never-treated municipalities outside Bavaria with at least 10,000 inhabitants in 2014. As in the matched design, I exclude districts with strong commuting ties to Bavaria, every municipality named in the relocation records, and the municipalities of the other relocation programs listed in Appendix A. The pre-treatment window is 2008–2014 for all sending municipalities; years between 2015 and a later first relocation are not used. Unit weights reproduce each sending municipality's pre-treatment outcome path up to a constant. Time weights select the pre-treatment years that best predict the donors' post-treatment outcomes; in practice they fall almost entirely on 2014, so effects are measured relative to the last year before the program. Standard errors come from placebo replications. In each replication, every sending municipality's timing is assigned to one of the 100 donors closest to it in 2014 population, and the models are re-estimated without the drawn donors. Placebo units of similar size matter because small municipalities have much noisier outcomes than large cities.

The donor weights match pre-treatment outcomes but not the characteristics used in the matching. To condition on these characteristics, I follow the two-step procedure of Kranz (2021). The covariates Z_i are the matching variables: the logs of 2014 population, unemployment, residence employment, population aged 18–30 and income per taxpayer; the 2009–2014 log changes in population, unemployment, residence employment and population aged 18–30; and the 2010–2014 log change in income per taxpayer. Because Z_i is fixed over time, it can only affect outcomes through year-specific coefficients. In the first step, I estimate

$$Y_{it} = \alpha_i + \delta_t + \sum_{s \neq 2014} \mathbb{1}[t = s]\, Z_i' \gamma_s + \varepsilon_{it} \tag{22}$$

by OLS on all untreated municipality-years: donor municipalities in every year and sending municipalities before their first relocation. Using only untreated observations keeps the relocations themselves out of the estimated covariate effects. In the second step, I remove the fitted covariate component,

$$\tilde{Y}_{it} = Y_{it} - \sum_{s \neq 2014} \mathbb{1}[t = s]\, Z_i' \hat{\gamma}_s, \tag{23}$$

and apply synthetic difference-in-differences to $\tilde{Y}_{it}$ with the same donors, time windows and placebo procedure. The adjustment allows, for example, larger municipalities to follow different trends than smaller ones. The placebo standard errors take $\hat{\gamma}_s$ as given and do not reflect estimation error from the first step (Kranz 2021).

As a falsification test, I backdate treatment to 2012 for every sending municipality and estimate the same models with 2008–2011 as pre-treatment years and 2012–2014 as placebo post-treatment years. Because the analyzed relocations start in 2015, these estimates should be close to zero. The dynamic version reports the treated-minus-synthetic gap in each year relative to the time-weighted pre-treatment baseline, with pointwise placebo bands.

Appendix D: Quantitative Model

This appendix derives the worker choices and baseline inversion for the model in Section 7, documents the quantification on all municipalities (2014), and reports validation checks and additional counterfactual results. It documents how wages, productivity, amenities, office requirements, housing demand, and the public budget combine to rationalize the baseline economy. Appendix E reports the model quantified on Census towns (2011).

Data preparation. The baseline uses employment subject to social-security contributions in 2014 by workplace and residence municipality from the IAB employment records. Public employment is employment in public administration, defense, and compulsory social security (WZ 2008 division 84); all other industries define private employment. These are industry classifications, and civil servants are not covered. Of the 10,742 municipalities in the data, 53 have no developable land in CORINE 2012 and 52 have no private workplace employment; the remaining 10,637 municipalities form the model. I rescale the workplace counts of each sector to that sector's residence total, with factors of 0.994 for private and 0.999 for public employment. Road travel times τ_{ij} are car travel times in minutes between municipalities from OSRM routing, with residence first and workplace second in worker choices. The commuting time within municipality i is the mean distance between two points in a disk of the municipality's area, $\frac{128}{45\pi}\sqrt{\text{area}_i/\pi}$ km, at 30 km per hour and with a circuity factor of 1.25. In the productivity and amenity kernels, the first index is the recipient and the second the source of density, and the travel time within a municipality is zero. Travel links across the policy-state boundary remain in the model.

Commuting parameters. A gravity equation for bilateral commuting flows of private employees between municipalities in 2014 identifies $\kappa\epsilon = 0.0397$ per minute (standard error 0.0022). It is estimated by Poisson pseudo-maximum likelihood with residence and workplace fixed effects and a control function for travel time. Given residence i, private conditional workplace choice is

$$\pi^P_{j|i} = \frac{D_{ij} E^P_j w_j^\epsilon}{\sum_s D_{is} E^P_s w_s^\epsilon}, \qquad H^P_{Mj} = \sum_i H^P_{Ri} \pi^P_{j|i}. \tag{24}$$

To calibrate ϵ, a preliminary private-sector inversion uses observed private workplace and private resident employment to recover transformed wages $\omega_j^{P,\text{cal}} = E^P_j (w_j^{\text{cal}})^\epsilon$ up to scale. Following Ahlfeldt, Redding, et al. (2015), I set $E^P_j = 1$, giving $w_j^{\text{cal}} = (\omega_j^{P,\text{cal}})^{1/\epsilon}$. Matching the cross-municipality variance of log private workplace wages in 2014 at this stage gives

$$\epsilon^2 = \frac{\text{Var}_j(\log \omega_j^{P,\text{cal}})}{\text{Var}_j(\log w_j^{\text{data}})}, \qquad \widehat{\epsilon} = 5.334, \qquad \widehat{\kappa} = \widehat{\kappa\epsilon}/\widehat{\epsilon} = 0.00745 \text{ per minute.}$$

Each municipality forms its own wage group, so the minimizer has this closed form.

Choice probabilities. Independence of the Fréchet draws gives the distribution of the best private alternative,

$$\Pr\Big(\max_{i,j} u^P_{ijo} \le u\Big) = \exp\Big(-\Phi^P u^{-\epsilon}\Big).$$

For a public worker assigned to j, the corresponding distribution is

$$\exp\Big(-[(1-t)w^G_j]^\epsilon \Psi^G_j u^{-\epsilon}\Big).$$

These distributions yield the probabilities and expected utilities in equations (3)–(5); expectations exist for $\epsilon > 1$. Bilateral flows satisfy

$$\sum_{i,j} N^P_{ij} = \bar{H}^P, \qquad \sum_i N^G_{ij} = \bar{H}^G_{Mj}, \qquad \sum_i H^m_{Ri} = \sum_j H^m_{Mj} = \bar{H}^m. \tag{25}$$

Private probabilities sum to one over all residence–workplace pairs. Public residence probabilities sum to one separately within each assigned workplace, preserving the government's workplace allocation.

Residential inversion. In the baseline inversion, I set $T^P_i = T^G_i = E^P_j = 1$ and recover a common effective amenity, giving $Z^P_i = Z^G_i = Z_i$. Define $\omega^{P,0}_j = (w^0_j)^\epsilon$, $\mathcal{W}^{P,0}_i = \sum_j D_{ij}\omega^{P,0}_j$, and $\Psi^{G,0}_j = \sum_r Z^0_r D_{rj}$. The inversion jointly solves for $(\omega^{P,0}, Z^0)$ using the private workplace and total residence margins:

$$\begin{aligned} H^{P,0}_{Mj} &= \bar{H}^P \omega^{P,0}_j \frac{\sum_i Z^0_i D_{ij}}{\sum_r Z^0_r \mathcal{W}^{P,0}_r}, \\ H^0_{Ri} &= \bar{H}^P \frac{Z^0_i \mathcal{W}^{P,0}_i}{\sum_r Z^0_r \mathcal{W}^{P,0}_r} + \sum_j \bar{H}^{G,0}_{Mj} \frac{Z^0_i D_{ij}}{\Psi^{G,0}_j}. \end{aligned} \tag{26}$$

Damped multiplicative updates converge to a largest absolute log gap below 10^{-10} on both margins. The common tax cancels from the employment probabilities. I scale the recovered nominal wages to the geometric mean of observed municipal private workplace wages in 2014 and set the baseline public wage to the local wage, $w^{G,0}_j = w^0_j$. The recovered residential index gives $B^0_i = (Q^{R,0}_i)^{1-\beta}(Z^0_i)^{1/\epsilon}$. Private and public resident employment are implied separately by the two choice rules. Their earnings enter resident income and housing demand, while their combined density enters amenities.

Production and space inversion. Wages and rents recover private productivity from zero profits and private floor space from factor payments:

$$A^{P,0}_j = \left(\frac{w^0_j}{\alpha}\right)^\alpha \left(\frac{q^0_j}{1-\alpha}\right)^{1-\alpha}, \qquad L^{P,0}_j = \frac{1-\alpha}{\alpha}\frac{w^0_j H^{P,0}_{Mj}}{q^0_j}. \tag{27}$$

Equation (12) gives baseline public office space. Payroll and office costs determine t^0, and bilateral flows allocate gross earnings to residences. Housing demand gives $L^{R,0}_i = (1-\beta)v^0_i/Q^{R,0}_i$. The effective stock is $L_i = L^{P,0}_i + \bar{L}^{G,0}_i + L^{R,0}_i$.

Fundamentals. For each candidate public elasticity λ, I remove the baseline density components:

$$a_j(\lambda) = \frac{A_j^{P,0}}{(\Upsilon_j^{P,0})^{\lambda_{PP}}(\Upsilon_j^{G,0})^{\lambda}}, \qquad b_i = \frac{B_i^0}{(\Omega_i^0)^{\eta}}. \tag{28}$$

Every candidate therefore rationalizes the same baseline productivity. Fundamentals remain fixed within each comparison of the baseline and relocation equilibria. For computation, kernel weights beyond 120 minutes (productivity) and 60 minutes (amenities) are set to zero. The dropped weights are below 2×10^{-19}, and the fundamentals are inverted with the truncated kernels, so the baseline is reproduced exactly. The observed rent ratio $\widehat{\xi}_i = q_i^{\text{data}}/Q_i^{R,\text{data}}$ imposes the mixed-use condition $q_i = \xi_i Q_i^R$ on the residual floor-space stock. I hold this wedge fixed in the counterfactuals. A comparison with the wedge implied by wage bills, resident income, and CORINE commercial land shares provides a separate land-use check.

Baseline fiscal identity. Summing income across residences gives $\sum_i I_i = W^P + W^G$. Combining disposable income with the public budget gives $\sum_i v_i = W^P - C^G$. At baseline, public jobs earn the local wage, and the office rule implies

$$C^{G,0} = \gamma_G \frac{1-\alpha}{\alpha} W^{G,0}, \qquad t^0 = \left(1 + \gamma_G \frac{1-\alpha}{\alpha}\right) \frac{W^{G,0}}{W^{P,0} + W^{G,0}}. \tag{29}$$

With $\gamma_G = 0.292$ and $\alpha = 0.80$, the baseline office bill equals 7.30% of public payroll. Public payroll is 4.94% of the national wage bill, and $t^0 = 5.30\%$. After a relocation, origin-specific office requirements remain fixed, and the government budget is evaluated at counterfactual wages, rents, and occupancy.

Counterfactual accounting. Gross departures satisfy $\sum_r n_{rs} + o_s \le \bar{H}_{Ms}^{G,0}$. Herrsching am Ammersee sends the largest share, 71 of its 81 baseline public jobs in the records. Moves within a municipality cancel in the municipal workplace shock. Each public job keeps its baseline wage, so the public wage bill at workplace j after the relocation is

$$WB_j^G = w_j^{G,0} \bar{H}_{Mj}^{G,0} + \sum_s n_{js} w_s^{G,0} + n_j^{\text{pool}} \bar{w}^G - \Big(\sum_r n_{rj} + o_j\Big) w_j^{G,0},$$

where $\bar{w}^G = \sum_j \bar{H}_{Mj}^{G,0} w_j^{G,0} / \bar{H}^G$ is the mean baseline public wage, and $\sum_j WB_j^G = W^{G,0}$. New state offices completed by 2024 add their office requirement to both public occupancy and the total destination stock. Offices accommodated in the existing stock use space that would otherwise be available to private firms and households. Private welfare follows from the expected-utility formula:

$$\Delta \log \bar{U}^P = \epsilon^{-1} \log\Big(\Phi^{P,\text{cf}}/\Phi^{P,0}\Big).$$

The tax factors inside Φ^P incorporate the change in disposable income. For a public cohort moved from s to r, the assignment-conditional utility ratio is

$$\frac{\bar{U}^{G|r,\text{cf}}}{\bar{U}^{G|s,0}} = \frac{1-t^{\text{cf}}}{1-t^0} \left(\frac{\Psi_r^{G,\text{cf}}}{\Psi_s^{G,0}}\right)^{1/\epsilon},$$

because the cohort keeps its pay. The job-weighted mean change across the relocated cohorts is -8.9% in the full model and -8.8% with all channels switched off. It measures the baseline difference in residential access between destinations and origins rather than an effect of the relocation channels (Figure 31).

Solution. For each candidate elasticity, I invert the baseline, solve a zero-shock equilibrium, solve the relocation counterfactual, and apply the auxiliary regressions. The solver iterates on wages, both rents, and the private share of the residual floor-space stock with the damped update $x \leftarrow x(x^{\text{new}}/x)^{0.5}$ until the largest absolute log change of prices, shares, and employment is below 10^{-10}. The damping weight lies below the stability bound of 0.78 implied by the feedback between wages and the private floor-space share at $\widehat{\epsilon}$. The solver never forms the bilateral flow matrices; it uses their row and column sums. The zero-shock solution reproduces the baseline: the largest absolute log deviation across wages, rents, employment, productivity, amenities, floor space, and income is 1.6×10^{-10}, and below 2.5×10^{-10} in every channel scenario with re-inverted fundamentals. At each solution, I verify national worker masses, public assignments, private factor payments, housing expenditure, floor-space use, resident income, and the government budget.

Model fit. The inversion matches private workplace and total resident employment exactly. The split of residents by sector is not targeted: across municipalities, the log correlation between model and data is 1.000 for private and 0.977 for public residents (Figure 25). Model wages correlate with observed private workplace wages at 0.635 in logs; the Fréchet shape matches only their variance (Figure 26a). Model commuting flows correlate at 0.388 in logs with the Federal Employment Agency flows of Bavarian residents in 2025 on 35,169 municipality pairs with at least 10 commuters. The model understates commuting within a municipality: 9% of residents work in their own municipality in the model, against roughly 57% in an approximate benchmark for 1,875 Bavarian municipalities (Figure 27). The targeted local private employment moment uses log changes, which do not depend on this level. The observed rent ratio is unrelated to the wedge implied by model wage bills, resident income, and CORINE commercial land shares: the Deming slope is -0.01 across 6,619 municipalities with commercial land shares between 0.05 and 0.95 (Figure 26b). CORINE measures land rather than floor space, so this check is only indicative.

Additional counterfactual results. Figure 28 reports the predictions of the model estimated from the receiving private workplace moment alone. Figure 29 reports the model effects on treated and control municipalities. Figure 30 compares the distance profile of the model with the reduced-form band estimates and the spatial decay of productivity spillovers with the decay estimated from residence employment. Figure 31 reports the assignment-conditional utility of each relocated cohort.

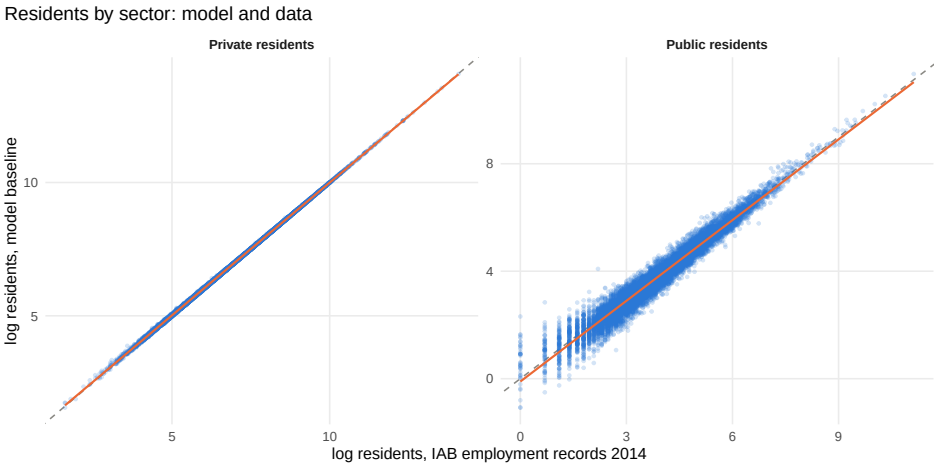
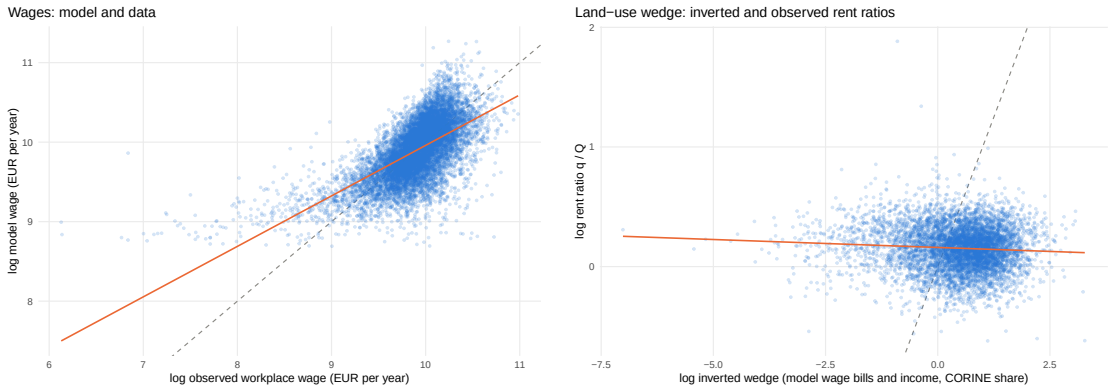
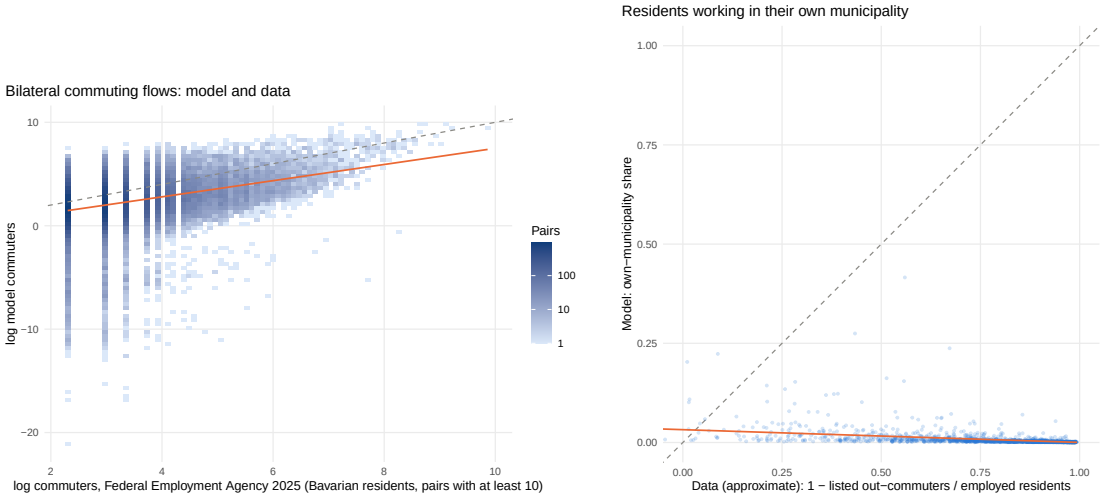


Figure 25. Residents by sector, model and data. Log baseline model resident employment against log resident employment in the 2014 employment records, private (left) and public (right), for municipalities with positive counts. The dashed line is the 45-degree line, the solid line an OLS fit. Not targeted. All municipalities (2014).



(a) Wages **(b)** Land-use wedge

Figure 26. Wages and the land-use wedge. Panel (a): log model wage against the log observed private workplace wage in 2014, in euros per year. Panel (b): log observed rent ratio q/Q^R against the log wedge inverted from model wage bills, resident income, and the CORINE commercial land share, for municipalities with commercial land shares between 0.05 and 0.95. Dashed lines are 45-degree lines, solid lines OLS fits. Not targeted. All municipalities (2014).



(a) Bilateral flows

(b) Residents working in their own municipality

Figure 27. Commuting, model and data. Panel (a): log model commuters $N_{ij}^P + N_{ij}^G$ against log Federal Employment Agency commuters of Bavarian residents in 2025, for municipality pairs with at least 10 commuters, as a two-dimensional histogram. Panel (b): share of residents who work in their own municipality; the data share is approximated as one minus listed out-commuters per employed resident. Dashed lines are 45-degree lines, solid lines OLS fits. Not targeted. All municipalities (2014).

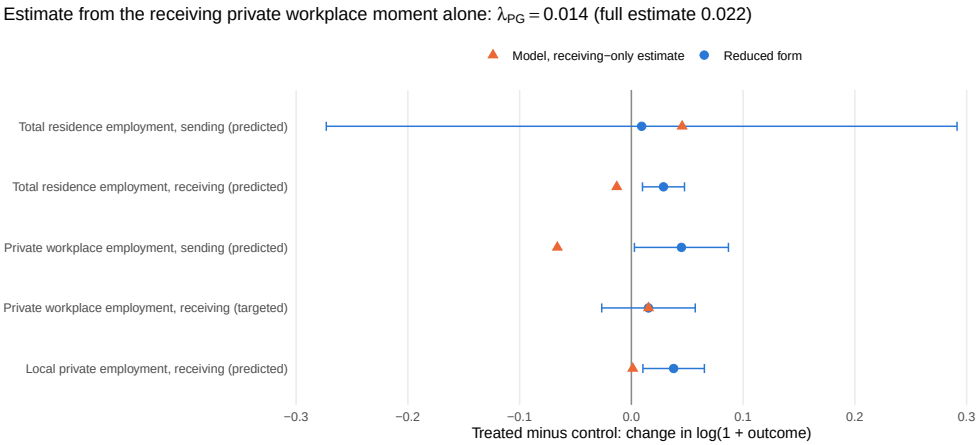


Figure 28. Receiving-only estimate. Model coefficients at $\lambda_{PG} = 0.0144$, estimated from the receiving private workplace moment alone, against the reduced form (circles, 95% intervals based on standard errors clustered by matched set). All other coefficients are predictions. Treated minus control, $\log(1 + E)$ changes 2014–2024. All municipalities (2014).

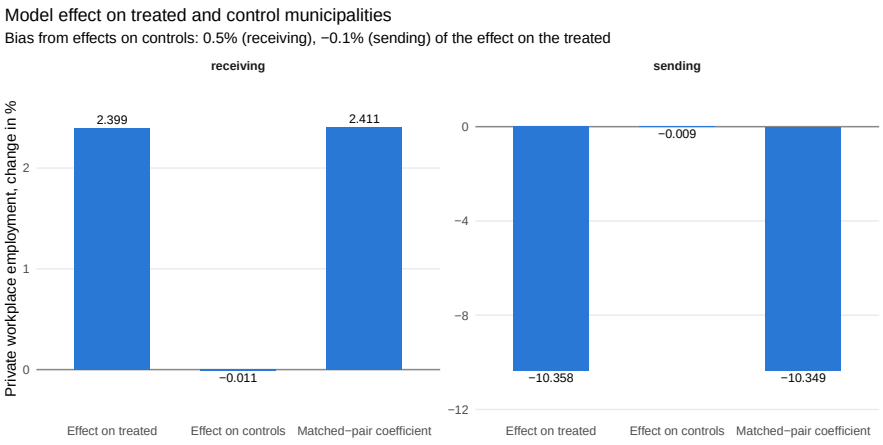
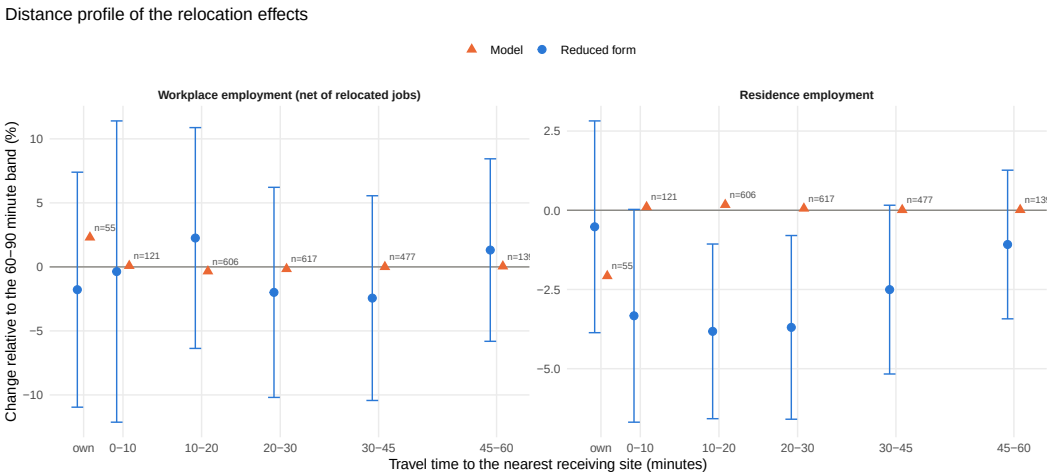
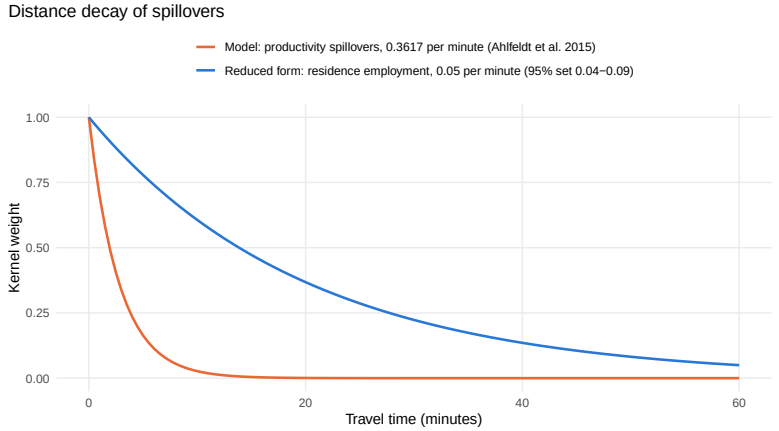


Figure 29. Model effects on treated and control municipalities. Mean model change of private workplace employment ($\log(1+E)$, times 100) in the treated municipalities and in their matched controls, and the matched-set coefficient (treated minus control). The difference between the first and the last bar is the bias that relocation effects on the controls add to the matched design. All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.



(a) Distance profile



(b) Spatial decay

Figure 30. Distance profile of the relocation effects. Panel (a): change at each band of car travel time to the nearest receiving site minus the change at 60 to 90 minutes, in percent. Reduced form (circles): Bavarian municipalities matched to municipalities outside Bavaria, long differences 2014–2024, 95% intervals from county-clustered standard errors. Model (triangles): mean change between the relocation and the zero-shock equilibrium of the Bavarian municipalities in the band; labels give their number. Workplace employment is net of relocated jobs. Panel (b): kernel weights $e^{-\delta t}$ against travel time in minutes for the productivity decay of Ahlfeldt, Redding, et al. (2015) and the decay estimated from residence employment around receiving sites. All municipalities (2014).

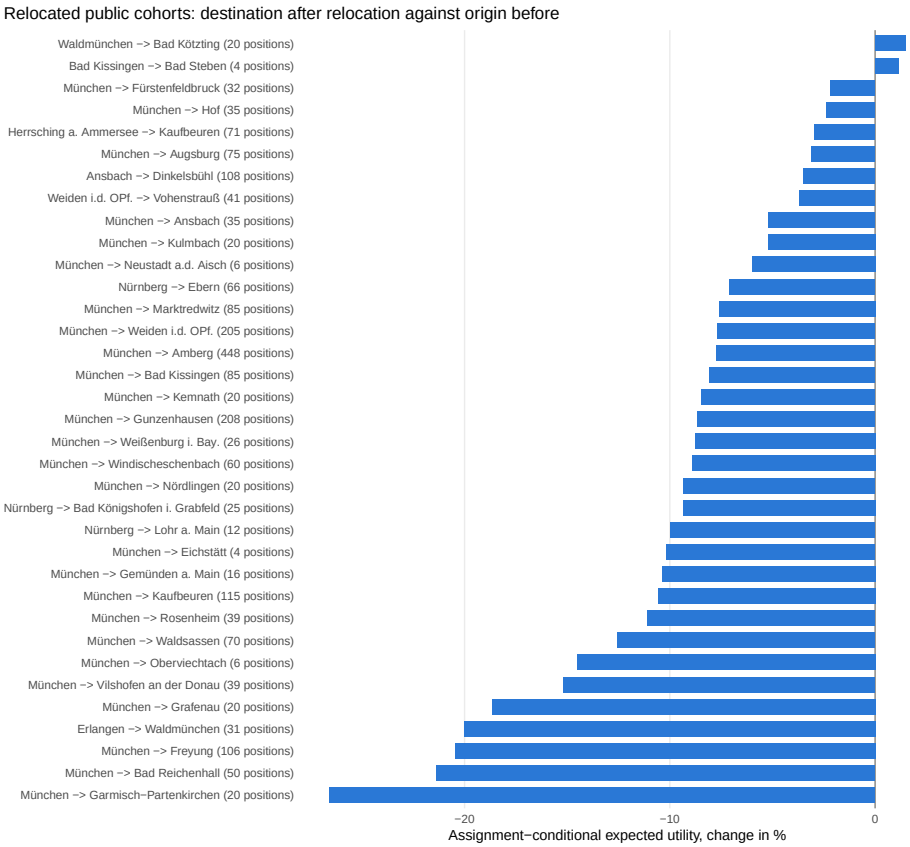


Figure 31. Relocated public cohorts. Change in assignment-conditional expected utility, in percent, of each cohort of named moves: destination after the relocation against origin before it. Cohorts keep their pay, so the change reflects the tax rate and the difference in residential access between destination and origin. With all channels switched off, the job-weighted mean change is almost the same as in the full model (-8.8% against -8.9%). All municipalities (2014), $\widehat{\lambda}_{PG} = 0.0222$.

Appendix E: The Model on Census Towns (2011)

This appendix reports the model quantified on 1,053 Census towns (2011) instead of all municipalities (2014). The baseline for Census towns (2011) uses employment by sector at workplaces and residences from the 2011 Census, which also covers civil servants, residential and office rents for 2011, and the all-employee workplace wage of 2011 from the Regional Atlas for the Fréchet shape. Public employment is Census industry O. The towns form a closed economy, so I rescale workplace counts to the residence totals of each sector. Travel times in minutes, the commuting decay, the maintained parameters, the reduced-form targets, and the relocation shock follow the quantification on all municipalities (2014). Table 20 compares the two quantifications.

Not every relocation and matched set falls inside the Census towns (2011). Four relocations with 182 positions go to Hallbergmoos, Ruhstorf an der Rott, Viechtach, and Waldkirchen, which are not in the sample, and have no named sending municipality; they drop out of the shock, which comprises 3,674 positions. For the same reason, 38 of the 55 receiving matched sets and 2 of the 3 sending matched sets have both members in the sample. The model coefficients use these sets, while the targets remain the reduced-form coefficients on all sets.

On Census towns (2011), the estimated public-to-private elasticity is 0.038 ($J = 3.96$), against 0.022 on all municipalities (2014); the receiving-only estimates are 0.017 and 0.014 (Figure 32). The model matches the targeted private workplace coefficient within one standard error (0.036 against 0.015) and comes closer to the local private employment coefficient (0.014 against 0.038, standard error 0.014) than on all municipalities (2014). As on all municipalities (2014), it predicts a decline of total residence employment at receiving towns (-0.017 against 0.029; Figure 33).

Private workplace employment in the 51 net receiving Census towns (2011) rises by 1.05%, against 0.72% in the 55 net receiving municipalities (2014), while employment among residents falls by 0.43% and 0.73%. National private output falls by 0.0136% on Census towns (2011) and by 0.0090% on all municipalities (2014); private expected utility rises by 0.0059% and 0.0020%. The larger responses on Census towns (2011) are in line with the larger estimated elasticity. Public-to-private spillovers again account for the receiving gain, with private agglomeration as the second channel (Figure 34). Choosing the highest-value destinations reduces the national output loss only slightly on Census towns (2011) as well, from 0.0136% to 0.0132%, and choosing the lowest-wage destinations more than doubles it, to 0.0301%. The baseline tax rate is higher on Census towns (2011), 7.46% against 5.30%, because the Census counts civil servants, so public payroll is a larger share of the wage bill.

Table 20. The model on Census towns (2011) and on all municipalities (2014)

	Census towns (2011)	All municipalities (2014)
Baseline		
Locations	1,053	10,637
Locations in Bavaria	215	2,055
Private workers $\bar{H}^P$ (million)	22.77	34.66
Public workers $\bar{H}^G$ (million)	1.72	1.79
Parameters		
Fréchet shape ϵ	5.312	5.334
Commuting decay $\kappa\epsilon$ (per minute)	0.0397	0.0397
Commuting cost κ (per minute)	0.00748	0.00745
Rent ratio ξ_i, median	1.151	1.200
Office intensity γ_G	0.218	0.292
Baseline tax rate t^0 (%)	7.46	5.30
Estimation		
Receiving matched sets in the sample	38 of 55	55 of 55
Sending matched sets in the sample	2 of 3	3 of 3
$\widehat{\lambda}_{PG}$	0.0383	0.0222
J at the minimum	3.96	6.86
λ_{PG}, receiving private workplace moment alone	0.0171	0.0144
Private workplace, receiving (reduced form 0.0153)	0.0356	0.0241
Local private employment, receiving (reduced form 0.0378)	0.0140	0.0023
Total residence, receiving, untargeted (reduced form 0.0287)	−0.0171	−0.0204
Relocation counterfactual		
Relocated positions	3,674	3,856
Net receiving / net sending locations	51 / 3	55 / 3
Private workplace employment, receiving (%)	1.05	0.72
Employed residents, receiving (%)	−0.43	−0.73
Private workplace employment, sending (%)	−0.56	−0.49
Private jobs per net position received	3.59	2.77
Private output, Germany (%)	−0.0136	−0.0090
Private output, Bavaria (%)	0.0643	0.0087
Private expected utility (%)	0.0059	0.0020
Concentration of private output, HHI (%)	−0.27	−0.14
Channels: receiving private workplace employment (percentage points)		
Public-to-private spillover	1.11	0.76
Private agglomeration	0.38	0.25
Office relocation	−0.014	−0.017
Allocation: private output, Germany (%)		
Actual destinations, no state-built floor space	−0.0136	−0.0090
Highest marginal value	−0.0132	−0.0089
Lowest baseline wage	−0.0301	−0.0198

Notes: Model coefficients are treated-minus-control differences in $\log(1+E)$, 2014–2024, on the matched sets inside the sample; reduced-form targets are the coefficients on all sets. Counterfactual changes compare the relocation equilibrium with the no-relocation baseline. Channel contributions are the full-model change minus the change without the channel. The allocation experiments use 51 destinations on Census towns (2011) and 55 on all municipalities (2014), drawn from 87 and 109 candidates.

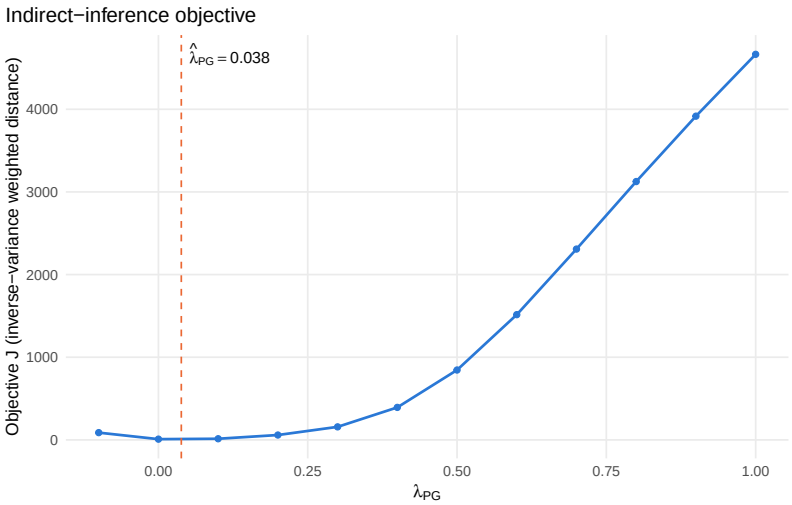


Figure 32. Indirect-inference objective, Census towns (2011). $J(\lambda_{PG})$ on a grid with step 0.1; the dashed line marks the minimum, $\widehat{\lambda}_{PG} = 0.0383$ with $J = 3.96$. Targets: receiving private workplace and local private employment, log changes 2014–2024, 38 matched sets inside the sample.

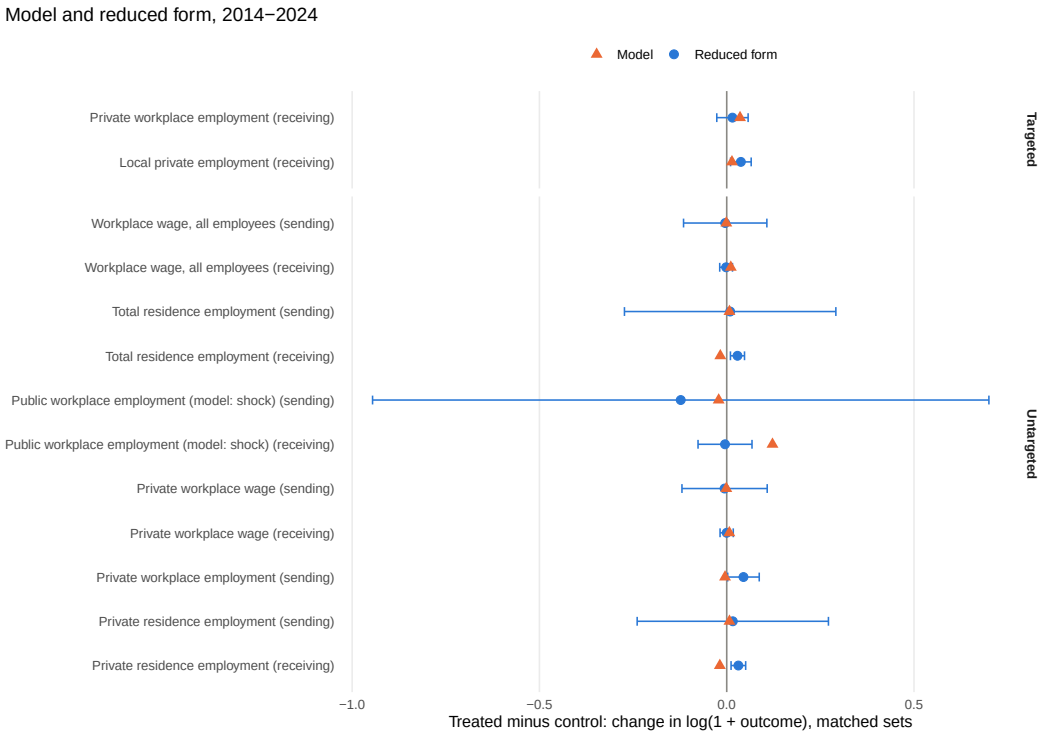


Figure 33. Model and reduced form, Census towns (2011). Reduced-form coefficients on all matched sets (circles) with 95% intervals based on standard errors clustered by matched set, and model coefficients on the matched sets inside the sample (triangles). Treated minus control, $\log(1 + E)$ changes 2014–2024. Targeted: receiving private workplace and local private employment.

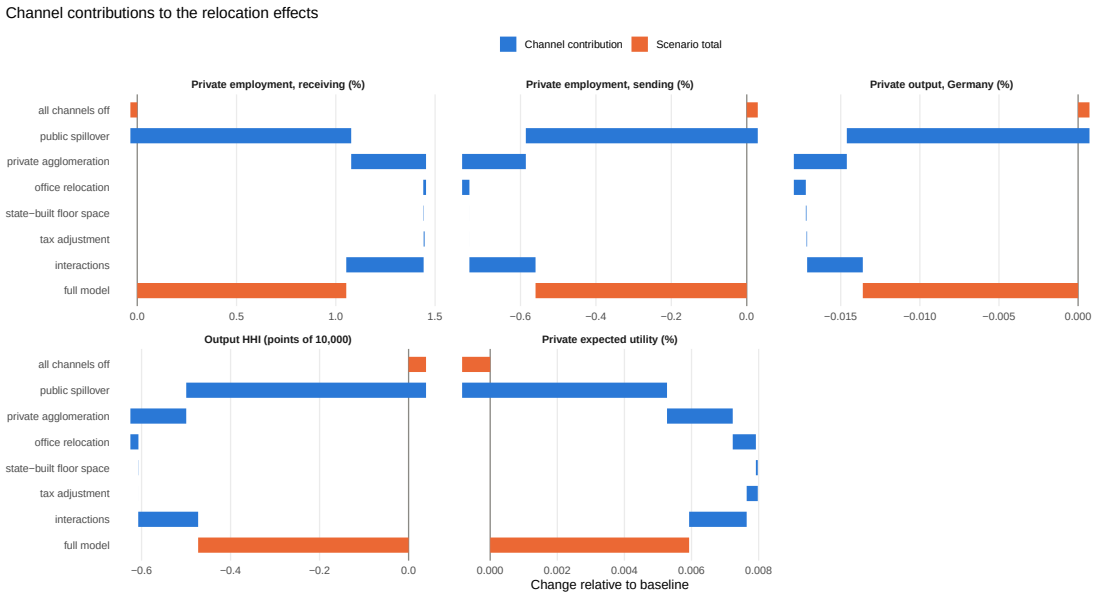


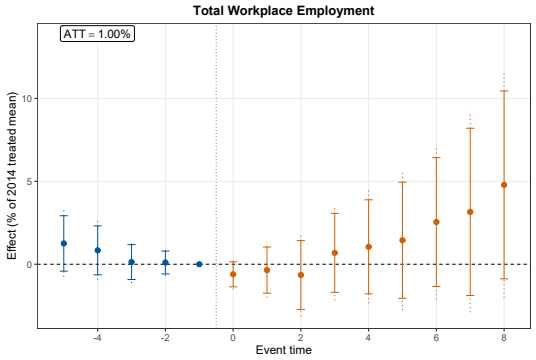
Figure 34. Channel contributions, Census towns (2011). The contribution of a channel is the full-model change minus the change without that channel. The bars run from the change with all channels off through each contribution and the interactions between channels to the full-model change.

Appendix F: Additional Robustness Checks

These checks change one comparison at a time and retain the distinction between employment at workplaces and at residences.

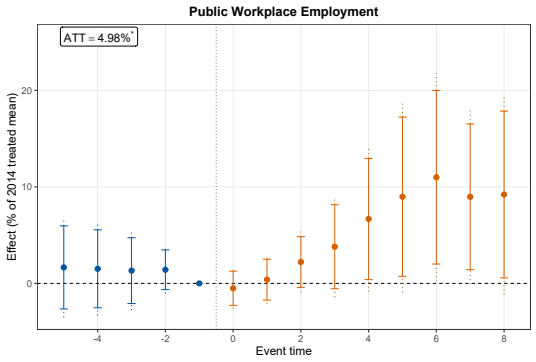
Regionalisierung von Verwaltung receivers

The main receiving estimates cover all relocations of the program. The following figures repeat the receiving event studies for the 40 receiving municipalities of the *Regionalisierung von Verwaltung* component, matched with the same procedure.



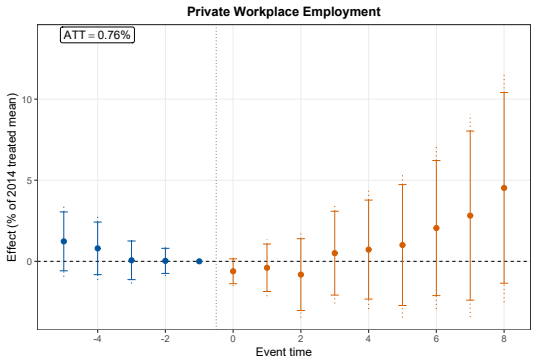
(a) Receiving. The 2014 treated mean is 9,624 employees.

Figure 35. Total workplace employment around the first relocation. Receiving municipalities of the *Regionalisierung von Verwaltung* component. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public and private employment. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



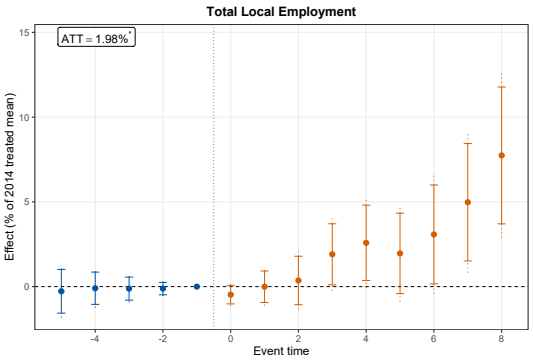
(a) Receiving. The 2014 treated mean is 531 employees.

Figure 36. Public workplace employment around the first relocation. Receiving municipalities of the *Regionalisierung von Verwaltung* component. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public administration, defense, and compulsory social security. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

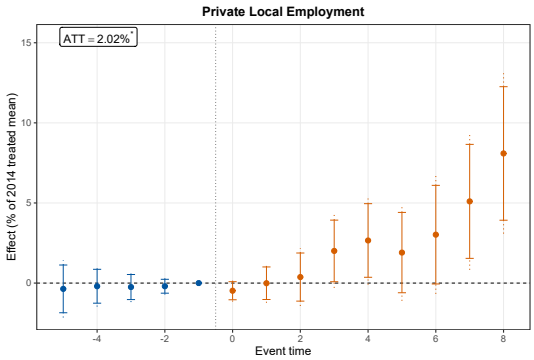


(a) Receiving. The 2014 treated mean is 9,093 employees.

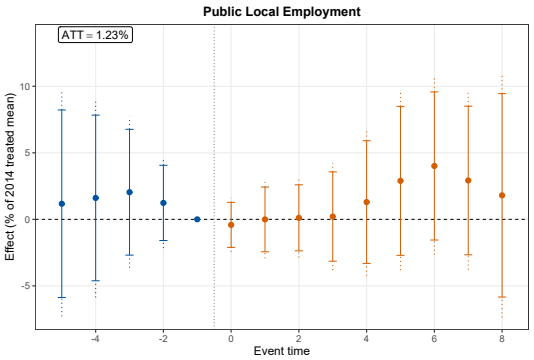
Figure 37. Private workplace employment around the first relocation. Receiving municipalities of the *Regionalisierung von Verwaltung* component. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in industries outside public administration. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Total: Receiving. The 2014 treated mean is 3,567 employees.



(b) Private: Receiving. The 2014 treated mean is 3,368 employees.

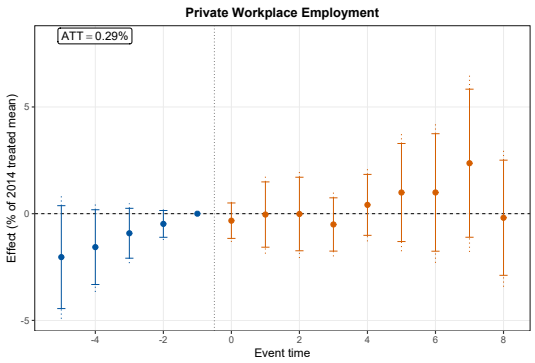


(c) Public: Receiving. The 2014 treated mean is 199 employees.

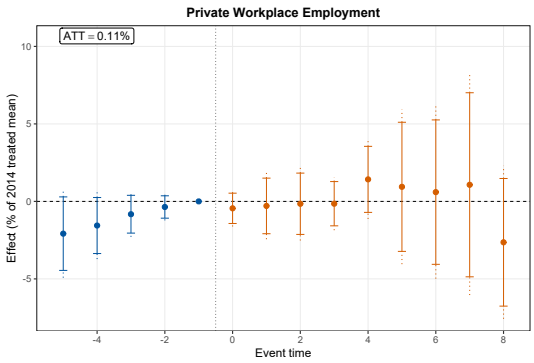
Figure 38. Local employment around the first relocation. Receiving municipalities of the *Regionalisierung von Verwaltung* component. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees who live and work in the same municipality. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Estimator and event-time support

Figures 39 and 40 compare Sun–Abraham and two-way fixed-effect event studies for receiving and sending municipalities. The estimators aggregate treatment effects differently.

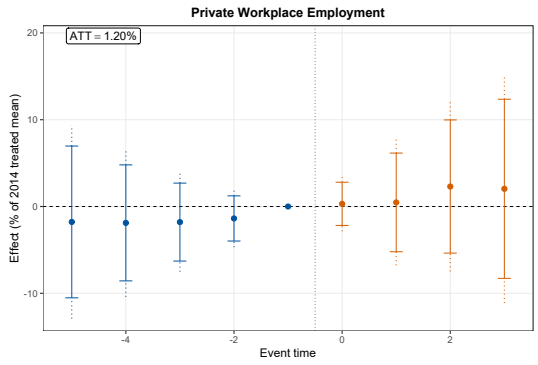


(a) Sun–Abraham: workplace. The 2014 treated mean is 24,691 employees.

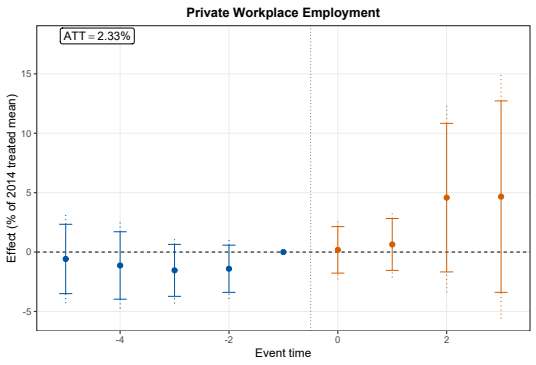


(b) TWFE: workplace. The 2014 treated mean is 24,691 employees.

Figure 39. Receiving private workplace employment dynamics under alternative estimators, using the outside-Bavaria matching design. Table 21 reports estimates and interval conventions at event year five. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.



(a) Sun–Abraham: workplace. The 2014 treated mean is 226,750 employees.



(b) TWFE: workplace. The 2014 treated mean is 226,750 employees.

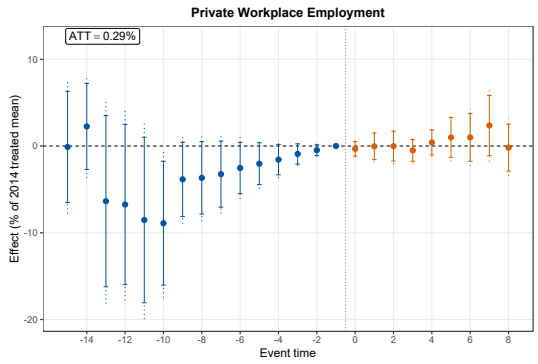
Figure 40. Sending private workplace employment dynamics under alternative estimators, using the outside-Bavaria matching design. Panel scales differ. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Municipalities	Measure	Estimator	Estimate (SE)	Interval
Receiving	Workplace	SA	4.28 (2.02)	[0.36, 8.21]
Receiving	Workplace	TWFE	4.28 (2.02)	[0.36, 8.21]
Receiving	Residence	SA	3.38 (1.18)	[1.01, 5.74]
Receiving	Residence	TWFE	3.04 (1.35)	[0.51, 5.57]
Sending	Workplace	SA	2.27 (1.61)	[-0.88, 5.42]
Sending	Workplace	TWFE	2.64 (1.73)	[-0.76, 6.04]
Sending	Residence	SA	0.83 (0.67)	[-0.48, 2.13]
Sending	Residence	TWFE	1.33 (1.53)	[-1.66, 4.33]

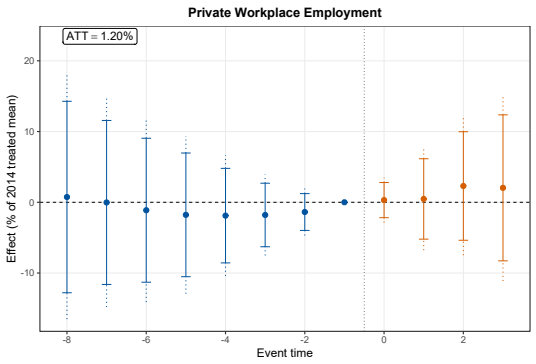
Table 21. Estimator comparison five years after first treatment. All entries are expressed as percentages of the corresponding 2014 treated mean. SA and TWFE intervals are pointwise 95% intervals based on municipality-clustered standard errors. Samples and specifications correspond to Figures 39 and 43.

For receiving workplace employment at event year five, the Sun–Abraham estimate is about 4.3% of 2014 private workplace employment.

Figure 41 reports the full Sun–Abraham event-time ranges, which extend the windows shown above. Cohorts enter each event time only where their outcomes are observed, so remote leads and lags rest on changing cohort support.



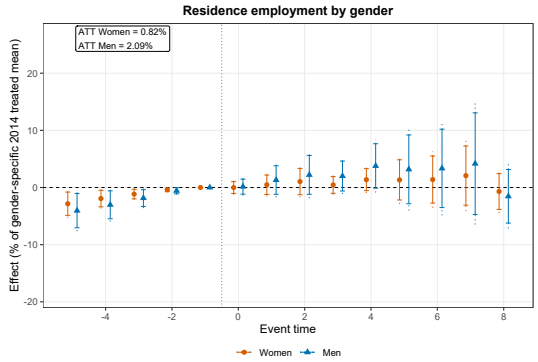
(a) Receiving: workplace. The 2014 treated mean is 24,691 employees.



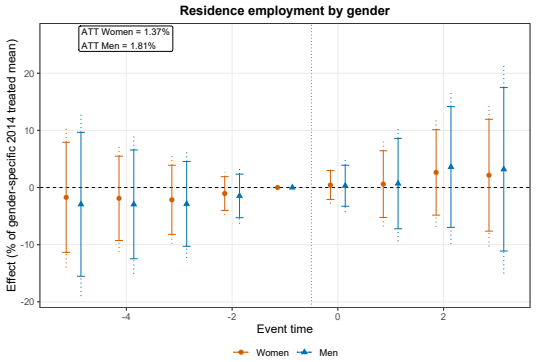
(b) Sending: workplace. The 2014 treated mean is 226,750 employees.

Figure 41. Full event-time support for private workplace employment. Sun–Abraham estimates for receiving municipalities in the left panel and sending municipalities in the right panel. The contributing cohorts change across horizons. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Appendix G: Residence and Local Employment

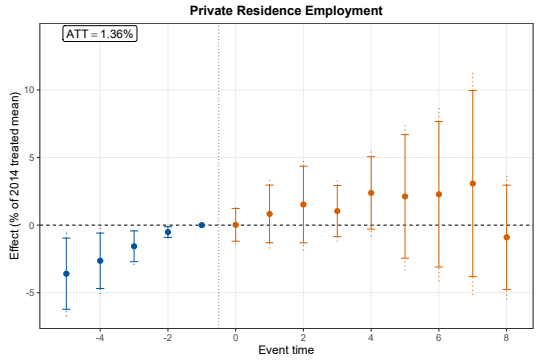


(a) Receiving. The 2014 treated means are 8,714 women and 9,017 men.

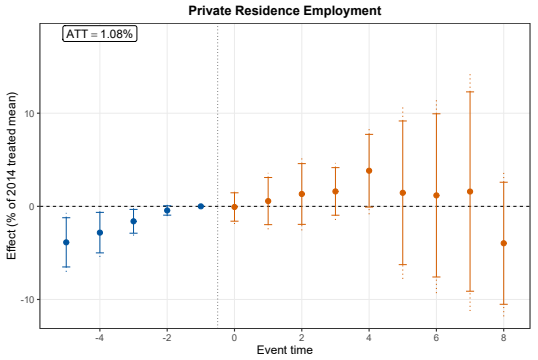


(b) Sending. The 2014 treated means are 84,978 women and 87,427 men.

Figure 42. Residence employment by gender around the first relocation. Sun–Abraham estimates relative to matched municipalities outside the policy state. The year before treatment is the reference. Estimates and confidence intervals are expressed as percentages of the respective gender's 2014 treated mean. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

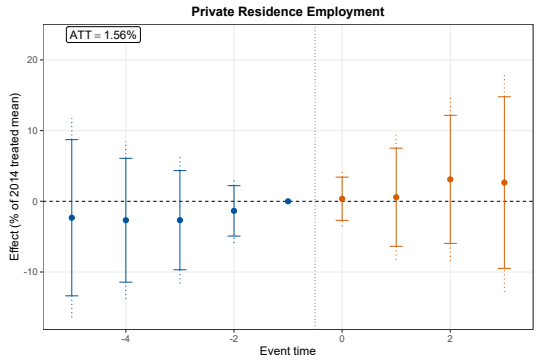


(a) Sun–Abraham: residence. The 2014 treated mean is 16,880 employees.

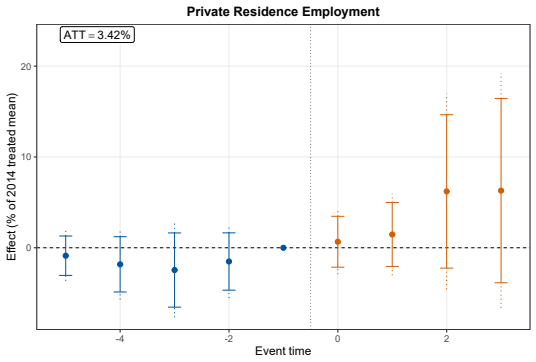


(b) TWFE: residence. The 2014 treated mean is 16,880 employees.

Figure 43. Receiving private residence employment dynamics under alternative estimators, using the outside-Bavaria matching design. Table 21 reports estimates and interval conventions at event year five. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

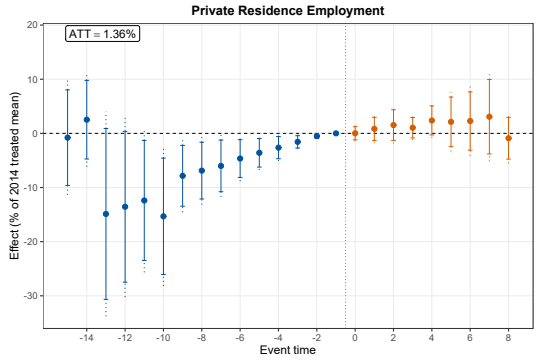


(a) Sun–Abraham: residence. The 2014 treated mean is 165,787 employees.

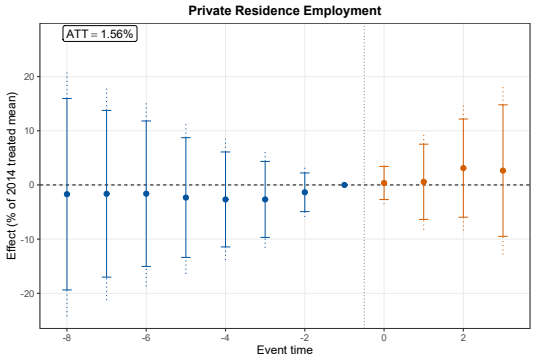


(b) TWFE: residence. The 2014 treated mean is 165,787 employees.

Figure 44. Sending private residence employment dynamics under alternative estimators, using the outside-Bavaria matching design. Panel scales differ. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.



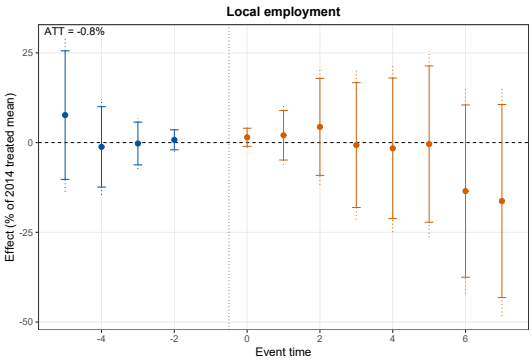
(a) Receiving: residence. The 2014 treated mean is 16,880 employees.



(b) Sending: residence. The 2014 treated mean is 165,787 employees.

Figure 45. Full event-time support for private residence employment. Sun–Abraham estimates for receiving municipalities in the left panel and sending municipalities in the right panel. The contributing cohorts change across horizons. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Figure 46 reports event studies for sending locations without Munich using total employment from municipal statistics, which includes public and private employees.



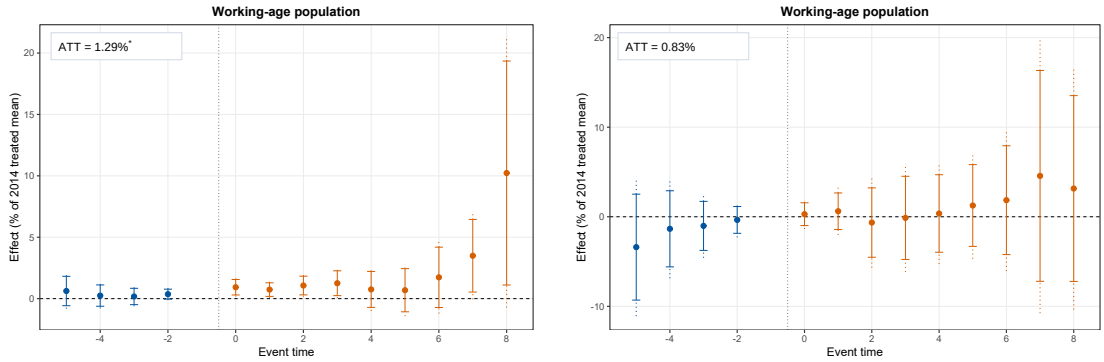
(a) Local employment. The 2014 treated mean is 33,674 employees.

Figure 46. Sending-location employment responses excluding Munich. Sun–Abraham estimates expressed as percentages of the 2014 treated mean using one-to-one outside-Bavaria matches on levels and pre-policy trends. Employment includes public and private employees. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Appendix H: Additional Place-Level Results

Population Adjustment

Receiving municipalities gain working-age residents, although the population response is modest. Their working-age population rises by 0.9% in the year of the first relocation and remains about 1% higher than controls over the subsequent three years (Figure 47). The average post-treatment gain is 1.3%. In sending municipalities, working-age population rises by 0.8%. However, the population response in the sending locations is not statistically significant.



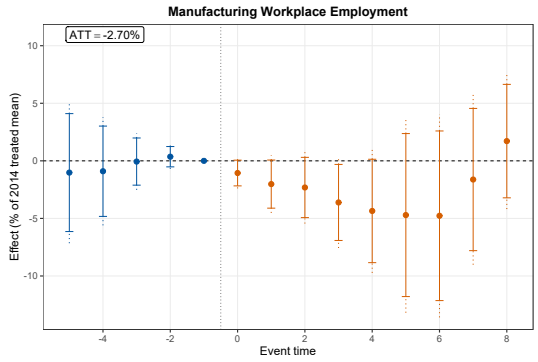
(a) Receiving. The 2014 treated mean is 9,831 residents. **(b)** Sending. The 2014 treated mean is 286,283 residents.

Figure 47. Working-age population around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Residents of working age. The series begin in 2011. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

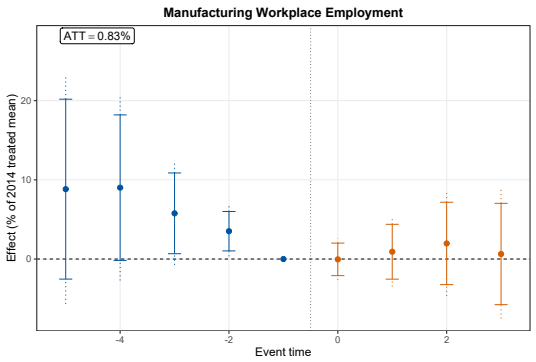
Sector Employment

I next examine how these employment responses differ across sectors. Manufacturing does not contribute to the employment expansion in receiving municipalities. Its response is negative over most of the post-treatment period: manufacturing employment falls by 3.6% after three years and by 4.7% after five years (Figure 48). The average post-treatment change is −2.7%. The expansion therefore does not extend to a sector that serves markets beyond the municipality, so the data do not point to a spillovers channel beyond local consumption. Sending municipalities have a small positive, but not statistically significant average manufacturing response of 0.8% that seems cyclical and unrelated to the policy implementation.

Aggregate service employment changes little in receiving municipalities and rises modestly in sending municipalities. The average responses are 0.2% and 1.4%, respectively (Figure 49). Neither aggregate change is statistically significant. Breaking down services by subsector reveals different sources of employment growth at the two ends of the



(a) Receiving. The 2014 treated mean is 5,513 employees.



(b) Sending. The 2014 treated mean is 32,723 employees.

Figure 48. Manufacturing employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

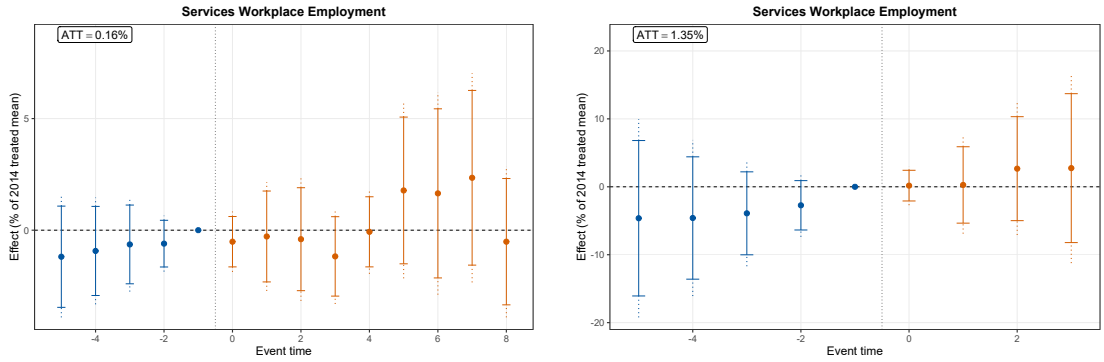
program: knowledge services expand in sending municipalities, while food and beverage services expand in receiving municipalities. Arts, entertainment, and recreation remain broadly unchanged in both groups.

Knowledge services expand in sending municipalities as public jobs move elsewhere. Knowledge-intensive services have an average post-treatment increase of 2.0%, while the narrower knowledge-intensive business services category increases by 2.3% (Figure 50). The larger proportional response in business services points to growth in activities that serve other firms. Knowledge services therefore explain most of the expansion of private employment in sending municipalities.

Food and beverage services expand in receiving municipalities, consistent with increased demand for local consumption. Employment in these services is slightly lower than in the controls during the first five years, and then rises by 6.4% after six years and 11.8% after seven years (Figure 51). The average post-treatment change is close to zero. This late response follows the build-up of public employment and suggests that local consumption contributes to the receiving municipalities' private employment growth as the program matures. Arts, entertainment, and recreation remain broadly unchanged in both receiving and sending municipalities, so the expansion of local consumption services is concentrated in food and beverage activities.

Employment by Gender

The relocations change the gender composition of public and private employment in opposite directions. The worker-level results in Section 4 show that women make up a larger share of employment in receiving agencies after the move. In the private sector of receiving municipalities, however, the female employment share decreases. The agencies become more female, while the composition of private employment shifts toward men.



(a) Receiving. The 2014 treated mean is 13,055 employees.

(b) Sending. The 2014 treated mean is 146,815 employees.

Figure 49. Service employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

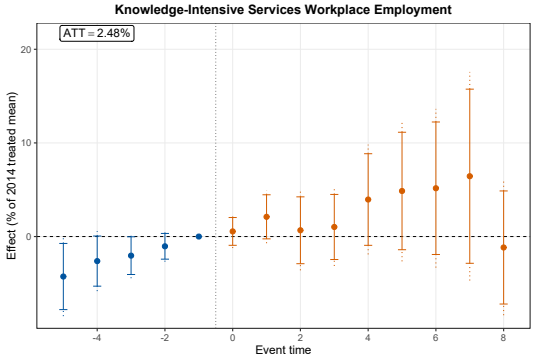
These patterns suggest that gender differences in sector specialization relocate with the agencies.

Factor Prices

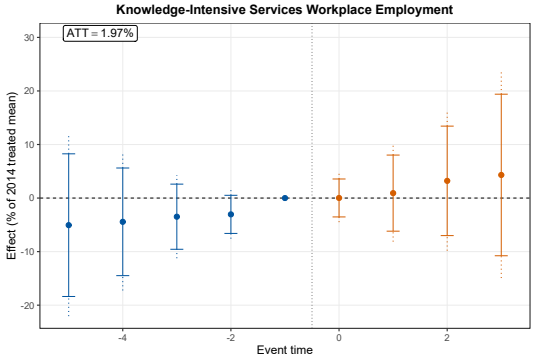
Private wages rise in receiving municipalities and decrease slightly in sending municipalities. The average post-treatment changes are approximately 0.5% and −0.7%, respectively (Figure 53). Receiving municipalities therefore gain private employment alongside a modest rise in average private pay, while private employment growth in sending municipalities is accompanied by slightly lower average pay. Neither *private* wage trajectory is statistically significant.

Public wages move in the opposite direction: average public workplace wages decrease in receiving municipalities and increase in sending municipalities. The average post-treatment changes are approximately −0.8% and 1.0%, respectively (Figure 54). Changes in the workforce offer a possible explanation. The receiving agencies hire younger workers who earn less than the former agency workforce, while many incumbent workers remain in the sending locations at other public agencies where their nominal wages rise. The municipal wage pattern is consistent with those descriptive findings. Average public wages can therefore change as the composition of employment changes, without every incumbent worker experiencing the same wage change. Neither average public wage response is statistically significant.

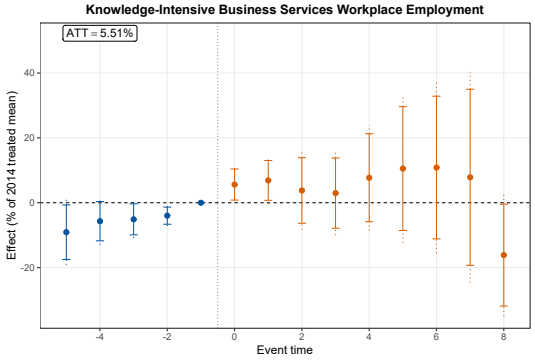
Residential rents in receiving municipalities initially fall modestly and rise at longer horizons, despite the earlier growth in local employment. Five years after the first relocation, the estimate is −3.3%, and reaches 9.8% after eight years (Figure 55). The average post-treatment response is a small decline of −0.6%. The trajectory suggests that employment expands before higher rents emerge in receiving municipalities. Sending



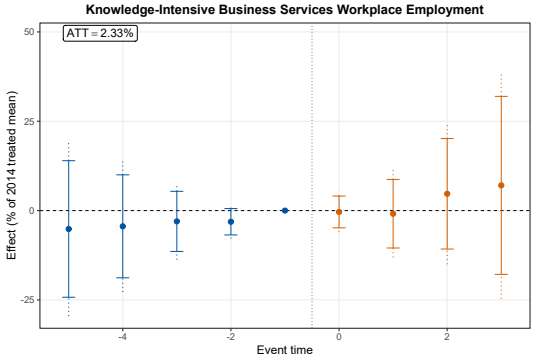
(a) Knowledge-intensive services: Receiving. The 2014 treated mean is 8,111 employees.



(b) Knowledge-intensive services: Sending. The 2014 treated mean is 95,630 employees.

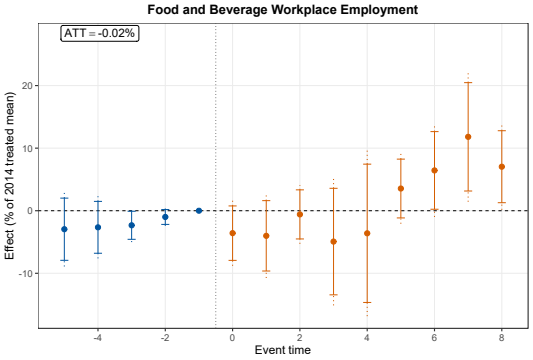


(c) Knowledge-intensive business services: Receiving. The 2014 treated mean is 1,973 employees.

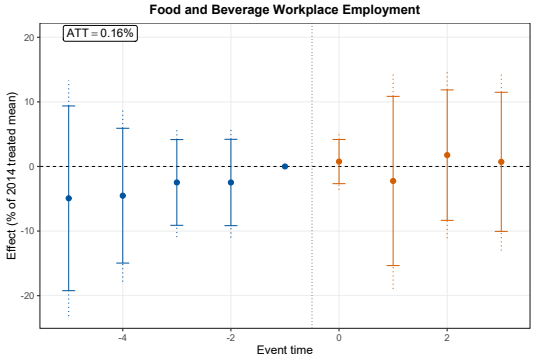


(d) Knowledge-intensive business services: Sending. The 2014 treated mean is 37,888 employees.

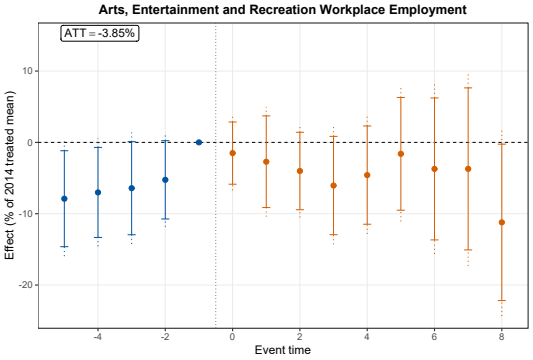
Figure 50. Knowledge services around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



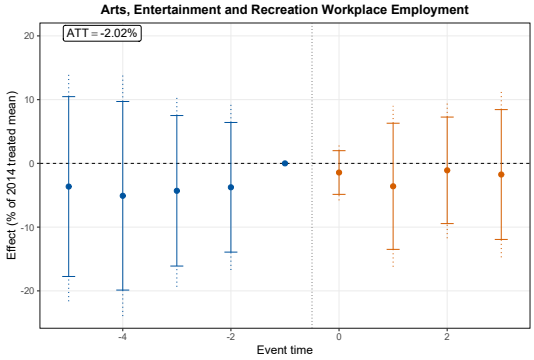
(a) Food and beverage: Receiving. The 2014 treated mean is 1,006 employees.



(b) Food and beverage: Sending. The 2014 treated mean is 9,918 employees.

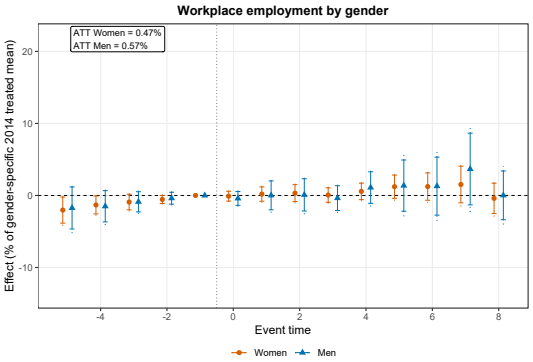


(c) Arts, entertainment, and recreation: Receiving. The 2014 treated mean is 272 employees.

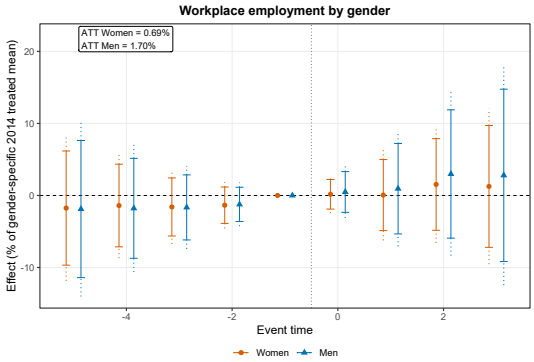


(d) Arts, entertainment, and recreation: Sending. The 2014 treated mean is 3,168 employees.

Figure 51. Local consumption services around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

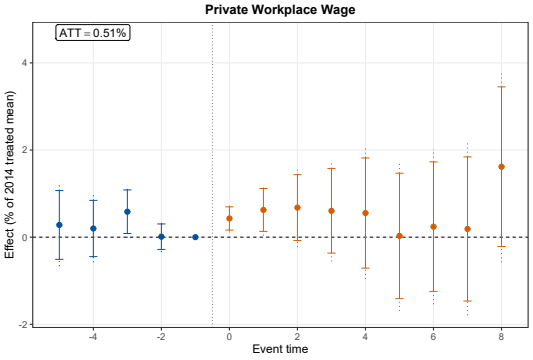


(a) Receiving. The 2014 treated means are 13,063 women and 13,130 men.

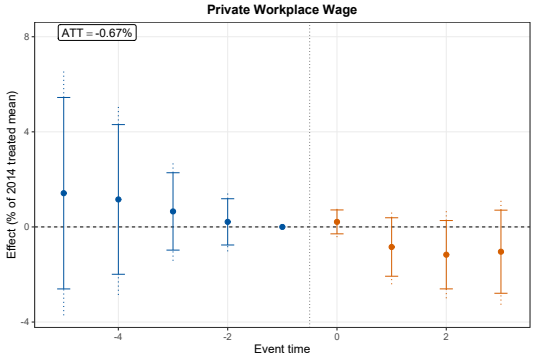


(b) Sending. The 2014 treated means are 115,899 women and 121,787 men.

Figure 52. Workplace employment by gender around the first relocation. Sun–Abraham estimates relative to matched municipalities outside the policy state. The year before treatment is the reference. Estimates and confidence intervals are expressed as percentages of the respective gender's 2014 treated mean. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

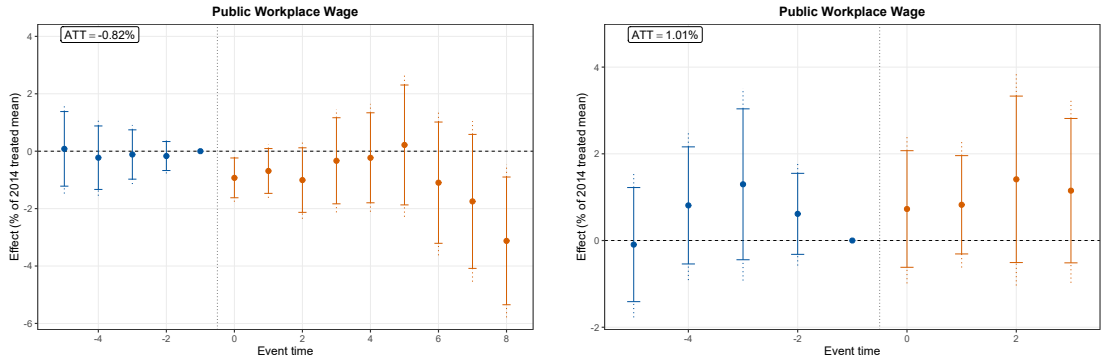


(a) Receiving. The 2014 treated mean is 27 thousand euros.



(b) Sending. The 2014 treated mean is 31 thousand euros.

Figure 53. Private workplace wages around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Nominal gross wages. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



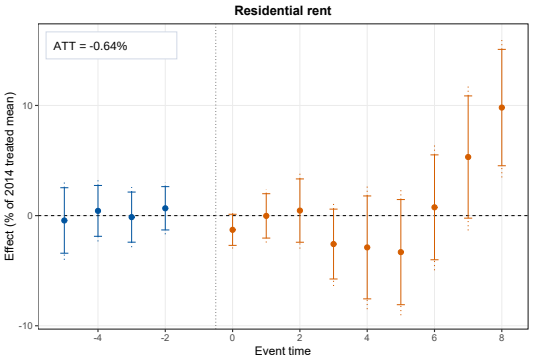
(a) Receiving. The 2014 treated mean is 33 thousand euros.

(b) Sending. The 2014 treated mean is 36 thousand euros.

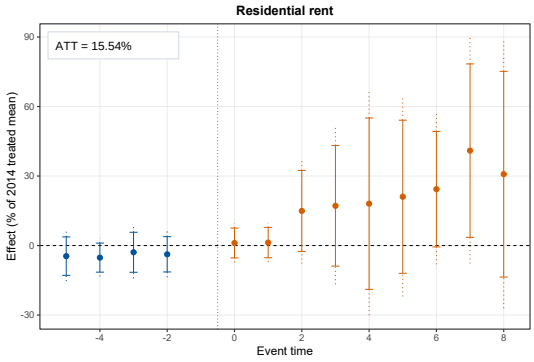
Figure 54. Public workplace wages around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Nominal gross wages. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

municipalities have an average residential rent increase of 15.5%, pointing to a substantial rise in housing costs as public jobs move elsewhere and private jobs increase.

Commercial rents rise modestly on average in receiving municipalities. The average post-treatment response is 2.4%, with a larger increase of 7.7% five years after the first relocation (Figure 56). The pattern is consistent with additional demand for office space from public agencies and private activity. Sending municipalities have a larger positive average response of 10.0%, pointing to rising office rents despite the release of public office space. Freeing public office space need not reduce rents if private firms take up that space and expand. The model in Section 7 accounts for both the release of floor space and the response of private firms.

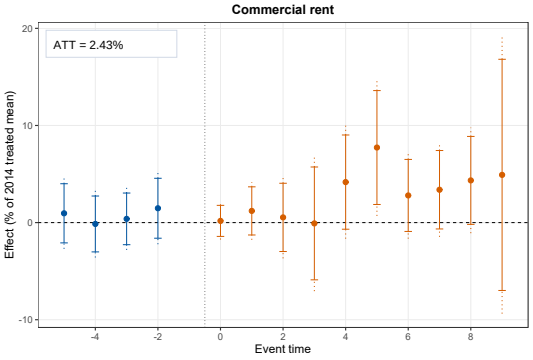


(a) Receiving. The 2014 treated mean is 5.59 euros per square meter.

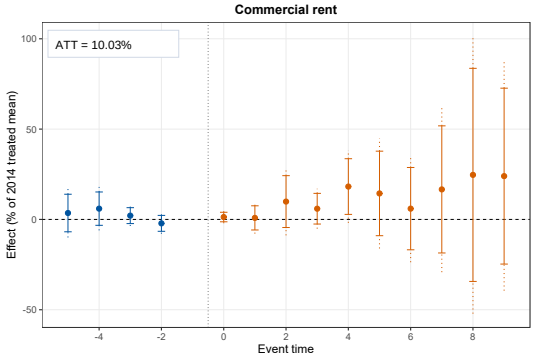


(b) Sending. The 2014 treated mean is 10.40 euros per square meter.

Figure 55. Residential rents around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Receiving. The 2014 treated mean is 6.71 euros per square meter.



(b) Sending. The 2014 treated mean is 9.36 euros per square meter.

Figure 56. Commercial rents around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.